Collection of exercises of nonlinear optimization

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1 Analysis of optimization problems, optimality conditions

1.1 Unconstrained optimization problems

1.1.1 First- and second-order necessary optimality conditions

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a twice continuously differentiable function. Let $\bar{x}$ be a local minimizer of f, that is to say, there exists $\varepsilon > 0$ such that for all $x \in \mathbb{R}^n$,

$$\|x - \bar{x}\| \leq \varepsilon \Longrightarrow f(x) \geq f(\bar{x}).$$

1. Prove that $Df(\bar{x}) = 0$ and that $D^2 f(\bar{x})$ is positive semi-definite.
2. Prove that the definition of a local minimizer is independent of the choice of the norm.
3. We consider now the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

   $$f(x, y) = \cos(x + y) + \sin(xy) + 2y^2.$$

   Is the point $(0, 0)$ a local minimizer?

Solution

1. Easy.
2. Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be two norms. There exists $\alpha > 0$ such that for all $x \in \mathbb{R}^n$,

   $$\|x\|_1 \leq \alpha \|x\|_2.$$

   Assume that $\bar{x}$ is a local minimizer of f for the norm $\|\cdot\|_1$. Let $\varepsilon_1 > 0$ be such that for all $x \in \mathbb{R}^n$,

   $$\|x - \bar{x}\|_1 \leq \varepsilon_1 \Longrightarrow f(x) \geq f(\bar{x}).$$

   Now, let us set $\varepsilon_2 = \varepsilon_1/\alpha$. For all $x \in \mathbb{R}^n$ such that $\|x - \bar{x}\|_2 \leq \varepsilon_2$, we have:

   $$\|x - \bar{x}\|_1 \leq \alpha \|x - \bar{x}\|_2 \leq \alpha \varepsilon_2 = \varepsilon_1$$

   and therefore, $f(x) \geq f(\bar{x})$, which proves that $\bar{x}$ is also a local minimizer of f for the norm $\|\cdot\|_2$.
3. We have:

   $$\nabla f(x) = \begin{pmatrix} -\sin(x + y) + y\cos(xy) \\ -\sin(x + y) + x\cos(xy) + 2y \end{pmatrix}$$

   and

   $$\nabla^2 f(x) = \begin{pmatrix} -\cos(x + y) - y^2 \sin(xy) & -\cos(x + y) + \cos(xy) - xy\sin(xy) \\ -\cos(x + y) + \cos(xy) - xy\sin(xy) & -\cos(x + y) - x^2 \sin(xy) + 4 \end{pmatrix}.$$

   Therefore,

   $$\nabla^2 f(x) = \begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix}.$$

   The second-order necessary optimality conditions are not satisfied. The point $(0, 0)$ is not a local minimizer.

1.1.2 Proving the existence of a minimizer

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a continuous function. We say that the function f is coercive if the following property holds: for all sequence $(x_k)_{k\in\mathbb{N}}$ in $\mathbb{R}^n$,

$$\|x_k\| \underset{k\to\infty}{\longrightarrow} +\infty \Longrightarrow f(x_k) \underset{k\to\infty}{\longrightarrow} +\infty.$$

1. Assume that f is coercive and prove the existence of a global minimizer. To this purpose, you can first prove that for a fixed $x_0 \in \mathbb{R}^n$, the following set A is non-empty and compact:

   $$A = \big\{x \in \mathbb{R}^n \,|\, f(x) \leq f(x_0)\big\}.$$

2. Prove that the coercivity of f is independent of the choice of the norm.
3. Compute a minimizer of $f(x, y) = x^4 + x^2 + y^2 - xy$. Justify carefully your answer.

Solution

1. The set A is also defined by:
$$A = f^{-1}\big((-\infty, f(x_0)]\big)$$
which is non-empty (it contains x_0) and closed (by theorem, since f is continuous and $(-\infty, f(x_0)]$ closed). If it is non bounded, we get a contradiction to the coercivity property. Thus it is compact. By theorem, $f(A)$ is a compact interval, thus, there exists $\bar{x} \in A$ such that for all $x \in A$, $f(x) \geq f(\bar{x})$. For all $x \notin A$, $f(x) > f(x_0) \geq f(\bar{x})$, thus, $\bar{x}$ is a global minimizer.

2. Let us consider two norms, say $\|\cdot\|_1$ and $\|\cdot\|_2$. Since $\mathbb{R}^n$ is a finite-dimensional space, these norms are equivalent: there exist $\alpha > 0$ and $\beta > 0$ such that for all $x \in \mathbb{R}^n$,
$$\alpha\|x\|_1 \leq \|x\|_2 \leq \beta\|x\|_1. \tag{1}$$
If $(x_k)_k$ is a sequence such that $\|x_k\|_1 \to +\infty$, then $\|x_k\|_2 \geq \alpha\|x_k\|_1 \to +\infty$, meaning that we can use any norm for the coercivity property.

3. We can write:
$$f(x,y) = x^4 + \frac{1}{2}(x^2+y^2) + \frac{1}{2}(x-y)^2 \geq \frac{1}{2}\|(x,y)\|^2,$$
where $\|\cdot\|$ is the Euclidean norm. The coercivity property is satisfied. Therefore, there exists an optimal solution.

   Any $(x,y) \in \mathbb{R}^2$ is a stationary point if and only if:
$$\begin{cases} 4x^3 + 2x - y = 0 \\ 2y - x = 0 \end{cases} \iff \begin{cases} y = \frac{1}{2}x \\ x(4x^2 + \frac{3}{2}) = 0 \end{cases} \iff x = y = 0.$$
There is a unique stationary point, $(0,0)$, which is therefore the unique optimal solution.

1.1.3 Optimization on the set of symmetric positive matrices (1)

In this exercise and in the following one, the set of symmetric matrices, the set of symmetric positive semi-definite matrices, and the set of symmetric positive definite matrices (of dimension n) are respectively denoted by $\mathcal{S}_n$, $\mathcal{S}_n^+$ and $\mathcal{S}_n^{++}$. The set $\mathcal{S}_n$ is identified with $\mathbb{R}^{n(n+1)/2}$.

Let A and $B \in \mathcal{S}_n^{++}$. We investigate the existence of a solution to the following optimization problem:
$$\inf_{X \in \mathcal{S}_n^{++}} f(X) := \mathrm{tr}(AX) + \mathrm{tr}(BX^{-1}).$$

1. Prove that $\mathcal{S}_n^{++}$ is an open subset of $\mathcal{S}_n$. Prove also that $\mathcal{S}_n^+$ is the closure of $\mathcal{S}_n^{++}$.

2. Prove that for all invertible matrices $P \in \mathbb{R}^{n\times n}$,
$$\forall X \in \mathcal{S}_n^+,\ PXP^\top \in \mathcal{S}_n^+ \quad \text{and} \quad \forall X \in \mathcal{S}_n^{++},\ PXP^\top \in \mathcal{S}_n^{++}.$$

3. Prove that for all $X \in \mathcal{S}_n^+$, $\mathrm{tr}(AX) \geq 0$. Prove also that for all $X \in \mathcal{S}_n^+$, $\mathrm{tr}(AX) = 0 \Longrightarrow X = 0$. Hint, for both results: consider first the case when A is diagonal, then extend the results with the spectral theorem.

4. Prove that there exists $\alpha > 0$ such that for all $X \in \mathcal{S}_n^{++}$, $\mathrm{tr}(AX) \geq \alpha\|X\|$.

5. Prove that f is continuous.

6. Prove that for any convergent sequence $(X_k)_{k\in\mathbb{N}}$ in S_n^{++} with limit $\bar{X} \in S_n^+ \backslash S_n^{++}$, $\|X_k^{-1}\| \underset{k\to\infty}{\longrightarrow} +\infty$.

7. Let $(X_k)_{k\in\mathbb{N}}$ be a minimizing sequence (i.e. a sequence in S_n^{++} such that $f(X_k) \underset{k\to\infty}{\longrightarrow} \inf_{X\in\mathcal{S}_n^{++}} f(X)$). Then, at least one of the following three statements is satisfied:
   (a) The sequence posseses a subsequence $(X_{k_\ell})_{\ell\in\mathbb{N}}$ such that $\|X_{k_\ell}\| \underset{\ell\to\infty}{\longrightarrow} +\infty$.
   (b) The sequence posseses an accumulation point in $\mathcal{S}_n^+ \backslash \mathcal{S}_n^{++}$.
   (c) The sequence posseses an accumulation point in $\mathcal{S}_n^{++}$.

   Prove that the two first statements cannot hold. Conclude.

Solution

1. Let $\phi : \mathcal{S}_n \to \mathbb{R}^n$ be the mapping associating with all X the determinants of the n upper-left square submatrices of X. By Sylvester's criterion, $\mathcal{S}_n^{++} = \phi^{-1}(\mathbb{R}_n^{++})$, thus is open since f is continuous and $\mathbb{R}_n^{++}$ is open.

   Let $(X_k)_{k\in\mathbb{N}}$ be a sequence converging to $\bar{X} \in \mathcal{S}_n$. Let $h \in \mathbb{R}^n$, then:
$$h^\top X h = \lim h^\top X_k h \geq 0.$$

2. Obvious.

3. Consider first the case A diagonal. It holds: $\mathrm{tr}(AB) = \sum_{i=1}^n A_{ii}B_{ii} \geq 0$. Moreover, if $\mathrm{tr}(AB) = 0$, then $B_{11} = ... = B_{nn} = 0$, and thus $\mathrm{tr}(B) = 0$ and then, $B = 0$. In the general case, A is diagonalizable. Let $R \in \mathbb{R}^{n\times n}$ and $D \in S_n^{++}$ be a diagonal matrix be such that:
$$A = RDR^\top, \quad RR^\top = I_n.$$
   Then, $\mathrm{tr}(AB) = \mathrm{tr}(RDR^\top B) = \mathrm{tr}(DR^\top BR) \geq 0$, since $R^\top BR \in \mathcal{S}_n^+$. Moreover, if $\mathrm{tr}(AB) = 0$, then $R^\top BR = 0$ and therefore, $B = 0$.

4. Proof by contradiction. Assume that for all $k \in \mathbb{N}\backslash\{0\}$, there exists X_k such that
$$\mathrm{tr}(AX_k) \leq \frac{1}{k}\|X_k\|.$$
   Setting $Y_k = X_k/\|X_k\|$, we obtain:
$$\mathrm{tr}(AY_k) \leq \frac{1}{k} \to 0.$$
   But the sequence $(Y_k)_{k\in\mathbb{N}}$ has an accumulation point $\bar{Y} \in \mathcal{S}_n^+$ such that $\|Y\| = 1$ and $\mathrm{tr}(AY) = 0$, which is a contradiction.

5. The function f can be written as a fractional polynomial.

6. Let $X_k \to \bar{X}$. Let $h \in \mathbb{R}^n\backslash\{0\}$ be such that $\bar{X}h = 0$. Set $y_k = X_k h$. We have: $h = X_k^{-1}y_k$ and therefore:
$$\|X_k^{-1}\| \geq \frac{\|h\|}{\|y_k\|} \to +\infty,$$
   since $y_k \to 0$.

7. In the first case, $f(X_{k_\ell}) \geq \alpha\|X_{k_\ell}\| \to +\infty$, contradiction. In the second case, $f(X_{k_\ell}) \geq \alpha\|X_{k_\ell}^{-1}\| \to +\infty$, contradition. Thus, only the third case is satisfied, and by continuity of f, $\bar{X}$ is a global minimizer.

1.1.4 Optimization on the set of symmetric positive matrices (2)

This exercise is the continuation of the previous one. We first recall the following result: for all $X \in \mathcal{S}_n^{++}$, there exists a unique matrix $Y \in \mathcal{S}_n^{++}$ such that $X = Y^2$. It is denoted $Y = X^{1/2}$ and called square root of X.

1. Let $h : X \in \mathcal{S}_n \to \mathbb{R}$ be a linear form. Prove that there exists a unique matrice $M \in \mathcal{S}_n$ such that for all $X \in \mathcal{S}_n$, $h(X) = \mathrm{tr}(MX)$.

2. Prove that f is differentiable on $\mathcal{S}_n^{++}$.

3. For all $X \in S_n^{++}$, the derivative $Df(X)$ can be identified with a symmetric matrix. This means that $Df(X)$ is the unique symmetric matrix such that: for all $Y \in \mathcal{S}_n$,
$$f(X + Y) = f(X) + \mathrm{tr}(Df(X)Y) + o(\|Y\|).$$
   Compute the derivative of f. Hint: for all $Y \in \mathbb{R}^{n\times n}$ small enough,
$$(I_n + Y)^{-1} = I_n - Y + o(\|Y\|),$$
   where I_n is the identity matrix.

4. Since S_n^{++} is open, any minimizer is a stationary point of f. Prove that the uniqueness of the minimizer.

Solution

1. We first prove the uniqueness. We have: $\mathrm{tr}(MX) = \sum_{i,j=1^n} M_{ij}X_{ij}$. Thus,

$$M_{ij} = h((e_{i,j} + e_{j,i})/2),$$

where $e_{i,j}$ is the matrix defined by:

$$(e_{i,j})_{k,\ell} = \delta_{i,k}\delta_{j,\ell}.$$

This proves the uniqueness. Let M be the matrix defined by (**??**). Then for all $X \in \mathcal{S}_n$,

$$X = \sum_{i,j} \frac{1}{2} X_{ij}(e_{ij} + e_{ji})$$

and therefore,

$$h(X) = \sum_{i,j} X_{ij} h(\frac{1}{2}(e_{ij} + e_{ji})) = \sum_{i,j} X_{ij} M_{ij} = \mathrm{tr}(MX).$$

2. Once again f is a fractional polynomial function, thus differentiable.

3. Observe that:

$$(X+h)^{-1} = (X(\mathrm{Id} + X^{-1}h))^{-1} = (\mathrm{Id} - X^{-1}h)X^{-1} + o(\|h\|) = X^{-1} - X^{-1}hX^{-1} + o(\|h\|).$$

Therefore,

$$f(X+h) = f(X) + \mathrm{tr}(Ah) - \mathrm{tr}(BX^{-1}hX^{-1}) + o(\|h\|) = f(X) + \mathrm{tr}((A - X^{-1}BX^{-1})h) + o(\|h\|).$$

Thus, $Df(X) = A - X^{-1}BX^{-1}$.

4. We already know that there exists a minimizer. Let us prove the uniqueness. Let X_1 be two minimizers, thus two stationary points. Thus,

$$X_1^{-1}BX_1^{-1} = X_2^{-1}BX_2^{-1},$$

thus

$$(B^{1/2}X_1^{-1}B^{1/2})(B^{1/2}X_1^{-1}B^{1/2}) = (B^{1/2}X_2^{-1}B^{1/2})(B^{1/2}X_2^{-1}B^{1/2})$$

thus

$$B^{1/2}X_1^{-1}B^{1/2} = B^{1/2}X_2^{-1}B^{1/2}$$

and finally, $X_1 = X_2$.

1.2 First-order optimality conditions for constrained problems

1.2.1 Projection on a hyperplane

Let $A \in \mathbb{R}^{m\times n}$ be a matrix of rank m, we denote by K the following plane: $\{x \in \mathbb{R}^n : Ax = b\}$, for a given $b \in \mathbb{R}^m$. Let $x_0 \in \mathbb{R}^n$, we consider the following problem:

$$\inf_{x\in K} \frac{1}{2}\|x - x_0\|^2, \tag{2}$$

where $\|\cdot\|$ is the Euclidean norm.

1. Prove that $AA^\top$ is invertible.

2. Prove the existence of a solution to the problem.

3. Write the optimality conditions. Prove that the problem has a unique solution $\bar{x}$, given by

$$\bar{x} = x_0 - A^\top(AA^\top)^{-1}(Ax_0 - b).$$

4. Consider the case $m = 1$. Give a geometrical interpretation of the expression of $\bar{x}$, using in particular the normal vector to the hyperplane K.

Solution

1. Since $A^\top$ is injective, one can easily check that $AA^\top$ is symmetric positive-definite, thus regular.

2. It is clear that all minimizing sequence is bounded, thus has a limit point, which is feasible since the constraints are linear. Therefore, the limit point is a global optimal solution. The derivative of the constraints is always the matrix A, which is of rank m, therefore all the points are qualified.

3. The Lagrange function is $L(x, \lambda) = \frac{1}{2}\|x - x_0\|^2 + \langle \lambda, Ax - b\rangle$. Therefore, the optimality conditions are:
$$(x - x_0) + A^\top \lambda = 0, \quad Ax = b. \tag{3}$$
Multiplying by A on the left, we find:
$$(b - Ax_0) + AA^\top \lambda = 0, \tag{4}$$
thus, $\lambda = (AA^\top)^{-1}(Ax_0 - b)$, and we find the value of x with:
$$x = x_0 - A^\top \lambda.$$

4. If $m = 1$, then A is a line vector, and we can write: $A = n^\top$, where n is the normal vector of the hyperplane. The value of $\bar{x}$ is then given by:
$$\bar{x} = x_0 - \frac{\langle n, x_0\rangle - b}{\|n\|^2} n. \tag{5}$$

1.2.2 Optimality conditions

Consider the following constrained problem:

$$\min_{x \in \mathbb{R}^n} f(x), \quad \text{subject to: } \|x\| \leq \delta,$$

where f is continuously differentiable, $\|\cdot\|$ is the Euclidean norm, and $\delta > 0$ is given. Let $\bar{x}$ be a local solution to the problem. The goal of the exercise is to derive optimality conditions for $\bar{x}$. If $\|\bar{x}\| < \delta$, then clearly $\nabla f(\bar{x}) = 0$, thus we assume from now on that $\|\bar{x}\| = \delta$.

1. Prove that for all $h \in \mathbb{R}^n$, if $\langle \bar{x}, h\rangle < 0$, then there exists $\bar{\theta} > 0$ (depending on h) such that for all $\theta \in [0, \bar{\theta}]$, $\|\bar{x} + \theta h\| \leq \delta$.

2. Prove that there exists $\bar{\lambda} \geq 0$ such that $\nabla f(\bar{x}) + \bar{\lambda}\bar{x} = 0$.

3. Assume now that f is convex. Prove that for all $x \in \mathbb{R}^n$, if $\|x\| = \delta$ and if there exists $\bar{\lambda} \geq 0$ such that $\nabla f(x) + \bar{\lambda}x = 0$, then x is a global solution to the problem.

4. Assume now that f is strongly convex. Prove that for all $\lambda \geq 0$, there exists a unique point $x(\lambda)$ such that $\nabla f(x(\lambda)) + \lambda x(\lambda) = 0$.

5. Prove that the mapping $\lambda \geq 0 \mapsto \|x(\lambda)\|$ is decreasing.

Solution

1. Let us fix h such that $\langle \bar{x}, h\rangle < 0$. Consider the function $\phi(\theta) = \|\bar{x} + \theta h\|$. Clearly, $\phi(0) = \delta^2$ and $\phi'(0) < 0$, thus for θ small enough, $\phi(\theta) \leq \phi(0)$.

2. We deduce the inequality: $\langle \nabla f(\bar{x}), h\rangle \geq 0$. By continuity, this inequality also holds true for vectors h such that $\langle \bar{x}, h\rangle = 0$. Therefore, $\nabla f(\bar{x}) \in (\bar{x}^\perp)^\perp$, proving that $\nabla f(\bar{x})$ and $\bar{x}$ are colinear. Thus, there exists $\lambda \in \mathbb{R}$ such that
$$\nabla f(\bar{x}) + \lambda \bar{x} = 0.$$
We use again the variational inequality with $h = -\bar{x}$. We obtain that
$$0 \leq \langle \nabla f(\bar{x}), -\bar{x}\rangle = \lambda \|\bar{x}\|^2,$$
which proves that $\lambda \geq 0$.

3. We have:

$$\begin{aligned} f(x) - f(\bar{x}) &\geq \langle \nabla f(\bar{x}), x - \bar{x} \rangle \\ &= -\lambda \langle \bar{x}, x - \bar{x} \rangle \\ &= \lambda(\delta^2 - \langle \bar{x}, x \rangle). \end{aligned}$$

By Cauchy-Schwarz, we have $\langle \bar{x}, x \rangle \leq \|barx\| \|x\| \leq \delta^2$. Therefore, $f(x) - f(\bar{x}) \geq 0$.

4. The point $x(\lambda)$ is obtained as unique minimizer of $x \mapsto \frac{\lambda}{2}\|x\|^2$.

5. Let $\lambda_1 < \lambda_2$. We have

$$\begin{aligned} f(x(\lambda_2)) + \frac{\lambda_1}{2}\|x(\lambda_2)\|^2 &\geq f(x(\lambda_1)) + \frac{\lambda_1}{2}\|x(\lambda_1)\|^2 \\ f(x(\lambda_1)) + \frac{\lambda_2}{2}\|x(\lambda_1)\|^2 &\geq f(x(\lambda_2)) + \frac{\lambda_2}{2}\|x(\lambda_2)\|^2. \end{aligned}$$

Summing up these inequalities, we obtain that

$$(\lambda_2 - \lambda_1)\|x(\lambda_1)\|^2 \geq (\lambda_2 - \lambda_1)\|x(\lambda_2)\|^2,$$

from which we deduce that $\|x(\lambda_1)\| \geq \|x(\lambda_2)\|$.

1.2.3 KKT conditions

Consider the following problem:

$$\inf_{(x,y)\in\mathbb{R}^2} (x-1)^2 - (y-2)^2, \quad \text{subject to: } x \geq 0,\ y \geq 0,\ 2x + y \leq 3.$$

Compute all the points satisfying the KKT conditions, and solve rigorously the problem.

Solution Let λ_1, λ_2, and λ_3 be the multipliers associated with the constraints:

$$-x \leq 0, \quad -y \leq 0, \quad \text{and} \quad 2x + y - 3 \leq 0,$$

respectively. The KKT conditions write: $\lambda \geq 0$,

$$\begin{cases} 2(x-1) - \lambda_1 + 2\lambda_3 = & 0 \\ -2(y-2) - \lambda_2 + \lambda_3 = & 0 \end{cases} \quad \text{and} \quad \begin{cases} -x < 0 & \Longrightarrow \quad \lambda_1 = 0 \\ -y < 0 & \Longrightarrow \quad \lambda_2 = 0 \\ 2x + y < 3 & \Longrightarrow \quad \lambda_3 = 0. \end{cases}$$

Let us look for the KKT points. There are 7 combinations to study (the constraints cannot all be active). Let (x, y, λ) satisfy the KKT conditions.

- Assume that $x = 0$.
  - If $y = 0$, then $2x + y = 0 < 3$, $\lambda_3 = 0$ and $\lambda_1 = -2 < 0$. Contradiction.
  - Assume that $y > 0$. Then $\lambda_2 = 0$.
    - If $2x + y = 3$, then $y = 3$, $\lambda_1 = 2$, and $\lambda_3 = 2$. KKT point.
    - If $2x + y < 3$, then $\lambda_3 = 0$ and $\lambda_1 = -2$. Contradiction.
- Assume that $x > 0$. Then, $\lambda_1 = 0$.
  - Assume that $y = 0$.
    - If $2x + y = 3$, then $x = 3/2$, $\lambda_3 = -1/2$. Contradiction.
    - If $2x + y < 3$, then $\lambda_3 = 0$, $x = 1$ and $\lambda_2 = 4$. KKT point.
  - Assume that $y > 0$. Then $\lambda_2 = 0$.
    - If $2x + y < 3$, then $\lambda = 0$, $x = 1$, $y = 2$. Contradiction ((x, y) is not feasible).

* If $2x + y = 3$, then (x, y, λ_3) solves the system:

$$2x + y = 3, \quad 2x + 2\lambda_3 = 2, \quad -2y + \lambda_3 = -4.$$

Solution: $x = 1/3$, $y = 7/3$, $\lambda_3 = 2/3$. KKT point.

Let us solve the problem. The feasible set is bounded, the cost function continuous, therefore, there exists an optimal solution. The constraints are linear, therefore the KKT conditions are satisfied. The global minimizer is one of the three points we have found. Comparing the value of the cost function for these three point, we obtain that $(1, 0)$ is the solution.

1.2.4 KKT conditions (2)

1. Solve rigorously the following problem:

$$\inf_{(x,y)\in\mathbb{R}^2} f(x, y) := \frac{1}{2}y^2 + 2xy + \frac{1}{2}x^3 - y, \quad \text{subject to: } x \leq 1,\ x \geq -1.$$

2. Consider the optimization problem:

$$\inf_{(x,y,z)\in\mathbb{R}^3} f(x, y, z) := \ln(2 + x^2 + y^2) - \frac{1}{2}xy + x^2z - z^2, \quad \text{subject to: } x \geq 0, y \geq 0, z \geq 0.$$

Prove that $(1, 1, 0)$ satisfies the KKT conditions. Is this point a local minimizer?

Solution

1. For all feasible (x, y), it holds:

$$f(x, y) \geq \frac{1}{2}y^2 - 2y - \frac{1}{2} - y.$$

A feasible sequence such that $\|(x_k, y_k)\| \to +\infty$ is necessarily such that $y_k \to +\infty$, since the variable x is bounded. The coercivity property is thus satisfied. The cost function is continuous, the feasible set closed, thus, there exists an optimal solution.

We define the Lagrangian:

$$L(x, y, z, \lambda, \mu) = \frac{1}{2}y^2 + 2xy + \frac{1}{2}x^3 - y + \lambda_1(x - 1) + \lambda_2(-1 - x).$$

The KKT conditions are: there exists $\lambda_1 \geq 0$ and $\lambda_2 \geq 0$ such that:

$$\begin{cases} 2y + \frac{3}{2}x^2 + \lambda_1 - \lambda_2 = 0 \\ y + 2x - 1 = 0 \\ x < 1 \Longrightarrow \lambda_1 = 0 \\ x > -1 \Longrightarrow \lambda_2 = 0. \end{cases}$$

We consider now three cases:

(a) $-1 < x < 1$ (thus, $\lambda_1 = \lambda_2 = 0$). We plug $y = 1 - 2x$ in the first equation and obtain:

$$3x^2 - 8x + 4 = 0.$$

This equation has two solutions: $x = 2$ (not feasible) or $x = 2/3$. The point $(2/3, -1/3)$ is a KKT point.

(b) $x = 1$, thus $\lambda_2 = 0$, $y = -1$ and $\lambda_1 = 1/2 \geq 0$. KKT point.

(c) $x = -1$, thus $\lambda_1 = 0$, $y = 3$, $\lambda_2 = 15/2 \geq 0$. KKT point.

2. We define the Lagrangian:

$$L(x, y, z, \lambda_1, \lambda_2, \lambda_3) = \ln(2 + x^2 + y^2) - \frac{1}{2}xy + x^2z - z^2 - \lambda_1 x - \lambda_2 y - \lambda_3 z.$$

The optimality conditions are: there exists $(\lambda_1, \lambda_2, \lambda_3) \in \mathbb{R}^3_+$ such that:

$$\begin{cases} D_x L = \frac{2x}{2+x^2+y^2} - \frac{1}{2}y + 2xz - \lambda_1 = 0 \\ D_y L = \frac{2y}{2+x^2+y^2} - \frac{1}{2}x - \lambda_2 = 0 \\ D_z L = x^2 - 2z - \lambda_3 = 0 \\ x > 0 \Rightarrow \lambda_1 = 0 \\ y > 0 \Rightarrow \lambda_2 = 0 \\ z > 0 \Rightarrow \lambda_3 = 0. \end{cases}$$

They are satisfied for $(x, y, z) = (1, 1, 0)$, with $\lambda_3 = 1 \geq 0$.

1.2.5 KKT conditions

We consider the following optimization problem:

$$\inf_{(x,y)\in\mathbb{R}^p\times\mathbb{R}^{n-p}} f(x, y), \quad \text{subject to: } e(x, y) = 0, \tag{P}$$

where $f\colon (\mathbb{R}^p \times \mathbb{R}^{n-p}) \to \mathbb{R}$ and $e\colon (\mathbb{R}^p \times \mathbb{R}^{n-p}) \to \mathbb{R}^p$ are continuously differentiable. Let $(\bar{x}, \bar{y})$ be a local solution to the problem. We assume that $D_x e(\bar{x}, \bar{y})$ is a square matrix (which implies that the LICQ holds at $(\bar{x}, \bar{y})$).

1. Write the KKT conditions for the point $(\bar{x}, \bar{y})$. Denoting by $\mu \in \mathbb{R}^p$ the Lagrange multiplier, show that
$$\mu^\top = -D_x f(\bar{x}, \bar{y})(D_x e(\bar{x}, \bar{y}))^{-1}. \tag{6}$$

2. Prove that there exist $\varepsilon > 0$ and a continuously differentiable mapping $X\colon \bar{B}(\bar{y}, \varepsilon) \to \mathbb{R}^p$ which is such that for all $y \in \bar{B}(\bar{y}, \varepsilon)$,
$$e(X(y), y) = 0.$$
Compute $DX(\bar{y})$.

3. We define now the following function:
$$\tilde{f}\colon y \in \bar{B}(\bar{y}, \varepsilon) \mapsto f(X(y), y).$$
Prove that $\bar{y}$ is a local minimizer of $\tilde{f}$.

4. We want now to prove the KKT conditions for problem (P), without making use of the results seen in the lecture. To this purpose, we define μ with (6). Explain why for proving the KKT conditions, it is sufficient to prove that
$$D_y f(\bar{x}, \bar{y}) + \mu^\top D_y e(\bar{x}, \bar{y}) = 0.$$
Prove this relation.

Solution

1. The KKT conditions are:
$$D_x f(\bar{x}, \bar{y}) + \mu^\top D_x e(\bar{x}, \bar{y}) = 0, \quad D_y f(\bar{x}, \bar{y}) + \mu^\top D_y e(\bar{x}, \bar{y}) = 0.$$
It follows from the first relation that $\mu^\top = -D_x f(\bar{x}, \bar{y})(D_x e(\bar{x}, \bar{y}))^{-1}$.

2. The existence of the mapping follows immediately from the implicit function theorem. Note that $X(\bar{y}) = \bar{x}$. Differentiating the relation $e(X(y), y) = 0$ with respect to y, we obtain at $y = \bar{y}$
$$D_x e(\bar{x}, \bar{y}) DX(\bar{y}) + D_y e(\bar{x}, \bar{y}) = 0.$$
Therefore,
$$DX(\bar{y}) = -(D_x e(\bar{x}, \bar{y}))^{-1} D_y e(\bar{x}, \bar{y}).$$

3. Let $\varepsilon > 0$ be such that for all $(x, y) \in \mathbb{R}^n$,

$$\|x - \bar{x}\| \leq \varepsilon \text{ and } \|y - \bar{y}\| \leq \varepsilon \text{ and } e(x, y) = 0 \Longrightarrow f(x, y) \geq f(\bar{x}, \bar{y}).$$

By continuity of X, there exists $\tilde{\varepsilon} \in (0, \varepsilon]$ such that for all $y \in \bar{B}(\bar{y}, \varepsilon)$, $\|X(y) - \bar{x}\| \leq \varepsilon$. It follows that for all $y \in \bar{B}(\bar{y}, \tilde{\varepsilon})$,

$$\tilde{f}(y) \geq \tilde{f}(\bar{y}).$$

4. Since $\bar{y}$ is a local minimizer of $\tilde{f}$, we have

$$\begin{aligned} 0 = D\tilde{f}(\bar{y}) &= D_x f(\bar{x}, \bar{y}) DX(\bar{y}) + D_y f(\bar{x}, \bar{y}) \\ &= -D_x f(\bar{x}, \bar{y})(D_x e(\bar{x}, \bar{y}))^{-1} D_y e(\bar{x}, \bar{y}) + D_y f(\bar{x}, \bar{y}) \\ &= \mu^\top D_y e(\bar{x}, \bar{y}) + D_y f(\bar{x}, \bar{y}). \end{aligned}$$

1.2.6 Proving inequalities with optimization

Let α_1,...,$\alpha_n > 0$, let $\alpha = \sum_{i=1}^n \alpha_i$. The weighted inequality of arithmetic and geometric means states that for all $x_1, ..., x_n \geq 0$,

$$\Big(\prod_{i=1}^n x_i^{\alpha_i}\Big)^{1/\alpha} \leq \frac{1}{\alpha} \sum_{i=1}^n \alpha_i x_i.$$

Consider the following problem:

$$\sup_{x \in \mathbb{R}^n} \prod_{i=1}^n x_i^{\alpha_i}, \quad \text{subject to: } \sum_{i=1}^n \alpha_i x_i = 1, x_i \geq 0, \ \forall i = 1, ..., n.$$

Prove the existence of an optimal solution. Write (and justify) the KKT conditions. In particular, explain why it is not necessary to introduce a Lagrange multiplier for the constraint: $x_i \geq 0, \forall i = 1, ..., n$. Compute explicitely the value of the problem. Deduce the weighted inequality of arithmetic and geometric means.

Solution

1. There exists an optimal solution $\bar{x} \in \mathbb{R}^n$ since the cost function is continuous and the feasible set non-empty and compact. Let us denote: $\bar{y} = \prod_{i=1}^n x_i^{\alpha_i}$. Moreover, for all i = 1,...,n, $\bar{x}_i > 0$. Indeed, if for some i, $\bar{x}_i = 0$, then $\bar{y} = 0$, which is not possible since there are feasible points with strictly positive coordinates, for which the cost function is strictly positive.

   Therefore, the cost function is differentiable at $\bar{x}$ and the constraints are qualified: the gradient is not equal to 0.

   The optimality conditions are therefore: there exists $\lambda \in \mathbb{R}$ such that for all $i = 1, ..., n$

$$\alpha_i \frac{\bar{y}}{x_i} - \alpha_i \lambda = 0.$$

   Therefore, $\lambda \neq 0$ and for all i = 1,...,n, $x_i = \frac{\bar{y}}{\lambda}$. Thus:

$$\alpha \frac{y}{\lambda} = 1 \quad \text{and} \quad \bar{y} = \frac{\bar{y}^\alpha}{\lambda^\alpha}.$$

   Combining the two equalities together, we obtain: $\bar{y} = \alpha^{-\alpha}$. Let us prove the inequality. Let x_1,...,$x_n \geq 0$. We exclude the trivial case $x = 0$. Let $z = x/(\sum_{i=1}^n \alpha_i x_i$, then:

$$\prod_{i=1}^n x_i^{\alpha_i} = \prod_{i=1}^n z_i^{\alpha_i} \Big(\sum_{i=1}^n \alpha_i x_i\Big)^\alpha \leq \alpha^{-\alpha} \Big(\sum_{i=1}^n \alpha_i x_i\Big)^\alpha.$$

   The inequality follows.

2. The cost function is continuous, the feasible set compact and non-empty. The gradient of the constraint is $2x$ and is not equal to zero, for all feasible point. The KKT conditions read: there exists $\lambda \in \mathbb{R}$ such that: $y - \lambda x = 0$. Since x and y have the same norm, $\lambda = -1$ or 1. The maximum is obtained for $\lambda = 1$. We deduce that for all x such that $\|x\| = 1$, $\langle x, y\rangle \leq 1$. Replacing y by $-y$, we obtain that $\langle x, y\rangle \geq -1$. Thus $|\langle x, y\rangle| \leq 1$.

   In the general cases, for all x and y in $\mathbb{R}^n \backslash \{0\}$,

$$\left|\left\langle \frac{x}{\|x\|}, \frac{y}{\|y\|} \right\rangle\right| \leq 1,$$

   and thus, $\langle x, y\rangle \leq \|x\| \cdot \|y\|$.

1.2.7 Ellipsoid

Let a_1,...,a_n be n nonzero real numbers. Consider the ellipsoid $\mathcal{E}$ defined by:

$$\mathcal{E} = \left\{ x \in \mathbb{R}^n \mid \sum_{i=1}^{n} \frac{x_i^2}{a_i^2} \leq 1 \right\}.$$

Let $y \notin \mathcal{E}$, let $\bar{x} = \mathrm{Proj}_{\mathcal{E}}(y)$.

1. Prove that there exists a unique real number $\bar{\lambda} \geq 0$ such that

$$\sum_{i=1}^{n} \frac{a_i^2 y_i^2}{(a_i^2 + \bar{\lambda})^2} = 1.$$

2. Prove that

$$\bar{x}_i = \frac{a_i^2 y_i}{a_i^2 + \bar{\lambda}}, \quad \forall i = 1, ..., n.$$

Solution

1. Define:

$$\phi \colon \lambda \geq 0 \mapsto \phi(\lambda) = \sum_{i=1}^{n} \frac{a_i^2 y_i^2}{(a_i^2 + \lambda)^2}.$$

   One can easily check that that ϕ is continuous and strictly decreasing. Since $\phi(0) > 1$ and $\lim_{\lambda \to \infty} \phi(\lambda) = 0$, the equation $\phi(\lambda) = 1$ has a unique solution in the interval $(0, \infty)$.

2. We recall that $\bar{x}$ is the solution to the following optimization problem:

$$\inf_{x \in \mathbb{R}^n} \frac{1}{2}\|y - x\|, \quad \text{subject to: } g(x) \leq 0,$$

   where $g(x) = \sum_{i=1}^{n} x_i^2/a_i^2 - 1$. If the constraint is not active at $\bar{x}$, we obtain that $y - \bar{x} = 0$, which is a contradiction, since $\bar{x} \in \mathbb{E}$ and $y \notin \mathcal{E}$. We have

$$Dg(x) = \begin{pmatrix} \frac{2x_1}{a_1^2} & \cdots & \frac{2x_n}{a_n^2} \end{pmatrix},$$

   thus $Dg(\bar{x}) \neq 0$, since at least one of the coordinates of $\bar{x}$ should be nonzero. This proves that $Dg(\bar{x})$ has full row rank (rank 1), therefore the KKT conditions are satisfied. There exists $\lambda \geq 0$ such that:

$$\forall i \in \{1, ..., n\}, \quad \bar{x}_i - y_i + \lambda \frac{\bar{x}_i}{a_i^2} = 0.$$

   Therefore, for all $i \in \{1, ..., n\}$,

$$\bar{x}_i = \frac{y_i}{1 + \lambda/a_i^2} = \frac{a_i^2 y_i}{a_i^2 + \lambda},$$

   and then,

$$\frac{\bar{x}_i^2}{a_i^2} = \frac{a_i^2 y_i^2}{(a_i^2 + \lambda)^2}.$$

Summing up these inequalities, we obtain:

$$\phi(\lambda) = \sum_{i=1}^{n} \frac{a_i^2 y_i^2}{(a_i^2 + \lambda)^2} = 1.$$

Therefore, $\lambda = \bar{\lambda}$.

1.2.8 Slater Qualification Condition

Let $g\colon \mathbb{R}^n \to \mathbb{R}^m$ be a twice continuously differentiable function. We assume that the m functions g_1,...,g_m are all convex. We denote

$$\mathcal{F} = \{x \in \mathbb{R}^n \mid g_i(x) \leq 0,\ \forall i = 1, ..., m\}.$$

We also consider a point $\bar{x}$ such that $g(\bar{x}) = 0$. In the lecture, we have proved that

$$\mathcal{T}_{\mathcal{F}}(\bar{x}) \subseteq \mathcal{T}'_{\mathcal{F}}(\bar{x}) := \{h \in \mathbb{R}^n \mid Dg(\bar{x})h \leq 0\}.$$

We have also proved that under the LICQ, $\mathcal{T}'_{\mathcal{F}}(\bar{x}) \subseteq \mathcal{T}_{\mathcal{F}}(\bar{x})$. The goal of the exercise is to prove this inclusion without the LICQ. Instead, we consider a weaker qualification: there exists $x_0 \in \mathcal{F}$ and $\varepsilon > 0$ such that $g_i(x_0) \leq -\varepsilon$, for all $i = 1$,...,m.

Let us fix a direction $h \in \mathcal{T}'_{\mathcal{F}}(\bar{x})$.

1. Prove that there exists a constant $A > 0$ such that for all $\theta \in [0, 1]$,

$$g_i(\bar{x} + \theta h) \leq A\theta^2, \quad \forall i = 1, ..., m.$$

2. Prove that for all $\theta \in [0, 1]$,

$$g\big((1 - \lambda(\theta))(\bar{x} + \theta h) + \lambda(\theta)x_0\big) \leq 0,$$

   where $\lambda(\theta) = \frac{A\theta^2}{\varepsilon + A\theta^2}$.

3. Deduce that $h \in \mathcal{T}_{\mathcal{F}}(\bar{x})$.

Solution

1. Let A be defined by

$$A := \max_{i=1,\dots,m} \max_{\theta \in [0,1]} \Big\{\frac{1}{2} h^\top \nabla^2 g_i(\bar{x} + \theta h) h\Big\}.$$

   Note that $A \geq 0$ since the mappings g_i are convex. For all $\theta \in [0, 1]$, we have

$$g_i(\bar{x} + \theta h) = g_i(\bar{x}) + Dg_i(\bar{x})h + \frac{1}{2}\theta^2 h^\top \nabla^2 g_i(\bar{x} + \xi\theta h)h \leq A\theta^2.$$

2. For all $\theta \in [0, 1]$, for all $i = 1, ..., m$, using the convexity of g_i and the fact that $\lambda(\theta) \in [0, 1]$, we obtain that

$$\begin{aligned} g_i((1 - \lambda(\theta))(\bar{x} + \theta h) + \lambda(\theta)x_0) &\leq (1 - \lambda(\theta))g_i(\bar{x} + \theta h) + \lambda(\theta)g_i(x_0) \\ &\leq \Big(1 - \frac{A\theta^2}{\varepsilon + A\theta^2}\Big)A\theta^2 - \frac{A\theta^2}{\varepsilon + A\theta^2}\varepsilon = 0. \end{aligned}$$

3. For all θ, we have

$$\begin{aligned} (1 - \lambda(\theta))(\bar{x} + \theta h) + \lambda(\theta)x_0 &= \bar{x} + \theta h - \lambda(\theta)(\bar{x} + \theta h) + \lambda(\theta)x_0 \\ &= \bar{x} + \theta\Big[h + \underbrace{\frac{\theta A}{\varepsilon + A\theta^2}(x_0 - \bar{x} - \theta h)}_{\xrightarrow[\theta \to 0]{} 0}\Big]. \end{aligned}$$

   Therefore, $h \in \mathcal{T}_{\mathcal{F}}(\bar{x})$.

1.2.9 Maximization of the area of a triangle

We consider the problem of maximizing the area of a triangle whose vertices A_1, A_2, and A_3 belong to a circle. Without loss of generality, we fix the radius of the circle to 1 and its center at $(0,0)$. Moreover, the problem is invariant by rotation, therefore, we can fix the coordinates of the point A_3 to $(1,0)$. We denote by (x_1, y_1) the coordinates of A_1 and by (x_2, y_2) the coordinates of A_2.

We recall that the area the triangle is given by: $\mathcal{A} = |\det(M)|$, where M is the matrix composed of the vectors $\overrightarrow{A_3A_1}$ and $\overrightarrow{A_3A_2}$. We denote by $f(x_1, y_1, x_2, y_2)$ the value of the determinant of M as a function of x_1, y_1, x_2, and y_2. Consider the following problem:

$$\sup_{(x_1,y_1,x_2,y_2)\in\mathbb{R}^3} f(x_1, y_1, x_2, y_2), \quad \text{subject to: } \begin{cases} x_1^2 + y_1^2 = 1 \\ x_2^2 + y_2^2 = 1. \end{cases}$$

Prove the existence of an optimal solution, write and justify the KKT conditions. Solve the problem (you can start by expressing the optimal solution as a function of the Lagrange multipliers). Justify that the solution provides a triangle of maximal area.

Solution The gradient of the constraints is given by:

$$2\begin{pmatrix} x_1 & y_1 & 0 & 0 \\ 0 & 0 & x_2 & y_2 \end{pmatrix}$$

and has rank 2 for any feasible point. The qualification condition is satisfied. The area is given by:

$$f(x_1, y_1, x_2) = \begin{vmatrix} x_1 - 1 & x_2 - 1 \\ y_1 & y_2 \end{vmatrix} \tag{7}$$

and the Lagrangian is given by:

$$L(x_1, y_1, x_2, y_2, \lambda_1, \lambda_2) = y_2(x_1 - 1) - y_1(x_2 - 1) + \lambda_1(x_1^2 + y_1^2 - 1) + \lambda_2(x_2^2 + y_2^2 - 1). \tag{8}$$

The optimality conditions read:

$$\begin{aligned} &y_2 + 2\lambda_1 x_1 = 0, && -y_1 + 2\lambda_2 x_2 = 0, \\ &-(x_2 - 1) + 2\lambda_1 y_1 = 0, && (x_1 - 1) + 2\lambda_2 y_2 = 0. \end{aligned}$$

This is a linear system with respect to the primal variables, that we can solve. We obtain:

$$x_1 = x_2 = \frac{1}{1 - 4\lambda_1\lambda_2}, \quad y_1 = \frac{2\lambda_2}{1 - 4\lambda_1\lambda_2}, \quad y_2 = \frac{-2\lambda_1}{1 - 4\lambda_1\lambda_2}.$$

Since $x_1 = x_2$, we deduce that $y_1 = -y_2$ or $y_1 = y_2$, but this last case is not interesting. We deduce that $\lambda_1 = \lambda_2$. Setting $X = 4\lambda_1^2$, we deduce from $x_1^2 + y_1^2 = 1$ that

$$\frac{1 + X}{(1 - X)^2} = 1, \tag{9}$$

and finally that $X = 3$ or $X = 0$. This last case is not interesting, and finally, $x_1 = x_2 = -1/2$ and $(y_1, y_2) = (\sqrt{3}/2, -\sqrt{3}/2)$ or $(-\sqrt{3}/2, \sqrt{3}/2)$.

1.2.10 Sensitivity analysis

Let $f : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}$, $e : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^m$ be two continuously twice-differentiable functions. We consider the following family of optimization problems: for all $u \in \mathbb{R}$,

$$\inf_{x\in\mathbb{R}^n} f(x, u) \text{ subject to: } e(x, u) = 0. \tag{$P(u)$}$$

Note that here, u is not an optimization variable but simply a parameter of the problem. We introduce now the Lagrangian:

$$L(x, \lambda, u) = f(x, u) + \langle \lambda, e(x, u)\rangle, \tag{10}$$

where $\lambda \in \mathbb{R}^m$. Let $u^* \in \mathbb{R}$, we assume that there exists a locally optimal solution x^* to the problem $(P(u))$ (for the value $u = u^*$) which is such that $\text{rank}(D_x(e(x^*, u^*))) = m$ and such that the second-order sufficient optimality conditions hold. We denote by λ^* the associated Lagrange multiplier.

1. Prove the existence of a neighborhood U of u^*, a neighborhood X of x^* and a neighborhood Λ of λ^* such that for all $u \in U$, there exists a unique pair $(x(u), \lambda(u))$ in (X, U) satisfying the KKT conditions for problem $P(u)$. Of course, $x(u^*) = x^*$ and $\lambda(u^*) = \lambda^*$. Prove also that this mapping is continuously differentiable. Hint: use the implicite function theorem and exercise 7.2.

2. Restricting if necessary the neighborhood U, prove that for all $u \in U$, $x(u)$ is a locally optimal solution to the perturbed problem. Hint: use the fact that if $x(u)$ is not locally optimal, the second-order sufficient conditions are not satisfied.

3. Compute a first-order Taylor expansion of $f(x(u))$ in the neighborhood of u^*. Hint: $f(x(u)) = L(x(u), \lambda^*)$.

Solution

1. Consider the mapping

$$F : (x, \lambda, u) \in \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R} \to \begin{pmatrix} \nabla_x L(x, \lambda, u) \\ e(x, u) \end{pmatrix}.$$

For any $u \in \mathbb{R}$, the pair $(x(u), \lambda(u))$ satisfies the KKT conditions at u if and only if

$$F(x(u), \lambda(u), u) = 0. \tag{11}$$

Equality (11) defines a set of $n + m$ (nonlinear) equations, with $n + m$ unknown variables. To apply the implicit function theorem, it suffices to check that $D_{(x,\lambda)}F(x^*, \lambda^*, u^*)$ is invertible. The derivative is given by:

$$D_{x,\lambda}F(x, \lambda, u) = \begin{pmatrix} D_{xx}L(x, \lambda, u) & D_x e(x, u)^\top \\ D_x e(x, u) & 0 \end{pmatrix}$$

and the invertibility at (x^*, λ^*, u^*) is a direct consequence of the sufficient conditions and question 3 of exercise 6 of the third sheet. The result of the question follows.

2. We prove the result by contradiction: we assume that there exists a sequence $(u_k)_{k\in\mathbb{N}}$ converging to u^* such that for all k, $x(u_k)$ is not locally optimal (for the problem with $u = u_k$). By continuity, for k large enough,

$$\operatorname{rank}(D_x(e(x(u_k), u_k))) = m.$$

Since the KKT conditions hold and since the full-rank condition is satisfied (at least for k large enough), the sufficient conditions cannot be satisfied: for k large enough, there exists $v_k \in \mathbb{R}^n$ with $\|v_k\| = 1$ such that:

$$D_x e(x(u_k), u_k)v_k = 0 \quad \text{and} \quad v_k^\top D_{xx}L(x(u_k), \lambda(u_k), u_k)v_k \leq 0. \tag{12}$$

Let v^* be a limit point of $(v_k)_k$, taking the corresponding subsequence and passing to the limit in (12), we obtain first that: $D_x e(x^*, \lambda^*, u^*)v^* = 0$ and $(v^*)^\top D_{xx}L(x^*, \lambda^*, u^*)v^* \leq 0$. Since $\|v^*\| = 1$, we obtain a contradiction with the sufficient conditions.

3. We derive the second-order expansion as follows:

$$\begin{aligned} & f(x(u)) - f(x^*) \\ = \; & L(x(u), \lambda^*, u) - L(x^*, \lambda^*, u^*) \\ = \; & \underbrace{D_x L(x^*, \lambda^*, u^*) D_u x(u^*)(u - u^*)}_{=0} + D_u L(x^*, \lambda^*, u^*)(u - u^*) + o(|u - u^*|). \end{aligned}$$

1.3 Convex analysis

Note. In the next three exercises, we consider functions $f : \mathbb{R} \to \bar{\mathbb{R}}$, where $\bar{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$. This simply means that the value $+\infty$ is allowed. In this case, the domain $\mathcal{D}(f)$ is defined as:

$$D(f) = \{x \in \mathbb{R}^n \mid f(x) < +\infty\}.$$

With the following conventions: $+\infty \leq +\infty$, $(+\infty) + (+\infty) = +\infty$, $a + (+\infty) = +\infty$, for all $a \in \mathbb{R}$, and $b \cdot (+\infty) = +\infty$, for all $b \geq 0$, we obtain a natural extension of the definition of convex functions for functions taking values in $\bar{\mathbb{R}}$.

1.3.1 Conjugate functions

For any function $f : \mathbb{R}^n \to \mathbb{R} \cup \{-\infty; +\infty\}$, we define its conjugate function $f^*(x)$ by:

$$f^* : \lambda \in \mathbb{R}^n \mapsto f^*(\lambda) = \sup_{x \in \mathbb{R}^n} \big\{\langle \lambda, x\rangle - f(x)\big\}.$$

The biconjugate f^{**} is the conjugate of f^*: for all $x \in \mathbb{R}^n$,

$$f^{**}(x) = \sup_{\lambda \in \mathbb{R}^n} \big\{\langle \lambda, x\rangle - f^*(\lambda)\big\}.$$

Part 1.

1. Prove that for f and $g : \mathbb{R}^n \to \mathbb{R} \cup \{-\infty; +\infty\}$, if $f \leq g$, then $f^* \geq g^*$ and $f^{**} \leq g^{**}$.
2. Prove that either: $f^*(\lambda) = -\infty,\ \forall \lambda \in \mathbb{R}^n$ or: $f^*(\lambda) > -\infty,\ \forall \lambda \in \mathbb{R}^n$. In this last case, prove that f^* is convex.
3. Prove that for all x, $f^{**}(x) \leq f(x)$.

Part 2. The Fenchel-Moreau-Rockafellar theorem ensures that for all convex functions $f : \mathbb{R}^n \to \bar{\mathbb{R}}$, if $\mathcal{D}(f)$ is non-empty and closed and if f is continuous on $\mathcal{D}(f)$, then for all $x \in \mathbb{R}^n$, $f(x) = f^{**}(x)$. This theorem was proved in the 60s. It has actually a more general form for convex functions on Hilbert spaces.

We aim at proving this theorem in the following case: f is such that $\mathcal{D}(f) = \mathbb{R}^n$ and f is differentiable.

1. Consider (just in the question) the case: $f(x) = \frac{1}{2}x^2$ $(x \in \mathbb{R})$. Compute f^*. Draw the graph of f and the graph of the line $\lambda x - f^*(\lambda)$ for a couple of values of λ of your choice.
2. Compute $f^*(Df(x))$ for $x \in \mathbb{R}^n$. Prove the Fenchel-Moreau-Rockafellar theorem.

Solution *Part 1.*

1. Obvious.
2. The first case happens if and only if for all x, $f(x) = +\infty$. Otherwise, let x_0 be such that $f(x_0) < +\infty$. Then, for all $\lambda \in \mathbb{R}^n$, $f^*(\lambda) \geq \langle \lambda, x_0\rangle - f(x_0) > -\infty$. Since for all x, $x \mapsto \langle \lambda, x\rangle - f(x)$ is convex (w.r.t. λ), $f^*(\lambda)$ is convex by question (...).
3. For all $x \in \mathbb{R}^n$, for all $\lambda \in \mathbb{R}^n$, $f^*(\lambda) \geq \langle \lambda, x\rangle - f(x)$. Therefore, $\langle \lambda, x\rangle - f^*(\lambda) \leq f(x)$. Maximizing w.r.t. λ, we obtain the result.

Part 2.

1. Here, $f^*(\lambda) = f(\lambda)$.
2. We already know that $f^{**} \leq f$. By definition,

$$f^*(Df(x)) = \sup_{y \in \mathbb{R}^n} \big\{Df(x)y - f(y)\big\} = Df(x)x - f(x).$$

   Indeed, for all $y \in \mathbb{R}^n$, $Df(x)y - f(y) \leq Df(x)x - f(x)$, since f is convex. Thus, for all x,

$$f^{**}(x) \geq Df(x)x - f^*(x) = f(x).$$

   The theorem is proved.

1.3.2 Convex hull of a function

1. Let us consider a non-empty set $\mathcal{F}$ of convex functions defined on $\mathbb{R}^n$. For all $x \in \mathbb{R}^n$, we define

$$\bar{f}(x) = \sup_{f \in \mathcal{F}} \{f(x)\} \in \bar{\mathbb{R}}.$$

   Prove that $\bar{f}$ is convex.

2. Let us fix now a function $f : \mathbb{R}^n \to \mathbb{R}$. We assume that f is bounded from below. We introduce the following order relation on functions from $\mathbb{R}^n$ to $\mathbb{R}$: $f_1 \leq f_2 \iff \big[f_1(x) \leq f_2(x)$, for all $x \in \mathbb{R}^n\big]$.

   (a) Prove that there exists a unique convex function, called convex hull of f and denoted $\mathrm{conv}(f)$, satisfying the following properties: (i) $\mathrm{conv}(f) \leq f$, (ii) for all convex functions $f_0 \colon \mathbb{R}^n \to \mathbb{R}$, if $f_0 \leq f$, then $f_0 \leq \mathrm{conv}(f)$.

   (b) A classical result from convex analysis states that convex functions are continuous on the inner set of their domain. Using this result and those of the previous exercise, prove that $f^{**} = \mathrm{conv}(f)$.

Solution

1. Let us fix x_1, x_2, and $\theta \in [0, 1]$. Let $f \in \mathcal{F}$. By convexity, we have

$$f(\theta x_1 + (1-\theta)x_2) \leq \theta f(x_1) + (1-\theta) f(x_2) \leq \theta \bar{f}(x_1) + (1-\theta)\bar{f}(x_2).$$

   Maximizing the left-hand side, we obtain that $\bar{f}(\theta x_1 + (1-\theta)x_2) \leq \theta \bar{f}(x_1) + (1-\theta)\bar{f}(x_2)$.

2. Denote by $\mathcal{F}$ the set of convex functions f_0 such that $f_0 \leq f$. This set is non-empty, it contains the constant function equal to the minimum of f. Then, $\mathrm{conv}(f)$ can be defined as the maximum of $\mathcal{F}$.

3. Clearly the domain of $\mathrm{conv}(f)$ is the whole space $\mathbb{R}^n$. Thus, $\mathrm{conv}(f)$ is continuous on $\mathbb{R}^n$ and thus one can apply Fenchel-Moreau-Rockafellar theorem: $\mathrm{conv}(f)^{**} = \mathrm{conv}(f)$, and finally, $f^{**} \geq \mathrm{conv}(f)$. We have proved that f^{**} is convex, smaller than f, and greater than $\mathrm{conv}(f)$. By definition of $\mathrm{conv}(f)$, $f^{**} = \mathrm{conv}(f)$.

1.3.3 Computation of conjugate functions

Compute the conjugate function of f in the following cases. In the last two cases, your answer should use the conjugate function of the functions involved in the definition of f.

1. $f(x) = \frac{1}{2}x^\top A x$, where $A \in \mathbb{R}^{n \times n}$ is a symmetric positive-definite matrix
2. $f(x) = \langle b, x \rangle$, where $b \in \mathbb{R}^n$
3. $f(x) = \exp(x)$
4. $f(x) = \inf_{y \in \mathbb{R}^n} f_1(x-y) + f_2(y)$, where $f_1 : \mathbb{R}^n \to \bar{\mathbb{R}}$ and $f_2 \to \bar{\mathbb{R}}$ are given
5. Given $g : \mathbb{R}^m \to \bar{\mathbb{R}}$ and $A \in \mathbb{R}^{n \times m}$,

$$f(x) = \inf_{y \in \mathbb{R}^m} g(y), \quad \text{subject to: } Ay = x.$$

Solution

1. Let $\lambda \in \mathbb{R}^n$, the maximizer x^* of $\langle \lambda, x \rangle - \frac{1}{2}x^\top A x$ is: $x^* = A^{-1}\lambda$. Thus,

$$f^*(\lambda) = \frac{1}{2}\lambda^\top A^{-1}\lambda.$$

2. Clearly, $f^*(\lambda) = 0$ if $\lambda = b$, $+\infty$ otherwise.

3. If $\lambda < 0$, then $f^*(\lambda) = +\infty$. If $\lambda = 0$, then $f^*(\lambda) = 0$. If $\lambda > 0$, then $f^*(\lambda) = \lambda \ln(\lambda) - \lambda$.

4. We have:

$$\begin{aligned} f^*(\lambda) &= \sup_{x,y\in\mathbb{R}^n} \langle \lambda, x\rangle - f_1(x-y) - f_2(y) \\ &= \sup_{y,z\in\mathbb{R}^n} \langle \lambda, y+z\rangle - f_1(z) - f_2(y) \\ &= f_1^*(\lambda) + f_2^*(\lambda). \end{aligned}$$

5. We have: $f^*(\lambda) = \sup_{y\in\mathbb{R}^m} \langle \lambda, Ay\rangle - g(y)\rangle = A^\top g^*(\lambda)$.

1.3.4 Conic duality

Let T be a closed and convex cone. Let T^0 be its polar cone and T^{00} be the polar cone of its polar cone.

1. Prove that the polar cone of a given set is closed and convex.
2. Prove that $T \subseteq T^{00}$.
3. Prove that $T^{00} \subseteq T$. Use a proof by contradiction, with similar ideas to the ones used in the proof of Farkas' lemma.

Solution

1. Let K be a subset of $\mathbb{R}^n$. Its polar cone is given by:

$$K^0 = \big\{h \in \mathbb{R}^n \,|\, \forall x \in K,\ \langle x, h\rangle \leq 0\big\}.$$

   One can write K^0 as an intersection of closed half-spaces:

$$K^0 = \bigcap_{x\in K} \big\{h \in \mathbb{R}^n \,|\, \langle x, h\rangle \leq 0\big\}.$$

   Therefore, K^0 is closed and convex (all half-spaces are convex).

2. Let $x \in T$. Then, for all $\lambda \in T^0$, $\langle \lambda, x\rangle \leq 0$. Therefore, $x \in T^{00}$.

3. We use a proof by contradiction. We assume that there exists $y \in T^{00}$ such that $y \in T$. Since by assumption, T is convex and closed, we can apply the theorem of Hahn-Banach. Let $\eta \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$ be such that for all $x \in T$,

$$\langle \eta, x\rangle \leq \alpha < \langle \eta, y\rangle.$$

   Since $0 \in T$, we have $\alpha \geq 0$. Let $x \in T$. For all $K > 0$, $Kx \in T$, therefore, $\langle \eta, Kx\rangle \leq \alpha$ and then

$$\langle \eta, x\rangle \leq \frac{\alpha}{K} \underset{K\to\infty}{\longrightarrow} 0.$$

   This proves that $\eta \in T^0$. Since $y \in T^{00}$, we have $\langle \eta, y\rangle \leq 0$, in contradiction with:

$$\langle \eta, y\rangle > \alpha \geq 0.$$

1.3.5 Tangent cones (1)

Given a closed set $\mathcal{F}$, we denote by $\mathrm{dist}_{\mathcal{F}}(x)$ the value of the following optimization problem:

$$\inf_{y\in\mathbb{R}^n} \|y - x\|, \quad \text{subject to: } y \in \mathcal{F},$$

for a given value of $x \in \mathbb{R}^n$. The existence of a solution to the above problem has already been discussed previously.

Let $x \in \mathcal{F}$, let $h \in \mathbb{R}^n$. Prove that $h \in \mathcal{T}_{\mathcal{F}}(x)$ if and only if there exists a sequence $(\theta_k)_{k\in\mathbb{N}}$ of positive numbers, converging to 0, such that

$$\frac{\mathrm{dist}_{\mathcal{F}}(x + \theta_k h)}{\theta_k} \underset{k\to\infty}{\longrightarrow} 0.$$

Solution Let us assume that $h \in \mathcal{T}_{\mathcal{F}}(x)$. Let $(h_k)_{k\in\mathbb{N}} \subseteq \mathbb{R}^n$ and $(\theta_k)_{k\in\mathbb{N}}$ be two sequences such that: $h_k \to h$, $\theta_k \to 0$, $\theta_k > 0$, $x + \theta_k h_k \in \mathcal{F}$. Then, we have:

$$\frac{\mathrm{dist}_{\mathcal{F}}(x+\theta_k h}{\theta_k} \leq \frac{\|(x+\theta_k h_k) - (x+\theta_k h)\|}{\theta_k} = \|h_k - h\| \to 0.$$

Now, let assume that there exists a sequence of positive numbers $(\theta_k)_{k\in\mathbb{N}}$, converging to 0, and such that

$$\varepsilon_k = \frac{\mathrm{dist}_{\mathcal{F}}(x+\theta_k h}{\theta_k} \underset{k\to\infty}{\longrightarrow} 0.$$

For all $k \in \mathbb{N}$, there exists $y_k \in \mathcal{F}$ such that

$$\theta_k \varepsilon_k = \|y_k - (x + \theta_k h)\|.$$

Let us set

$$h_k = \frac{y_k - x}{\theta_k},$$

so that $y_k = x + \theta_k h_k$. To prove that $h \in \mathcal{T}_{\mathcal{F}}(x)$, it suffices to prove that $h_k \to h$. We have:

$$\|h_k - h\| = \left\| \frac{y_k - x}{\theta_k} - h \right\| = \frac{1}{\theta_k}\|y_k - (x+\theta_k h)\| = \varepsilon_k \to 0.$$

The equivalence is proved.

1.3.6 Tangent cones (2)

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a continuously differentiable function. We consider the graph G_f of f, defined by:

$$G_f = \{(x, y) \in \mathbb{R}^{n+1} \,|\, y = f(x)\}.$$

Let $\bar{x} \in \mathbb{R}^n$. Compute the tangent cone of G_f at $(\bar{x}, f(\bar{x}))$. Justify your answer, without using any theorem seen in the lecture.

Solution Let us prove that

$$\mathcal{T}_{G_f}(\bar{x}, f(\bar{x})) = \{(u, v) \in \mathbb{R}^{n+1} \,|\, v = Df(\bar{x})u\}.$$

Let $(u, v) \in \mathbb{R}^{n+1}$ be such that $v = Df(\bar{x})u$. Let $\theta_k = 1/(k+1)$, let

$$u_k = u \quad \text{and} \quad v_k = \frac{f(\bar{x} + \theta_k u_k) - f(\bar{x})}{\theta_k}.$$

Clearly, $u_k \to u$, $v_k \to Df(\bar{x})u = v$ and for all $k \in \mathbb{N}$, $(\bar{x} + \theta_k u_k, f(\bar{x}) + \theta_k v_k) \in G_f$, therefore, $(u, v) \in \mathcal{T}_{G_f}(\bar{x}, f(\bar{x}))$.

Now, let $(u, v) \in \mathcal{T}_{G_f}(\bar{x}, f(\bar{x})$. Let $(u_k)_{k\in\mathbb{N}}$, $(v_k)_{k\in\mathbb{N}}$, and $(\theta_k)_{k\in\mathbb{N}}$ be such that:

$$u_k \to u, \quad v_k \to v, \quad \theta_k \to 0, \quad f(\bar{x} + \theta_k u_k) = \bar{y} + \theta_k v_k.$$

Therefore,

$$v_k = \frac{f(\bar{x} + \theta_k u_k) - f(\bar{x})}{\theta_k} = Df(\bar{x} + \xi_k \theta_k u_k)u_k,$$

for some $\xi_k \in [0, 1]$. Passing to the limit in both sides, we obtain that $v = Df(\bar{x})u$.

1.3.7 Tangent cones (3)

Let $\mathcal{F}$ be a non-empty subset of $\mathbb{R}^n$, let $x \in \mathcal{F}$.

1. Prove that the tangent cone $T_{\mathcal{F}}(x)$ is closed.
2. Prove that if $\mathcal{F}$ is convex, then $T_{\mathcal{F}}(x)$ is also convex.

1.3.8 Sufficiency of the KKT conditions in the convex case

The goal of this exercise is to prove that in the case of a convex problem, the KKT conditions are not only necessary conditions, but also sufficient conditions for optimality.

Let $g : \mathbb{R}^n \to \mathbb{R}^m$ be such that for all $i = 1,...,m$, g_i is a differentiable convex function, let $e : \mathbb{R}^n \to \mathbb{R}^p$ be an affine function (i.e. there exists $A \in \mathbb{R}^{p\times n}$ and $b \in \mathbb{R}^p$ such that for all x, $e(x) = Ax + b$). We consider the problem P:

$$\min_{x\in\mathbb{R}^n} \; f(x), \text{ subject to: } g(x) \leq 0, \; e(x) = 0. \tag{P}$$

Let x^* be a feasible point satisfying the KKT conditions. We denote by $L(x, \lambda, \mu) = f(x) + \langle \lambda, g(x)\rangle + \langle \mu, e(x)\rangle$ the Lagrangian, let (λ^*, μ^*) be the Lagrange multipliers associated with x^*. We also define:

$$N(x^*) = \big\{(\lambda, \mu) \in \mathbb{R}^m_+ \times \mathbb{R}^n \text{ such that } \lambda^\top g(x^*) = 0\big\}. \tag{13}$$

1. Prove that x^* is a global minimizer of $x \mapsto L(x, \lambda^*, \mu^*)$.
2. Prove the following inequality : for all feasible point x of problem P, for all $(\lambda, \mu) \in N(x^*)$,
$$f(x) - f(x^*) \geq L(x, \lambda, \mu) - L(x^*, \lambda, \mu). \tag{14}$$
3. Deduce that x^* is a global minimizer of problem P.
4. We consider now the following family of optimization problems, parameterized by $a \in \mathbb{R}^m$ and $b \in \mathbb{R}^p$:
$$V(a, b) = \min_{x\in\mathbb{R}^n} \; f(x), \text{ subject to: } g(x) + a \leq 0, \; e(x) + b = 0.$$
We assume that f is lower-bounded, so that V is also lower-bounded.
   (a) Prove the convexity of V.
   (b) Compute the conjugate function $V^*(\lambda, \mu)$.
   Hint: for some values of (λ, μ), the expected expression of $V^*(\lambda, \mu)$ is the value of a parameterized optimization problem, with optimization variable x and with parameters λ and μ, for the others values, $V^*(\lambda, \mu)$ is equal to $+\infty$.

Solution

1. By the convexity assumptions, $x \mapsto L(x, \lambda^*, \mu^*)$ is convex. By the KKT conditions, x^* is a stationary point of this function, thus, a global minimizer.
2. For all feasible x, for all $\lambda \in N(x^*)$,
$$\begin{aligned} L(x, \lambda, \mu) - L(x^*, \lambda, \mu) &= f(x) - f(x^*) \\ &+ \Big(\sum_{i\in I} \lambda_i(g_i(x) - g_i(\bar{x}))\Big) + \Big(\sum_{i\notin I} \lambda_i(g_i(x) - g_i(\bar{x}))\Big) + \mu(e(x) - e(\bar{x})). \end{aligned}$$
The result follows, since for $i \in I$, $\lambda_i \geq 0$ and $g_i(x) - g_i(\bar{i}) = g_i(x) \leq 0$, for all $i \notin I$, $\lambda_i = 0$ and $e(x) - e(\bar{x}) = 0$.
3. Since $(\lambda^*, \mu^*) \in N(x^*)$, the optimality of x^* follows from the inequality of question 2 and the optimality of x^* (for the function $L(\cdot, \lambda^*, \mu^*)$.
4. (a) Let (a_1, b_1) and (a_2, b_2) in $\mathbb{R}^{m+p}$. Let us assume that the problem is feasible for these two points. Let $\varepsilon > 0$, let x_1 and x_2 be two corresponding ε-optimal solutions. Then, using the linearity of e and the convexity of g,
$$e\Big(\frac{x_1 + x_2}{2}\Big) = \frac{e(x_1) + e(x_2)}{2} = \frac{b_1 + b_2}{2}, \quad g\Big(\frac{x_1 + x_2}{2}\Big) \leq \frac{g(x_1) + g(x_2)}{2} \leq \frac{a_1 + a_2}{2}$$
Therefore, $\frac{x_1+x_2}{2}$ is feasible for the problem with parameters $\frac{a_1+a_2}{2}$ and $\frac{b_1+b_2}{2}$. Thus, using the convexity of f,
$$V\Big(\frac{a_1 + a_2}{2}, \frac{b_1 + b_2}{2}\Big) \leq f\Big(\frac{x_1 + x_2}{2}\Big) \leq \frac{f(x_1) + f(x_2)}{2} \leq \frac{V(a) + V(b)}{2} + \varepsilon.$$
Taking the limit when ε converges to 0, we obtain the convexity of V.

(b) Let $(\lambda, \mu) \in \mathbb{R}^{m+p}$. We have:

$$\begin{aligned} V^*(\lambda, \mu) &= \sup_{x \in \mathbb{R}^n,\, (a,b) \in \mathbb{R}^{m+p}} \langle \lambda, a \rangle + \langle \mu, b \rangle - f(x), \quad \text{subject to: } e(x) = -b,\ g(x) \leq -a \\ &= \begin{cases} \sup_{x \in \mathbb{R}^n} -L(x, \lambda, \mu) & \text{if } \lambda \geq 0 \\ +\infty & \text{otherwise.} \end{cases} \end{aligned}$$

Indeed, the variable b can be directly replaced by $e(x)$ and the result follows by maximizing with respect to a.

1.3.9 Min-max duality

Consider the following optimization problem

$$\inf_{x \in \mathbb{R}^n} \frac{1}{2} x^\top Q x + \langle c, x \rangle, \quad \text{subject to: } Ax = b,\ Gx \leq d, \tag{P}$$

where $Q \in \mathbb{R}^{n \times n}$ is symmetric positive definite, $c \in \mathbb{R}^n$, $A \in \mathbb{R}^{p \times n}$, $b \in \mathbb{R}^p$, $G \in \mathbb{R}^{m \times n}$, $d \in \mathbb{R}^m$. We also assume that the feasible set is non-empty.

1. Prove the existence of a unique solution to the problem. Justify that the KKT conditions are satisfied, write them.

2. For a given pair of Lagrange multipliers $(\lambda, \mu) \in \mathbb{R}^{m+p}$, compute the unique solution and the value of the following problem:
$$H(\lambda, \mu) = \inf_{x \in \mathbb{R}^n} L(x, \lambda, \mu).$$
Compute $\nabla_\lambda H(\lambda, \mu)$ and $\nabla_\mu H(\lambda, \mu)$.

3. Consider the dual problem to (P), given by
$$\sup_{(\lambda, \mu) \in \mathbb{R}^{m+p}} H(\lambda, \mu), \quad \text{subject to: } \lambda \geq 0. \tag{D}$$
Let $(\tilde{\lambda}, \tilde{\mu})$ be a solution to the dual problem. Prove that for an appropriate value of $\tilde{x} \in \mathbb{R}^n$, the triplet $(\tilde{x}, \tilde{\lambda}, \tilde{\mu})$ satisfies the KKT conditions for the primal problem (P).

Solution

1. The cost function is coercive and the feasible set closed, therefore, there exists a solution $\bar{x}$. The KKT conditions are satisfied, since the constraints are linear. The Lagrangian is given by
$$L(x, \lambda, \mu) = \frac{1}{2} x^\top Q x + \langle c + A^\top \mu + G^\top \lambda, x \rangle - \langle b, \mu \rangle - \langle d, \lambda \rangle.$$
The KKT conditions are
$$\nabla_x L(\bar{x}, \lambda, \mu) = Q\bar{x} + c + A^\top \mu + G^\top \lambda = 0.$$

2. The solution is given by
$$x = -Q^{-1}(c + A^\top \mu + G^\top \lambda).$$
Therefore,
$$H(\lambda, \mu) = -\frac{1}{2}(c + A^\top \mu + G^\top \lambda)^\top Q^{-1}(c + A^\top \mu + G^\top \lambda) - \langle b, \mu \rangle - \langle d, \lambda \rangle.$$
We have
$$\nabla_\lambda D(\lambda, \mu) = -GQ^{-1}(c + A^\top \mu + G^\top \lambda) - d \quad \text{and} \quad \nabla_\mu D(\lambda, \mu) = -AQ^{-1}(c + A^\top \mu + G^\top \lambda)^\top - b.$$

3. Let $(\tilde{\lambda}, \tilde{\mu})$ be a solution to the dual problem. Let us write the corresponding optimality conditions, denoting by z the multiplier associated with the constraint $-\lambda \leq 0$. We obtain:

$$\nabla_\mu D(\tilde{\lambda}, \tilde{\mu}) = 0 \tag{15}$$

and

$$-\nabla_\lambda D(\tilde{\lambda}, \tilde{\mu}) - z = 0, \quad z \geq 0, \quad \big[\tilde{\lambda}_i > 0 \Longrightarrow z_i = 0, \quad \forall i = 1, ..., m\big]. \tag{16}$$

Setting $\tilde{x} = -Q^{-1}(c + A^\top \tilde{\mu} + G^\top \tilde{\lambda})$, we obtain first that

$$Q\tilde{x} = -c - A^\top \tilde{\mu} - G^\top \tilde{\mu}.$$

We deduce from (15) that $A\tilde{x} = b$ and from (16) that $-(G\tilde{x} - d) = z \geq 0$. Therefore, $\tilde{x}$ is feasible (for the primal problem). Moreover, if $(G\tilde{x})_i < d_i$, then $z_i > 0$ and therefore $\tilde{\lambda}_i = 0$. The triplet $(\tilde{x}, \tilde{\lambda}, \tilde{\mu})$ satisfies the KKT conditions for the primal problem.

1.3.10 Hahn-Banach with 2 sets

We consider in this exercise a generalization of the version of the Hahn-Banach theorem seen in the lecture. Let K_0 and K_1 be two closed, non-empty, and convex subsets of $\mathbb{R}^n$. Their intersection is assumed to be empty, K_0 is assumed to be bounded (it is therefore compact).

1. Prove that the following optimization problem has a solution:

$$\inf_{(x_0,x_1)\in\mathbb{R}^{2n}} \|x_1 - x_0\|, \quad \text{subject to: } x_0 \in K_0,\ x_1 \in K_1.$$

2. Let $(\bar{x}_0, \bar{x}_1)$ be a solution to the above problem. Prove that for all $x_0 \in K_0$ and $x_1 \in K_1$,

$$\langle \bar{x}_1 - \bar{x}_0, x_0 - \bar{x}_0 \rangle \leq 0 \quad \text{and} \quad \langle \bar{x}_0 - \bar{x}_1, x_1 - \bar{x}_1 \rangle \leq 0.$$

3. Prove the existence of $\eta \in \mathbb{R}^n$, $\alpha_1 \in \mathbb{R}$ and $\alpha_2 \in \mathbb{R}$ such that for all $x_0 \in K_0$ and for all $x_1 \in K_1$,

$$\langle \eta, x_0 \rangle \leq \alpha_1 < \alpha_2 \leq \langle \eta, x_1 \rangle.$$

Give a geometrical interpretation.

Solution

1. Let C_0 be such that $K_0 \subseteq \bar{B}(0, C_0)$. Let $\tilde{x}_0$ and $\tilde{x}_1$ be two points in K_0 and K_1, respectively. Consider the optimization problem:

$$\inf_{(x_0,x_1)\in\mathbb{R}^{2n}} \|x_1 - x_0\|, \quad \text{subject to: } x_0 \in K_0,\ x_1 \in K_1 \cap \bar{B}(0, C_0 + \|\tilde{x}_1 - \tilde{x}_0\|).$$

The set $K_1 \cap \bar{B}(0, C_0 + \|\tilde{x}_1 - \tilde{x}_0\|)$ is the intersection of a compact set with a closed set, therefore, it is compact. Therefore, the feasible set of the above problem is compact, as a cartesian product of two compact sets. Since the cost function is continuous, the problem has a solution $(\bar{x}_0, \bar{x}_1)$. Observe that $(\tilde{x}_0, \tilde{x}_1)$ is feasible for the reduced problem, thus $\|\bar{x}_1 - \bar{x}_0\| \leq \|\tilde{x}_1 - \tilde{x}_0\|$.

Let us check that $(\bar{x}_0, \bar{x}_1)$ is a solution to the original problem. Let $(x_0, x_1) \in K_0 \times K_1$. If $\|x_1\| \leq C_0 + \|\tilde{x}_1 - \tilde{x}_0\|$, then $\|x_1 - x_0\| \geq \|\bar{x}_1 - \bar{x}_0\|$. Otherwise,

$$\|x_1 - x_0\| \geq \|x_1\| - \|x_0\| > C_0 + \|\tilde{x}_1 - \tilde{x}_0\| - C_0 \geq \|\bar{x}_1 - \bar{x}_0\|.$$

Thus $(\bar{x}_0, \bar{x}_1)$ is a solution to the original problem.

2. Let $x_1 \in K_0$. For all $\theta \in [0, 1]$, $\bar{x}_1 + \theta(x_1 - \bar{x}_1) \in K_1$, therefore,

$$\|\bar{x}_1 - \bar{x}_0\|^2 \leq \|\bar{x}_1 + \theta(x_1 - \bar{x}_1) - \bar{x}_0\|^2.$$

Expanding the right-hand side and dividing by $2\theta > 0$, we obtain:

$$0 \leq \langle \bar{x}_1 - \bar{x}_0, x_1 - \bar{x}_1 \rangle + \frac{1}{2}\theta\|x_1 - \bar{x}_1\|^2.$$

To the limit when $\theta \to 0$, we obtain

$$0 \leq \langle \bar{x}_1 - \bar{x}_0, x_1 - \bar{x}_1 \rangle,$$

which proves the second inequality. The first inequality can be proved similarly.

3. Observe first that $x_0 \neq x_1$, since $K_0 \cap K_1 = \emptyset$. Let us set: $\eta = \bar{x}_1 - \bar{x}_0$ and

$$\alpha_0 = \langle \eta, \bar{x}_0 \rangle \quad \text{and} \quad \alpha_1 = \langle \eta, \bar{x}_1 \rangle.$$

We have:

$$\alpha_1 - \alpha_0 = \langle \eta, \bar{x}_1 - \bar{x}_0 \rangle = \|\bar{x}_1 - \bar{x}_0\|^2 > 0.$$

Let $x_0 \in K_0$. We have

$$\langle \eta, x_0 \rangle - \alpha_0 = \langle \eta, x_0 - \bar{x}_0 \rangle = \langle \bar{x}_1 - \bar{x}_0, x_0 - \bar{x}_0 \rangle \leq 0.$$

The first inequality is proved. Similarly, for $x_1 \in K_1$, we have

$$\langle \eta, x_1 \rangle - \alpha_1 = \langle \eta, x_1 - \bar{x}_1 \rangle = \langle \bar{x}_1 - \bar{x}_0, x_1 - \bar{x}_1 \rangle.$$

Interpretation: we have two (parallel) separating hyperplanes, passing through $\bar{x}_0$ and $\bar{x}_1$ respectively and orthogonal to $\bar{x}_1 - \bar{x}_0$.

1.3.11 Optimality conditions for convex feasible sets

1. Let $K \subseteq \mathbb{R}^n$ be a non-empty, closed and convex set. Let $f : \mathbb{R}^n \to \mathbb{R}$ be a continuously differentiable and convex function. Let $\bar{x} \in K$ be such that

$$\mathrm{Proj}_K(\bar{x} - \nabla f(\bar{x})) = \bar{x}.$$

Prove that $\bar{x}$ is a solution to: $\inf_{x \in K} f(x)$.

2. Consider now the following optimization problem:

$$\inf_{x \in \mathbb{R}^n} f(x), \quad \text{subject to: } \|x\|^2 \leq 1. \tag{P}$$

The function f is assumed to be continuously differentiable, but not necessarily convex. Let K be the feasible set of the problem.

(a) Give an explicit formula (without justification) for $\mathrm{Proj}_K(x)$, for any $x \in \mathbb{R}^n$.

(b) Prove that a feasible point x satisfies the KKT conditions for problem (P) if and only if

$$x = \mathrm{Proj}_K(x - \nabla f(x)).$$

Solution

1. For all $y \in K$,

$$0 \geq \langle (\bar{x} - \nabla f(\bar{x})) - \mathrm{Proj}_K(\bar{x} - \nabla f(\bar{x})), y - \bar{x} \rangle = \langle \bar{x} - \nabla f(\bar{x}) - \bar{x}, y - \bar{x} \rangle = \langle -\nabla f(\bar{x}), y - x \rangle.$$

Therefore, for all $y \in K$,

$$\langle \nabla f(\bar{x}), y - x \rangle \geq 0$$

and finally, by convexity of f,

$$f(y) - f(x) \geq \langle \nabla f(\bar{x}), y - x \rangle \geq 0,$$

which proves the optimality of $\bar{x}$.

2. (a) The projection formula is given by

$$\mathrm{Proj}_K(x) = \begin{cases} x/\|x\| & \text{if } \|x\| \geq 1 \\ x & \text{otherwise.} \end{cases}$$

(b) A feasible point x satisfies the KKT conditions iff there exists $\lambda \geq 0$ such that

$$\nabla f(\bar{x}) + 2\lambda x = 0,$$

and such that $\|x\| < 1 \Longrightarrow \lambda = 0$.

Assume that the KKT conditions hold.

- If $\|x\| < 1$, then by the complementarity condition, $\nabla f(x) = 0$. Therefore,
$$\mathrm{Proj}_K(x - \nabla f(x)) = \mathrm{Proj}_K(x) = x.$$
- If $\|x\| = 1$, then $x - \nabla f(x) = (1 + 2\lambda)x$, therefore, $\|x - \nabla f(x)\| = 1 + 2\lambda$ and
$$\mathrm{Proj}_K(x - \nabla f(x)) = \frac{(1 + 2\lambda)x}{1 + 2\lambda} = x.$$

In both cases, $x = \mathrm{Proj}_K(x - \nabla f(x))$.

Now, let us assume that $x = \mathrm{Proj}_K(x - \nabla f(x))$.

- According to the projection formula, we have $\|\mathrm{Proj}_K(x')\| < 1$ only if $\|x'\| < 1$, in this case we have $x' = \mathrm{Proj}_K(x')$. Therefore, if $\|x\| < 1$, then $x = x - \nabla f(x)$ and thus, $\nabla f(x) = 0$. The KKT conditions are satisfied (with $\lambda = 0$).
- Otherwise, if $\|x\| = 1$, then
$$x = \frac{x - \nabla f(x)}{\|x - \nabla f(x)\|}.$$
It follows that
$$x(\|x - \nabla f(x)\| - 1) + \nabla f(x) = 0,$$
thus the KKT conditions are satisfied with $\lambda = (\|x - \nabla f(x)\| - 1)/2 \geq 0$.

1.4 Second-order optimality conditions for constrained problems

1.4.1 Second-order optimality conditions

Consider the following optimization problem:
$$\inf_{(x,y,z)\in\mathbb{R}^3} f(x, y, z), \quad \text{subject to } e(x, y, z) = 0,$$
where
$$f(x, y, z) = \exp(xy + z) + x + y \quad \text{and} \quad e(x, y, z) = \sin(x + y + z).$$
Prove that the point $(0, 0, 0)$ satisfies the KKT conditions. Is this point a local solution to the problem?

Solution The gradients of f and e are given by
$$\nabla f(x, y, z) = \begin{pmatrix} y\exp(xy + z) + 1 \\ x\exp(xy + z) + 1 \\ \exp(xy + z) \end{pmatrix} \quad \text{and} \quad \nabla e(x, y, z) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \cos(x + y + z).$$
The Hessian matrices of f and e are given by
$$\nabla^2 f(x, y, z) = \begin{pmatrix} y^2 & (1 + xy) & y \\ (1 + xy) & x^2 & x \\ y & x & 1 \end{pmatrix} \exp(xy + z) \quad \text{and} \quad \nabla^2 e(x, y, z) = -\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \sin(x + y + z).$$
Therefore, at $(0, 0, 0)$,
$$\nabla f(0, 0, 0) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \nabla e(0, 0, 0) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \nabla^2 f(0, 0, 0) = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \nabla^2 e(0, 0, 0) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$
The KKT conditions are satisfied with $\mu = -1$. The LICQ holds (since $De(0, 0, 0)$ has rank 1), therefore
$$C(0, 0, 0) = \{h \in \mathbb{R}^3 \mid h_x + h_y + h_z = 1\}.$$
A basis of $C(0, 0, 0)$ is given by the two vectors $(-1, 1, 0)$ and $(-1, 0, 1)$. Finally, we have:
$$X^\top \nabla^2 f(0, 0, 0) X = \begin{pmatrix} -2 & -1 \\ -1 & 1 \end{pmatrix}.$$
The obtained matrix is not positive semi-definite (since it has a negative diagonal term), thus $(0, 0, 0)$ is not a local minizer.

1.4.2 Second-order optimality conditions

Consider the following optimization problem:

$$\inf_{x\in\mathbb{R}^4} x_1^2 + x_2^2 - x_3^2 + x_4^2 + 4x_2x_3 - 2x_1x_4$$
$$\text{such that } \begin{cases} x_1 + x_2 + x_3 = 4, \\ x_1 - 9x_2 + x_4 = -16. \end{cases}$$

Compute the (unique) feasible point satisfying the KKT conditions. Are the sufficient second-order optimality conditions satisfied ? All the computations can be done with the help of a computer.

Solution The optimality conditions consist in the following linear system:

$$\begin{pmatrix} 2 & 0 & 0 & -2 & 1 & 1 \\ 0 & 2 & 4 & 0 & 1 & -9 \\ 0 & 4 & -2 & 0 & 1 & 0 \\ -2 & 0 & 0 & 2 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & -9 & 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ \lambda_1 \\ \lambda_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 4 \\ -16 \end{pmatrix}$$

The solution is:

$$x = \frac{1}{7}(0, 12, 16, -4, -16, 8) \approx (0, 1.71, 2.29, -0.57, -2.29, 1.14).$$

The tangential cone of the constraints is described by $\{Bh \,|\, h \in \mathbb{R}^4\}$, where

$$B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \\ -1 & 9 \end{pmatrix}.$$

The Hessian is given by:

$$A = \begin{pmatrix} 2 & 0 & 0 & -2 \\ 0 & 2 & 4 & 0 \\ 0 & 4 & -1 & 0 \\ -2 & 0 & 0 & 2 \end{pmatrix}$$

Finally, we compute:

$$B^\top AB = \begin{pmatrix} 6 & -42 \\ -42 & 154 \end{pmatrix}$$

which is neither semi-definite positive, neither semi-definite negative.

1.5 Convexity and positive-definite matrices

1.5.1 Sylvester's criterion

1. Prove that for all symmetric matrix S, for all invertible matrix Q (both of same size), S is positive definite if and only if $Q^\top SQ$ is positive definite.

2. Let A be a symmetric positive-definite matrix in $\mathbb{R}^{n\times n}$, let $b \in \mathbb{R}^n$, let $c \in \mathbb{R}$. Let $u = -A^{-1}b$. Let us define:

$$M = \left(\begin{array}{c|c} A & b \\ \hline b^\top & c \end{array}\right), \quad B = \left(\begin{array}{c|c} \mathrm{Id}_n & u \\ \hline 0 & 1 \end{array}\right). \tag{17}$$

   (a) Compute $B^\top MB$ and its determinant.

   (b) Prove that if $\det(M) > 0$, then M is positive-definite, by using the first question.

3. Finally, let S be a symmetric matrix of size n. For all $k = 1,...n$, we denote by S_k the submatrix of S formed by the first k rows and first k columns. Prove by induction Sylvester's criterion: S is positive-definite if and only if for all $k = 1,...n$, $\det(S_k) > 0$.

Solution

1. Let us assume that S is symmetric positive definite. Then, for all $x \in \mathbb{R}^n$,

$$x^\top Q^\top SQx = (Qx)^\top SQx \geq 0,$$

thus, $Q^\top SQ$ is positive semi-definite. Moreover,

$$x^\top Q^\top SQx = 0 \quad \Longleftrightarrow \quad Qx = 0 \quad \Longleftrightarrow x = 0.$$

Therefore, $Q^\top SQ$ is positive definite. Conversely, assume that $Q^\top SQ$ is positive definite. Then, $S = (Q^{-1})^\top (Q^\top SQ)Q^{-1}$ is positive definite.

2. (a) The matrix $B^\top MB$ is given by:

$$B^\top MB = \begin{pmatrix} A & 0 \\ 0 & c - b^\top Ab \end{pmatrix}.$$

Thus, $\det(B^\top MB) = \det(A)(c - b^\top Ab)$, and from the definition: $\det(B^\top MB) = \det(B)^2 \det(M)$.

(b) If $\det(M) > 0$, then $c - b^\top Ab = \frac{\det(M)\det(B)^2}{\det(A)} > 0$. The matrix $B^\top MB$ is block diagonal with positive definite blocks, thus it is itself positive definite.

3. If S is positive definite then for all k, for all $x \in \mathbb{R}^k$, setting $y = (x^\top, 0, ..., 0)^\top$,

$$x^\top S_k x = y^\top Sy \geq 0, \tag{18}$$

therefore, S_k is positive semi-definite. Then, $x^\top S_k x = 0$ if and only if $y = 0$, and $y = 0$ if and only if $x = 0$, thus, S_k is positive definite. As a positive definite matrix, it has only positive eigenvalues (by the spectral theorem), and therefore, $\det(S_k) = 0$.

Let us prove the converse property by induction. Let n, let us assume to have proved the property for this value. Let S be of size $n+1$, such that for $k = 1, ..., n+1$, $\det(S_k) > 0$. By the induction assumption, we obtain that S_n is positive definite, and since $\det(S) > 0$, we deduce that $S(= S_{n+1})$ is positive definite with question 2.

4. A symmetric matrix S is negative definite if and only if $-S$ is positive definite, thus if and anly if for all k, $\det(-S^k) = (-1)^{-k}\det(S_k) > 0$.

1.5.2 Characterizations of convexity

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a continuously differentiable function. In this exercise, $\|\cdot\|$ denotes the Euclidean norm. Consider the following statements:

(a) For all x and y in $\mathbb{R}^n$, for all $\lambda \in [0,1]$, $f((1-\lambda)x + \lambda y) \leq (1-\lambda)f(x) + \lambda f(y) - \frac{1}{2}\lambda(1-\lambda)m\|y-x\|^2$.

(b) For all x and y in $\mathbb{R}^n$, $f(y) \geq f(x) + Df(x)(y-x) + \frac{1}{2}m\|y-x\|^2$.

(c) For all x and y in $\mathbb{R}^n$, $(Df(y) - Df(x))(y-x) \geq m\|y-x\|^2$.

Prove the following equivalences.

1. (a) $\Longleftrightarrow$ (b)
2. (b) $\Longleftrightarrow$ (c).

Solution

1. (a) $\Longrightarrow$ (b). For all x and $y \in \mathbb{R}^n$, for all $\lambda \in (0,1]$, we have:

$$\frac{f(x + \lambda(y-x)) - f(x)}{\lambda} \leq f(y) - f(x) + \frac{1}{2}(1-\lambda)m\|y-x\|^2.$$

Passing to the limit when λ tends to 0, we obtain:

$$Df(x)(y-x) \leq f(y) - f(x) + \frac{1}{2}m\|y-x\|^2,$$

which proves (b).

(b) $\Longrightarrow$ (a). Let x and y in $\mathbb{R}^n$, let $\lambda \in [0,1]$. Le $z = (1-\lambda)x + \lambda y$. Applying statement (b) to z and x and to z and y, we obtain:

$$f(x) \geq f(z) + Df(z)(x-z) + \frac{1}{2}m\|x-z\|^2 = f(z) - \lambda Df(z)(y-x) + \frac{1}{2}\lambda^2\|y-x\|^2$$
$$f(y) \geq f(z) + Df(z)(y-z) + \frac{1}{2}m\|y-z\|^2 = f(z) + (1-\lambda)Df(z)(y-x) + \frac{1}{2}m(1-\lambda)^2\|y-x\|^2.$$

Multiplying these two inequalities by respectively $(1-\lambda)$ and λ and summing them, we obtain:

$$(1-\lambda)f(x) + \lambda f(y) \geq f(z) + \frac{1}{2}\underbrace{((1-\lambda)\lambda^2 + \lambda(1-\lambda)^2)}_{=\lambda(1-\lambda)} m\|y-x\|^2,$$

which proves (a).

2. Let us prove (b) $\Longrightarrow$ (c). We have:

$$f(y) \geq f(x) + Df(x)(y-x) + \frac{1}{2}m\|y-x\|^2$$
$$f(x) \geq f(y) + Df(y)(x-y) + \frac{1}{2}m\|y-x\|^2.$$

Summing these inequalities, we obtain:

$$0 \geq -((Df(y) - Df(x))(y-x)) + m\|y-x\|^2,$$

which proves (c). The implication (c) $\Longrightarrow$ (b) has already been proved.

1.5.3 Strongly convex functions

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a continuously differentiable function. We assume that f is strongly convex, that is to say, that there exists $m > 0$ such that for all x and $y \in \mathbb{R}^n$,

$$\langle \nabla f(y) - \nabla f(x), y - x\rangle \geq m\|y-x\|^2.$$

In this exercise, the norm $\|\cdot\|$ denotes the Euclidean norm.

1. Let $\tilde{x} \in \mathbb{R}^n$. Prove the following inequality: for all $x \in \mathbb{R}^n$,

$$f(x) \geq g(x) := f(\tilde{x}) + \langle \nabla f(\tilde{x}), x - \tilde{x}\rangle + \frac{1}{2}m\|x-\tilde{x}\|^2.$$

2. Prove that g possesses a unique minimizer. Calculate it. Calculate also $\min_{x\in\mathbb{R}^n} g(x)$.
3. Deduce from the above inequality that f possesses a unique minimizer, that we denote by $\bar{x}$.
4. Prove that

$$\|\tilde{x} - \bar{x}\| \leq \frac{\|\nabla f(\tilde{x})\|}{m} \quad \text{and} \quad f(\tilde{x}) - f(\bar{x}) \leq \frac{1}{2m}\|\nabla f(\tilde{x})\|^2.$$

Solution

1. It holds:

$$\begin{aligned}
f(x) - f(\tilde{x}) &= \int_0^1 \langle \nabla f(\tilde{x} + \theta(x-\tilde{x})), x - \tilde{x}\rangle \,\mathrm{d}\theta \\
&= \int_0^1 \frac{1}{\theta}(\langle \nabla f(\tilde{x} + \theta(x-\tilde{x})) - \nabla f(\tilde{x}), \theta(x-\tilde{x})\rangle)\,\mathrm{d}\theta + \int_0^1 \langle \nabla f(\tilde{x}), x - \tilde{x}\rangle \,\mathrm{d}\theta \\
&= \int_0^1 \frac{1}{\theta} m\|\theta(x-\tilde{x})\|^2 \,\mathrm{d}\theta + \langle \nabla f(\tilde{x}), (x-\tilde{x})\rangle \\
&= \underbrace{\Big(\int_0^1 \theta \,\mathrm{d}\theta\Big)}_{=1/2} m\|x-\tilde{x}\|^2 + \langle \nabla f(\tilde{x}), (x-\tilde{x})\rangle.
\end{aligned}$$

2. The function g is obviously convex. Its gradient is given by:

$$\nabla g(x) = \nabla f(\tilde{x}) + m(x - \tilde{x}).$$

Therefore, g has a unique stationary point, given by

$$\hat{x} = \tilde{x} - \frac{\nabla f(\tilde{x})}{m},$$

which is a global minimizer. We have:

$$g(\hat{x}) = f(\tilde{x}) - \frac{1}{2m}\|\nabla f(\tilde{x})\|^2.$$

3. We deduce that for all x,

$$f(x) \geq f(0) + \langle \nabla f(0), x\rangle + \frac{1}{2}m\|x\|^2 \geq f(0) - \|\nabla f(0)\| \cdot \|x\| + \frac{1}{2}m\|x\|^2.$$

The right-hand side is obviously a coercive function, therefore, f is also coercive. Since f is coercive and continuous, it possesses a minimizer. Let x_1 and x_2 be two minimizers. Then, $\nabla f(x_1) = \nabla f(x_2) = 0$ and therefore,

$$0 = \langle \nabla f(x_2) - \nabla f(x_1), x_2 - x_1\rangle \geq m\|x_2 - x_1\|^2,$$

thus, $x_1 = x_2$, which proves the uniqueness.

4. By the Cauchy-Schwarz inequality, we have:

$$m\|\tilde{x} - \bar{x}\|^2 \leq \langle \nabla f(\tilde{x}) - \nabla f(\bar{x}), \tilde{x} - \bar{x}\rangle = \langle \nabla f(\tilde{x}), \tilde{x} - \bar{x}\rangle \leq \|\nabla f(\tilde{x})\| \cdot \|\tilde{x} - \bar{x}\|.$$

It follows that $\|\tilde{x} - \bar{x}\| \leq \|\nabla f(\tilde{x})\|/m$. We also have:

$$f(\bar{x}) \geq g(\bar{x}) \geq g(\hat{x}) = f(\tilde{x}) - \frac{1}{2m}\|\nabla f(\tilde{x})\|^2,$$

which proves the second inequality.

1.6 Analysis of linear-quadratic problems

1.6.1 Linear regression

For given data points (x_1, y_1),...,(x_n, y_n) in $\mathbb{R}^2$, the linear regression problem problem consists in finding real numbers d and k solution to:

$$\inf_{(d,k)\in\mathbb{R}^2} \frac{1}{2}\sum_{i=1}^{n}(d + kx_i - y_i)^2.$$

We assume that there exists indices i and j for which $x_i \neq x_j$.

1. Motivate the problem.

2. Rewrite the problem in the form of the following unconstrained problem:

$$\inf_{z\in\mathbb{R}^2} \frac{1}{2}(Az - b)^\top(Az - b),$$

where $A \in \mathbb{R}^{n\times 2}$, $b \in \mathbb{R}^n$.

3. Prove that $A^\top A$ is a positive definite symmetric matrix.

4. Prove the existence and uniqueness of the optimal solution. Compute it explicitely. To this purpose, you can use the results of exercise 3 and the following notations:

$$S_{xx} = \frac{1}{n}\sum x_i^2, \quad S_{xy} = \frac{1}{n}\sum x_i y_i, \quad \bar{x} = \frac{1}{n}\sum x_i, \quad \bar{y} = \frac{1}{n}\sum y_i.$$

Solution

1. We want to approximate the set of points with a line of equation $y = kx + d$. The term to minimize is the sum of the square of the distances of the points to a line (described with the parameters k and d).

2. Setting $z = (d, k)^\top$, the matrix A and the vector b are given by:

$$A = \begin{pmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}, \quad b = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}. \tag{19}$$

3. The matrix $A^\top A$ is clearly symmetric, it is also positive semi-definite since for all $z \in \mathbb{R}^2$, $z^\top A^\top A z = \|Az\|^2 \geq 0$. Let us compute it explicitely:

$$A^\top A = \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}.$$

Let $z \in \mathbb{R}^2$ be such that $z^\top A^\top A z = 0$, then $Az = 0$. Then, $z = 0$, because A is injective. Indeed, it has 2 columns and posseses a invertible submatrix of size 2:

$$\begin{pmatrix} 1 & x_i \\ 1 & x_j \end{pmatrix},$$

where i and j are the indices such that $x_i \neq x_j$.

4. The function to be minimize is convex and differentiable, thus point x is a global solution if and only if $\nabla f(z) = 0$, where:

$$f(z) = \frac{1}{2}(Az - b)^\top (Az - b) = \frac{1}{2} z^\top (A^\top A) z - b^\top A z + \frac{1}{2}\|b\|^2.$$

Since $\nabla f(z) = A^\top A z - A^\top b$, we have:

$$\nabla f(d, k) = 0 \iff \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix} \begin{pmatrix} d \\ k \end{pmatrix} = \begin{pmatrix} \sum y_i \\ \sum x_i y_i \end{pmatrix}$$

$$\iff \begin{pmatrix} 1 & \bar{x} \\ \bar{x} & S_{xx} \end{pmatrix} \begin{pmatrix} d \\ k \end{pmatrix} = \begin{pmatrix} \bar{y} \\ S_{xy} \end{pmatrix} \iff \begin{cases} k = \frac{S_{xy} - \bar{x}\bar{y}}{S_{xx} - \bar{x}^2} \\ d = \bar{y} - k\bar{x}. \end{cases}$$

The proves the existence and uniqueness and provides the explicit solution.

1.6.2 Polynomial interpolation

Let $t_1 < ... < t_m$ and $y_1, ..., y_m$ be $2m$ real numbers. In this exercise, the coordinates of a vector $x \in \mathbb{R}^{n+1}$ are denoted x_0,x_1,...,x_n. For all $x \in \mathbb{R}^{n+1}$, we denote by P_x the polynomial function defined by:

$$P_x(t) = \sum_{i=0}^{n} x_i t^i.$$

We consider the following optimization problem:

$$\inf_{x \in \mathbb{R}^n} f(x) := \frac{1}{2} \sum_{j=1}^{m} \big(P_x(t_j) - y_j\big)^2.$$

1. Motivate the problem.

2. Find a matrix $C \in \mathbb{R}^{m \times (n+1)}$ such that for all $x \in \mathbb{R}^{n+1}$,

$$Cx = (P_x(t_1), ..., P_x(t_m)).$$

3. Find a matrix $A \in \mathbb{R}^{(n+1)\times(n+1)}$, a vector $b \in \mathbb{R}^{n+1}$, and a real number c such that for all $x \in \mathbb{R}^n$,

$$f(x) = \frac{1}{2} x^\top A x + \langle b, x \rangle + c.$$

4. It can be proved that for $m \geq n + 1$, the matrix C has full column rank (i.e. the mapping $x \mapsto Cx$ is injective). Using this property, prove that A symmetric positive definite and that f is strongly convex.

Solution

1. The goal is to identify a polynomial function of degree n such that $P_x(t_j) \approx y_j$. If $n = m + 1$, we are in the situation of Lagrange interpolation.

2. The matrix C is given by:
$$C = \begin{pmatrix} 1 & t_1 & t_1^2 & \dots & t_1^n \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & t_m & t_m^2 & \dots & t_m^n \end{pmatrix}.$$

3. We have:
$$f(x) = \frac{1}{2}\|Cx - y\|^2 = \frac{1}{2}(Cx - y)^\top (Cx - y) = \frac{1}{2}x^\top C^\top Cx + \langle C^\top y, x + \frac{1}{2}\|y\|^2.$$

4. Clearly, A is symmetric and positive semi-definite. Let x be such that $Ax = 0$. Then, $x^\top Ax = x^\top C^\top Cx = \|Cx\| = 0$ and thus $Cx = 0$. Since C has full column rank, $x = 0$, which proves that A is positive definite. We have:
$$\langle \nabla f(y) - \nabla f(x), y - x\rangle = \langle A(y - x), y - x\rangle \geq m\|y - x\|^2,$$
where $m > 0$ is the small eigenvalue of A.

1.6.3 Distance between lines

Given x_1, x_2, v_1, and v_2 in $\mathbb{R}^n$, we consider two lines in $\mathbb{R}^n$, parameterized as follows:
$$\mathcal{L}_1 = \{x_1 + tv_1 \,|\, t \in \mathbb{R}\} \quad \text{and} \quad \mathcal{L}_2 = \{x_2 + tv_2 \,|\, t \in \mathbb{R}\}.$$
We assume that v_1 and v_2 are linearly independent. Compute explicitely the solution to:
$$\inf_{(x_1,x_2)\in\mathcal{L}_1\times\mathcal{L}_2} \|x_2 - x_1\|^2,$$
where $\|\cdot\|$ is the Euclidean norm.

Solution To simplify, we formulate the problem as follows:
$$\inf_{(x_1,x_2)\in\mathcal{L}_1\times\mathcal{L}_2} \frac{1}{2}\|x_2 - x_1\|^2.$$
Setting: $x = (x_2 - x_1)$, $t = (t_1, t_2)^\top$, $V = (-v_1\, v_2)$, we observe that:
$$(x_2 + t_2v_2) - (x_1 + t_1v_1) = x + Vt.$$
The problem is reformulated as follows:
$$\inf_{t\in\mathbb{R}^2} \frac{1}{2}t^\top(V^\top V)t + x^\top Vt + \frac{1}{2}\|x\|^2.$$
We already know that $V^\top V$ is symmetric positive-semidefinite, therefore, we minimize a convex function. For all $t \in \mathbb{R}^2$, t is an optimal solution if and only if it a stationary point, that is to say, if:
$$(V^\top V)t + V^\top x = 0.$$
Note that:
$$V^\top V = \begin{pmatrix} \|v_1\|^2 & -\langle v_1, v_2\rangle \\ -\langle v_1, v_2\rangle & \|v_2\|^2 \end{pmatrix}, \quad V^\top x = \begin{pmatrix} -\langle v_1, x\rangle \\ \langle v_2, x\rangle \end{pmatrix}.$$
We consider different cases.

1. If v_1 and v_2 are not colinear, then V is injective and thus $V^\top V$ positive-definite (and thus regular), there exists a unique solution, given by:
$$t = -(V^\top V)^{-1}V^\top x = \frac{1}{\|v_1\|^2\|v_2\|^2 - \langle v_1, v_2\rangle^2} \begin{pmatrix} \|v_2\|^2 & \langle v_1, v_2\rangle \\ \langle v_1, v_2\rangle & \|v_1\|^2 \end{pmatrix} \begin{pmatrix} \langle v_1, x\rangle \\ -\langle v_2, x\rangle \end{pmatrix}.$$

2. We assume that $v_1 \neq 0$ and $v_2 = \beta v_1$, with $\beta \in \mathbb{R}$. The optimality conditions reads:

$$\|v_1\|^2 \begin{pmatrix} 1 & -\beta \\ -\beta & \beta^2 \end{pmatrix} \begin{pmatrix} t_1 \\ t_2 \end{pmatrix} + \langle v_1, x \rangle \begin{pmatrix} 1 \\ -\beta \end{pmatrix} = 0.$$

The second line is equal to $-\beta$ multiplied by the first line: it is redundant. The set of solutions is:

$$\Big\{ (t_1, t_2) \in \mathbb{R}^2 \,|\, t_1 = \frac{-\langle v_1, x \rangle}{\|v_1\|^2} + \beta t_2 \Big\}.$$

Observe that if $\beta = 0$, t_1 is uniquely defined.

3. If $v_1 = v_2 = 0$, the problem is trivial and all $(t_1, t_2) \in \mathbb{R}^2$ are solutions.

1.6.4 Signal processing

A signal, taking real values, is measured at times $t_1 < ... < t_n$. The corresponding measures are denoted y_1,...,y_n. Because of error measurements, a denoising process may be required in order to smoothen the signal. A (very basic) approach consists in solving the following problem:

$$\inf_{x \in \mathbb{R}^n} f(x) := \frac{1}{2} \sum_{i=1}^{n} (x_i - y_i)^2 + \frac{\alpha}{2} \sum_{i=1}^{n-1} (x_{i+1} - x_i)^2,$$

for some regularization parameter $\alpha > 0$. Find a symmetric matrix $A \in \mathbb{R}^{n \times n}$, a vector $b \in \mathbb{R}^n$ and a real number c such that for all $x \in \mathbb{R}^n$, $f(x) = \frac{1}{2} x^\top A x + \langle b, x \rangle + c$. Prove the existence and uniqueness of a solution to the optimization problem.

Solution Consider the matrix $C \in \mathbb{R}^{(n-1) \times n}$ defined as follows:

$$C = \begin{pmatrix} -1 & 1 & & \\ & \ddots & \ddots & \\ & & -1 & 1 \end{pmatrix}.$$

We have:

$$f(x) = \frac{1}{2} \|x - y\|^2 + \frac{\alpha}{2} \|Cx\|^2 = \frac{1}{2} x^\top (\mathrm{Id} + C^\top C) x + \langle y, x \rangle + \frac{1}{2} \|y\|^2.$$

Clearly, $\mathrm{Id} + C^\top C$ is symmetric positive definite, which proves that f has a unique minimizer. We also have:

$$C^\top C = \begin{pmatrix} -1 & & \\ 1 & \ddots & \\ & \ddots & -1 \\ & & 1 \end{pmatrix} \begin{pmatrix} -1 & 1 & & \\ & \ddots & \ddots & \\ & & -1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 1 \end{pmatrix}.$$

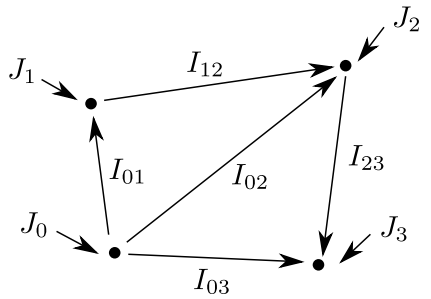
1.7 Various problems

1.7.1 Electrical networks

We consider an electrical network made of N nodes, 1,...,N, connected by lines. The set of lines is described by a subset E of $N \times N$: any pair of nodes (k, ℓ) with $k < \ell$ is connected if and only if $(k, \ell) \in E$. We assume that the network is connected, which means that for all $k \neq \ell$, there exists a sequence of nodes $k = k_1, k_2, ..., k_{p-1}, k_p = \ell$ such that $(k_1, k_2) \in E$ or $(k_2, k_1) \in E$,..., and such that $(k_{p-1}, k_p) \in E$ or $(k_p, k_{p-1}) \in E$.

Let $(k, \ell) \in E$, the resistance of the line connecting k and ℓ is denoted by $R_{k\ell} > 0$. We denote by $I_{k\ell}$ the intensity of the current between k and ℓ, supposed to be oriented from k to ℓ. We also denote: $I = (I_{ij})_{(i,j) \in E} \in \mathbb{R}^E$. At each node k of the network, a current of intensity J_k is injected. The variable J_k can be negative (meaning that some current is withdrawn from the network). An example is shown Figure 1.7.1.

Our model is a direct current model, described by the following two laws:



$$E = \{(0,1),(0,2),(0,3),(1,2),(2,3)\}$$

Figure 1: Example of electrical network

- Kirchhoff's law at node k: $h_k(I) := \Big(\sum_{(\ell,k)\in E} I_{\ell k}\Big) - \Big(\sum_{(k,\ell)\in E} I_{k\ell}\Big) = -J_k$.
- Ohm's law: there exist real numbers V_1,...,V_N such that $V_\ell - V_k = R_{k\ell} I_{k\ell}$, $\forall (k,\ell) \in E$. The variable V_k is called voltage (or electrical tension) at the node k.

Our goal is to prove that for prescribed injected currents J_1,...,J_N, there exists $I \in \mathbb{R}^E$ and $V \in \mathbb{R}^{N+1}$ such that Kirchhoff's and Ohm's laws are satisfied.

We consider the following optimization problem:

$$\inf_{I\in\mathbb{R}^E} \ \frac{1}{2} \sum_{(i,j)\in E} R_{(i,j)} I_{(i,j)}^2, \quad \text{subject to: } h_k(I) = -J_k,\ \forall k = 1,...,N. \tag{P}$$

1. Let the injected currents J_1,...,J_N be given. Prove that if the feasible set of (P) is non-empty, then $\sum_{k=1}^N J_k = 0$.
2. *Optional.* Prove that if $\sum_{k=1}^N J_k = 0$, then the feasible set of (P) is non-empty.
3. We assume now that the feasible set of (P) is non-empty. Prove the existence of a solution to the optimization problem, justify and write the KKT conditions. Conclude.
4. Is the voltage uniquely defined? Does the LICQ hold at the solution of Problem (P)?

Solution

1. If Kirchhoff's law holds at all the nodes, we have:

$$-\sum_{k=1}^N J_k = \underbrace{\sum_{k=1}^N \Big(\sum_{(\ell,k)\in E} I_{\ell k}\Big)}_{A} - \underbrace{\sum_{k=1}^N \Big(\sum_{(k,\ell)\in E} I_{k\ell}\Big)}_{B} = 0.$$

   Indeed, each current I_{ij} is counted exactly once in A (for $k = j$ and $\ell = i$) and exactly once in B (for $k = i$ and $\ell = j$).

2. Let J_1,...,J_N be given injected currents. Algorithm 1 below provides a feasible flow of currents.
3. The cost function is coercive and continuous. The feasible set is non-empty, closed (since described by linear equalities). Therefore, there exists a solution. The KKT conditions are satisfied, since the constraints are linear. Let us denote by $V \in \mathbb{R}^N$ the Lagrange multiplier associated with Kirchhoff's law. The corresponding Lagrangian is then:

$$L(I,V) = \frac{1}{2}\sum_{k=1}^N \sum_{(k,\ell)\in E} R_{k\ell} I_{k\ell}^2 - \sum_{k=1}^N V_k \Big[\Big(\sum_{(\ell,k)\in E} I_{\ell k}\Big) - \Big(\sum_{(k,\ell)\in E} I_{k\ell}\Big) + J_k\Big].$$

Deriving L with respect to I, we obtain that

$$D_{I_{ij}} L(I, V) = R_{ij} I_{ij} - V_j + V_i = 0,$$

and Ohm's law follows.

4. If Ohm's law holds for V, then it also holds for $V + c$, where c is a real constant. The Lagrange multiplier is not uniquely defined, which implies that the LICQ does not hold. This can be understood as follows: there are too many constraints. Since $\sum_{k=1}^{N} J_k = 0$, if Kirchhoff's law holds at $N - 1$ nodes, then it also holds at the last remaining node.

Algorithm 1: Construction of a feasible current assignment

Set $I^0 = 0$ (in $\mathbb{R}^{|E|}$) and $\tilde{J}^0 = 0$ (in $\mathbb{R}^N$);
for $k = 2$ *to* N **do**
  Set $I^k = I^{k-1}$, $\tilde{J}^k = \tilde{J}^{k-1}$, $\tilde{J}^k = J_k$;
  Find a non-cycling path $k = k_1,...,k_p = 1$ connecting the node k to the node 1;
  for $\ell = 1$ *to* $p - 1$ **do**
    If $k_\ell < k_{\ell+1}$, set $I^k_{k_\ell,k_{\ell+1}} = I^k_{k_\ell,k_{\ell+1}} - J_k$;
    If $k_\ell < k_{\ell+1}$, set $I^k_{k_\ell,k_{\ell+1}} = I^k_{k_\ell,k_{\ell+1}} + J_k$;
  end
  At this end of this loop, I^k and $\tilde{J}^k$ satisfy Kirchhoff's law at the nodes 1,...,N;
end
At the end of the algorithm, $\tilde{J}^N = J$. Moreover, I^N and J satisfy Kirchhoff's law.

1.7.2 Modelization of a transport network

In this problem, we study the equilibrium arising in a transport network when all the users try to use optimal paths. The network is modelized by an oriented graph, described by a set of nodes of cardinal N (denoted by $\mathcal{N} = \{1, ..., N\}$), and by a set of edges (denoted by $\mathcal{E}$, with cardinal M). An edge is described by its origin i and its destination j and is denoted by (i, j). For a given node i, we denote by $\sigma(i)$ (resp. $\rho(i)$) the set of nodes j such that the edge (i, j) (resp. (j, i)) exists.

We also consider a matrix A of size N describing for all $k \neq l$ the number of users of the network willing to go from node k (the origin) to node l (the destination). To this purpose, a user with origin k and destination l will travel on a path

$$k = k_0,\ k_1,...,k_q = l, \text{ where } k_1 \in \sigma(k_0),...,k_q \in \sigma(k_{q-1}). \tag{20}$$

We assume that for all k and l in $\mathcal{N}$, there exists a path connecting these two nodes. The diagonal terms of the matrix A are set to:

$$A_{kk} = -\sum_{q \neq k} A_{kq}, \quad \forall k = 1, ..., N.$$

An assignement is a variable $x = (x_{(i,j),k})_{(i,j)\in\mathcal{E},k\in\mathcal{N}}$ in $\mathbb{R}_+^{MN}$. For all $(i, j) \in \mathcal{E}$, for all $k \in \mathcal{N}$, $x_{(i,j),k}$ is the (real) number of users having the same origin k (not necessarily the same destination) using the edge (i, j). The number of users is not necessarily an integer number, and can be any non-negative real number (our model is "continuous", with infinitesimaly small users). An example of a feasible assignement, for a given network and a given matrix A is given in figure 2.

A continuous and non-decreasing function $f_{ij} : \mathbb{R}_+ \to \mathbb{R}_+$ is associated to each edge (i, j). The time needed to travel on a given edge (i, j) is a function of the total amount of users on the edge:

$$f_{ij}\Big(\sum_{k\in\mathcal{N}} x_{(i,j),k}\Big).$$

The time to travel on a path like (20) is simply the sum of the time needed for all the edges (k_0, k_1),...(k_{q-1}, k_q). Finally, we denote by F_{ij} a fixed primitive of f_{ij}.

To summarize, the data of the model is: the nodes $\mathcal{N}$, the edges $\mathcal{E}$, the functions $(f_{ij})_{(i,j)\in\mathcal{E}}$ and the matrix A. The goal of the problem is to prove the existence of an assignement x for which all the users take an optimal path.

$$A = \begin{pmatrix} -3 & 1 & 1 & 1 \\ 0 & -1 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \begin{cases} x_{(1,2),1} = & 3, \quad x_{(2,3),2} = & 1, \\ x_{(2,3),1} = & 1, \quad x_{(3,2),3} = & 1, \\ x_{(2,4),1} = & 1, \quad x_{(i,j),k} = & 0, \text{ otherwise.} \end{cases}$$

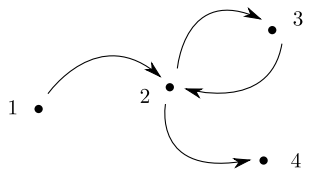


Figure 2: Example of a network with a feasible assignement

1. We denote by X the subset $\mathbb{R}^{NM}$ of feasible assignements. Give a mathematical description of X with $N^2 + NM$ linear constraints.

2. Prove the existence of an optimal solution x^* to the problem:

$$\min_{x \in \mathbb{R}^{NM}} \sum_{(i,j)\in\mathcal{E}} F_{ij}\Big(\sum_{k\in\mathcal{N}} x_{(i,j),k}\Big), \quad \text{subject to: } x \in X$$

   For such a problem with linear equality constraints, it can be shown that the optimal solutions satisfies the KKT conditions.

3. Write the KKT conditions.

4. Prove that any x^* satisfying the KKT conditions is optimal.

5. Consider now a path of the form (20), and assume that the path is visited by the users with origin k, that is to say: $x_{(k_0,k_1),k} > 0$,...,$x_{(k_{q-1},k_q),k} > 0$. Find a relation between the time to travel on the path and the Lagrange multipliers.

6. Deduce from the KKT conditions that a path of the form (20) is visited only if it is the shortest path connecting k and l.

Solution 1. The set X is described by:

$$\sum_{j\in\rho(i)} x_{(j,i),k} - \sum_{j\in\sigma(i)} x_{(i,j),k} = A_{ki}, \ \forall (i,k) \in \mathcal{N}^2, \quad x_{(i,j),k} \geq 0, \quad \forall ((i,j),k) \in \mathcal{E} \times \mathcal{N}.$$

2. The feasible set is non-empty, since by assumption, there always exists a path to go from a point to another point. Moreover it is bounded: for all $(i,j) \in \mathcal{E}$, for all $k \in \mathcal{N}$, $0 \leq x_{(i,j),k} \leq -A_{kk}$, and clearly closed. The cost function is continuous, therefore, there exists an optimal solution.

3. We denote by $\lambda \in \mathbb{R}^{N^2}$ and $\mu \in \mathbb{R}^{NM}$ the dual variables. The Lagrangian is:

$$\begin{aligned} L(x,\lambda,\mu) = & \sum_{(i,j)\in\mathcal{E}} F_{ij}\Big(\sum_{k\in\mathcal{N}} x_{(i,j),k}\Big) - \sum_{(i,j)\in\mathcal{E},\, k\in\mathcal{N}} \mu_{(i,j),k} x_{(i,j),k} \\ & + \sum_{(i,k)\in\mathcal{N}^2} \lambda_{k,i}\Big(\sum_{j\in\rho(i)} x_{(j,i),k} - \sum_{j\in\sigma(i)} x_{(i,j),k} - A_{ki}\Big). \end{aligned}$$

The KKT conditions are: there exist $\lambda \in \mathbb{R}^{NN}$ and $\mu \in \mathbb{R}_+^{NM}$ such that:

$$D_{x_{(i,j),k}} L(x,\lambda,\mu) = f_{ij}\Big(\sum_{k\in\mathcal{N}} x_{(i,j),k}\Big) - \lambda_{k,i} + \lambda_{k,j} - \mu_{(i,j),k}, \quad \mu x = 0.$$

4. Since the functions f_{ij} are non-decreasing, the functions F_{ij} are convex, and the cost function is convex. The constraints are all linear, therefore the problem is convex, and any KKT point is an optimal solution.

5. Let us consider the path of the form $k = k_0, k_1, ..., k_q = l$. By the optimality conditions, the time needed to travel on an edge (k_r, k_{r+1}) is: $-\lambda_{k,k_{r+1}} + \lambda_{k,k_r} - \mu_{(k_r,k_{r+1}),k}$. However, $\mu_{(k_r,k_{r+1}),k} = 0$, since $x_{(k_r,k_{r+1}),k} > 0$. The time to travel on the path is then: $-\lambda_{k,l} + \lambda_{k,k}$.

6. Consider one of the quickest paths from k to l, that we denote $k = k'_0, k'_1, ..., k'_{q'} = l$. The associated time, (given in the left-hand side of (21)), is then longer or equal to the one of any visited path:

$$-\lambda_{k,l} + \lambda_{k,k} + \sum_{r=1}^{q'} \mu_{(k'_{r+1},k'_r),k} \geq -\lambda_{k,l} + \lambda_{k,k}, \tag{21}$$

meaning that a visited path is necessarily the quickest.

1.7.3 Nash equilibria

We consider in this exercise the following idealized economical situation: the textile market is divided in N submarkets (men, women, and children's wear, sportswear...), each one them being dominated by a single company. Each company undertakes a prodution strategy and its sales performance can be expressed as function of the amount of cotton bought to a producer. In this somewhat simple model, there is a unique cotton producer. The unitary price of cotton is an increasing function of the total demand in cotton (of all textile companies).

The situation is therefore a typical from game theory: each company minimizes a certain cost function, which itself depends on the production strategies of the other companies (through the price of cotton). The goal of the exercise is to prove the existence of an equilibrium, under some convexity assumptions.

We denote by $x_i \in \mathbb{R}$ the quantity of cotton bought by the company i (to simplify, we do not consider restrictions on x_i). The opposite of the sales revenue is given by a function $L_i(x_i)$, assumed to be continuously differentiable and strongly convex. The unitary price of cotton is given by $P = \phi\big(\sum_{i=1}^N x_i\big)$. The function $\phi\colon \mathbb{R} \to \mathbb{R}$ is assumed to be continuous and increasing.

A vector $\bar{x} \in \mathbb{R}^N$ is called Nash equilibrium if the following holds true:

$$\bar{x}_i \in \operatorname{argmin}_{x_i \in \mathbb{R}} \big\{Px_i + L_i(x_i)\big\}, \quad \forall i = 1, ..., N$$

for $P = \phi\big(\sum_{i=1}^N \bar{x}_i\big)$.

1. Find a function $f\colon \mathbb{R}^n \to \mathbb{R}$ which is such that for all $x \in \mathbb{R}^n$, x is a Nash equilibrium if and only if x minimizes f.

2. Deduce that there exists a unique Nash equilibrium.

3. Compute explicitely the Nash equilibrium in the following situation: $\phi(y) = 1 + y$, $L_i(x_i) = \frac{1}{2a_i}(x_i - y_i)^2$, where the coefficients a_1,...,a_n are all positive and y_1,...,y_n are given real numbers.

Solution

1. Observe first that the mapping $Px_i + L_i(x_i)$ is convex. Therefore, $\bar{x}$ is a Nash equilibrium if and only if
$$P + L'_i(\bar{x}_i) = 0.$$
Let Φ be defined by $\Phi(y) = \int_0^y \phi(z)dz$. We define:
$$f\colon x \in \mathbb{R}^n \mapsto \Phi\Big(\sum_{i=1} x_i\Big) + \sum_{i=1}^n L_i(x_i).$$
We have
$$\nabla f(x) = \begin{pmatrix} \phi\big(\sum_{i=1} x_i\big) + L'_1(x_1) \\ \vdots \\ \phi\big(\sum_{i=1} x_i\big) + L'_N(x_N) \end{pmatrix}.$$
We deduce that $\bar{x}$ is a Nash equilibrium if and only if $\nabla f(\bar{x}) = 0$. Obviously f is convex, therefore $\bar{x}$ is a Nash equilibrium if and only if $\bar{x}$ minimizes f.

2. Let us denote by m_i the convexity parameter of the functions L_i. Let $m = \min_{i=1,\dots,N} m_i > 0$. For all x and $y \in \mathbb{R}^N$, we have

$$\begin{aligned}\langle \nabla f(y) - \nabla f(x), y - x\rangle &= (\phi(Y) - \phi(X))(Y - X) + \sum_{i=1}^{N} \big(L_i'(y_i) - L_i'(x_i)\big)(y_i - x_i)\\ &\geq \sum_{i=1}^{N} m_i (y_i - x_i)^2\\ &\geq m\|y - x\|^2,\end{aligned}$$

where $X = \sum_{i=1}^N x_i$ and $Y = \sum_{i=1}^N y_i$. Thus f is strongly convex and possesses a unique minimizer.

3. We have in this case:

$$P + (x_i - y_i)/a_i = 0 \quad P = 1 + \sum_{i=1}^{n} x_i.$$

Multiplying the first relation by a_i and summing up, we find:

$$\Big(\sum_{i=1}^{n} a_i\Big) P + \sum_{i=1}^{n} x_i = \sum_{i=1}^{n} y_i.$$

Therefore, setting $A = \sum_{i=1}^n a_i$ and $Y = \sum_{i=1}^n y_i$, we successively obtain that

$$A + (1 + A)\sum_{i=1}^{n} x_i = Y, \quad \sum_{i=1}^{n} x_i = \frac{Y - A}{1 + A}, \quad P = 1 + \frac{Y - A}{1 + A} = \frac{1 + Y}{1 + A}, \quad x_i = y_i - a_i P.$$

1.7.4 Optimal control problems

We consider in this exercise a system which evolves over time under the control of a "manager". We denote by $x_0 \in \mathbb{R}^n$ the state of the system at time 0, by $x_1 \in \mathbb{R}^n$ the state at time 1, and so on until time N. The variable x_i is called *state variable*. In our model, we consider that x_{i+1} depends on the previous value of the state x_i and an additional variable, $u_{i+1} \in \mathbb{R}^m$, called *control variable*:

$$x_{i+1} = f_i(x_i, u_{i+1}), \quad \text{for all } i = 0, \dots, N - 1.$$

The function $f\colon \{0, \dots, N - 1\} \times \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^n$ is known and assumed to be continuously differentiable with respect to (x, u). For example, the manager can be a gas company: the variable x_i models the level of stocks of gas at a given time and the variable u_i the quantity of gas sold or bought on a market.

Let us consider now the following optimization problem:

$$\inf_{\substack{x_1, x_2, \dots, x_N \in \mathbb{R}^n\\ u_1, u_2, \dots, u_N \in \mathbb{R}^m}} \sum_{i=0}^{N-1} \ell_i(x_i, u_{i+1}) + \phi(x_N), \quad \text{subject to: } x_{i+1} = f_i(x_i, u_{i+1}), \quad \forall i = 0, \dots, N - 1.$$

The function $\ell\colon \{0, \dots, N - 1\} \times \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ is given and has the same regularity as f. The function ϕ is given and is continuously differentiable. The initial value of the state x_0 is also known and fixed.

The problem under investigation can be recasted as a *reduced* unconstrained optimization problem: it suffices to eliminate the variables x_1,...,x_N. Indeed, given the parameter x_0 and the controls u_1,...,u_N, the variable x_1 is uniquely defined by $x_1 = f_0(x_0, u_1)$, the variable x_2 by $x_2 = f_1(f_0(x_0, u_1), u_2)$, and so on.

The goal of the exercise is to calculate the gradient of the cost function of the reduced problem. To this purpose, we introduce the following function, for $k \in \{0, \dots, N\}$:

$$J_k(x_k, u_{k+1}, \dots, u_N) = \Big(\sum_{i=k}^{N-1} \ell_i(x_i, u_{i+1})\Big) + \phi(x_N),$$

where x_{k+1},...,x_N are defined by: $x_{i+1} = f_i(x_i, u_{i+1})$, for $i = k, \dots, N - 1$. Note that $J_N(x_N) = \phi(x_N)$. The goal is now to compute the gradient of $J_0(x_0, u_1, \dots, u_N)$ with respect to the control variables.

1. Find a relation between J_{k-1} and J_k.
2. Let us fix u_1,...,u_N. Let x_1,...,x_N be such that $x_{i+1} = f_i(x_i, u_{i+1})$, for all $i = 0, ..., N-1$. We set:

$$p_k = \nabla_{x_k} J_k(x_k, u_{k+1}, ..., u_N),$$

for all $k = 0, ..., N$. Note that $p_N = \nabla\phi(x_N)$. Find a relation between p_{k-1} and p_k.

3. Prove the following formula:

$$\nabla_{u_k} J_0(x_0, u_1, ..., u_N) = \nabla_u \ell_{k-1}(x_{k-1}, u_k) + \nabla_u f_{k-1}(x_{k-1}, u_k) p_k.$$

Solution

1. We have

$$J_{k-1}(x_{k-1}, u_k, ..., u_N) = \ell_{k-1}(x_{k-1}, u_k) + J_k(f_{k-1}(x_{k-1}, u_k), u_{k+1}, ..., u_N).$$

2. Differentiating this relation with respect to x_{k-1}, we obtain with the chain rule that

$$p_{k-1} = \nabla_x \ell_{k-1}(x_{k-1}, u_k) + \nabla_x f_{k-1}(x_{k-1}, u_k) p_k.$$

3. We have

$$\nabla_{u_k} J_k(x_{k-1}, u_k, ..., u_N) = \nabla_u \ell_{k-1}(x_{k-1}, u_k) + \nabla_u f_{k-1}(x_{k-1}, u_k) p_k.$$

It is then easy to show by induction that

$$\nabla_{u_k} J_0(x_0, u_1, ..., u_N) = \nabla_{u_k} J_k(x_{k-1}, u_k, ..., u_N),$$

which proves the formula.

1.7.5 Optimal control

Let us consider the optimal control problem described in exercise 6.4, with the same regularity assumptions of f, ℓ, and ϕ:

$$\inf_{\substack{x_1, x_2, ..., x_N \in \mathbb{R}^n \\ u_1, u_2, ..., u_N \in \mathbb{R}^m}} \sum_{i=0}^{N-1} \ell_i(x_i, u_{i+1}) + \phi(x_N), \quad \text{subject to: } x_{i+1} = f_i(x_i, u_{i+1}), \quad \forall i = 0, ..., N-1.$$

We recall that x_0 is given. We regard this problem as a problem with constraints and do not follow anymore the approach consisting in eliminating the variables $(x_1, ..., x_N)$. Consider a solution $(x_1, ..., x_N, u_1, ..., u_N)$ to the problem. Prove that the LICQ is satisfied and write the KKT conditions (denote by p_1 the Lagrange multiplier associated with the constraint $f_0(x_0, u_1) - x_1 = 0$, by p_2 the Lagrange multiplier associated with the constraint $f_1(x_1, u_2) - x_2 = 0$, and so on).

1.7.6 Optimal control: application

We analyse an application of the previous exercise. We consider a kind of necklace made of $N+1$ balls of identical masses, connected by lines of identical lengths (say 1). The coordinates of the i-th ball is $x_i = (x_{i,1}, x_{i,2})$. The whole system is hanged and submitted to gravity (in the opposite direction of the second coordinate), more precisely, the balls 0 and N are fixed to the positions $\bar{x}_0$ and $\bar{x}_N$, see figure 3.

The equilibrium position of the system is such that the potential energy: $\sum_{i=0}^{N} x_{i,2}$ is minimized (under the constraints of position for the balls 0 and N, and under the constraint that the distance between two consecutive balls is smaller than 1).

Formulate the problem as an optimal control problem and write the KKT conditions. Give a physical interpretation of the KKT conditions. We do not discuss the existence of a solution and do not discuss the qualification condition.

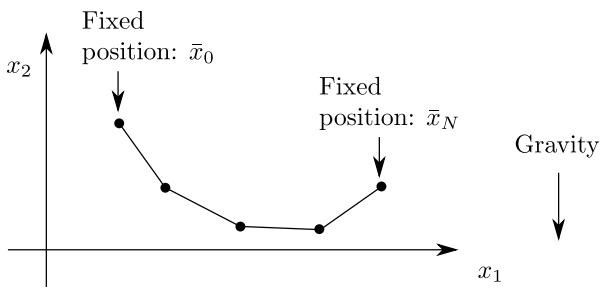


Figure 3: Hanged necklace

Solution We define: $f(x_i, u_i) = x + u$, $\ell(x_i, u_i) = x_{2,i}$, $c(x, u) = \|u\|^2$, $\phi(x_N) = x_{2,N}$, $\psi(x_N) = x_N - \bar{x}_N$ and apply the KKT conditions. We obtain:

$$\begin{aligned} &p_i = p_{i+1} + (0, 1) \\ &p_{i+1} + 2\alpha_i u_i = 0,\ \alpha_i \geq 0,\ \alpha_i(\|u_i\|^2 - 1) = 0. \end{aligned}$$

Interpretation: p_i is the force of the ball i on the ball $i+1$. The vector $(0, -1)$ is the gravity.

1.7.7 Optimal control: fishing problem

In this problem, we modelize the evolution of a population of fishes and optimize the management of the resource. Let $t_0 \leq T$ be two integers, defining the time interval of the problem. The dynamics of the population is given by a vector $(x_{t_0}, x_{t_0+1}, ..., x_T)$ satisfying:

$$x_{t+1} = \gamma x_t - u_t, \quad \text{for all } t \in [t_0, T-1].$$

The parameter $\gamma > 1$ describes the natural increase of the population, the variable u_t is the number of fishes which are fished in the time interval $[t, t+1]$, bringing to the manager a benefit of $c(u_t)$.

Finally, we consider the optimization problem for the manager:

$$\sup_{\substack{(u_t)_{t=t_0,\dots,T-1} \\ (x_t)_{t=t_0+1,\dots,T}}} \sum_{t=t_0}^{T-1} c(u_t) \quad \text{subject to:} \quad \begin{cases} x_{t+1} = \gamma x_t - u_t, \text{ for all } t = t_0, ..., T-1 \\ x_T \geq \bar{x} \\ u_t \geq 0, \text{ for all } t = t_0, ..., T-1. \end{cases}$$

The parameters of the problem are: γ, the cost function c (supposed to be continuous), the initial time t_0, the initial population x_{t_0}, the final time T and the minimal population at the final time $\bar{x}$.

We denote by $V(t_0, x_0)$ the value of the problem as a function of the initial time and population.

1. Give a condition on the parameters for the existence of an optimal solution.
2. We fix c, T, $\bar{x}$, and γ. Prove rigorously[1] that for all $x_0 \in \mathbb{R}$, for all $t_0 \leq T-1$,
$$V(t_0, x_0) = \sup_{u \geq 0} \big\{c(u) + V(t_0 + 1, \gamma x_0 - u)\big\}, \quad V(T, x_0) = \begin{cases} 0 & \text{if } x_0 \geq \bar{x} \\ -\infty & \text{if } x_0 < \bar{x}. \end{cases}$$
3. We consider from now on that $c(u) = u^\alpha$, where $\alpha \in (0, 1)$ is a given parameter. We fix $t_0 = 0$. Prove the uniqueness of the optimal solution (we assume that the condition of question 1 is fulfilled), we denote it (u^*, x^*). Prove that for all $t = t_0, ..., T-1$, $u_t^* > 0$ and that $x_T^* = \bar{x}$, or that for all t, $u_t = 0$.
4. Write the KKT conditions.
5. Solve explicitely the problem (justify carefully your answer).
Hint: Any sequence $(v_t)_t$ satisfying $v_{t+1} = \gamma v_t - \beta \xi^t$ for all $t \geq 0$ and for some $\beta \in \mathbb{R}$ and $\xi \neq \gamma$ can be described explicitely by
$$v_t = v_0 \gamma^t - \beta \frac{\gamma^t - \xi^t}{\gamma - \xi}.$$

[1] A short proof will be sufficient, actually.

Solution 1. We observe that, if there is no fishing, the population of the fishes is equal to $x_{t_0}\gamma^{t-t_0}$ at any time t. We deduce that:

$$0 \leq u_t \leq x_{t_0}\gamma^{t-t_0+1} \quad \text{and} \quad 0 \leq x_t \leq x_{t_0}\gamma^{t-t_0}.$$

Therefore, the feasible set is bounded, it is also clearly closed. It is feasible if: $x_{t_0}\gamma^{T-t_0} \geq \bar{x}$, and then, there exists an optimal solution, the cost function being continuous.

2. The second equation is obvious: the cost function is 0. Therefore, the problem is either feasible with value 0, or infeasible with value $-\infty$. Now, let $t_0 < T$ and x_0.

$$\begin{aligned}
V(t_0, x_0) &= \sup_{\substack{(u_t)_{t=t_0,\dots,T-1} \\ (x_t)_{t=t_0+1,\dots,T}}} \left\{ c(u_{t_0}) + \sum_{t=t_0+1}^{T-1} c(u_t) \,\middle|\, \begin{array}{l} u_t \geq 0, \\ x_{t+1} = \gamma x_t - u_t, \end{array} \forall t = t_0, ..., T-1 \right\} \\
&= \sup_{u_{t_0} \geq 0} \left\{ c(u_{t_0}) + \sup_{\substack{(u_t)_{t=t_0+1,\dots,T-1} \\ (x_t)_{t=t_0+1,\dots,T}}} \left\{ \sum_{t=t_0+1}^{T-1} c(u_t) \,\middle|\, \begin{array}{l} u_t \geq 0, \\ x_{t+1} = \gamma x_t - u_t, \\ x_{t_0+1} = \gamma x_{t_0} - u_{t_0}, \end{array} \forall t = t_0+1, ..., T-1 \right\} \right\} \\
&= \sup_{u_{t_0} \geq 0} \left\{ c(u_{t_0}) + V(t_0+1, \gamma x_{t_0} - u_{t_0}) \right\}.
\end{aligned}$$

3. Obviously, the optimal solution is $(u_t = 0, x_t = x_0\gamma^t)$ if and only if $x_0\gamma^T = \bar{x}$. Let us assume that $x_0\gamma^T > \bar{x}$. Let $(\bar{u}, \bar{x})$ be a feasible point. Let us assume that there exist two time indices t and $t+1$ so that $\bar{u}_t = 0$ and $\bar{u}_{t+1} > 0$. Let $\varepsilon > 0$ be very small. Consider the new control u with:

$$u_t = \varepsilon, \quad u_{t+1} = \bar{u}_{t+1} - \gamma\varepsilon, \quad u_i = \bar{u}_i \ \text{ otherwise},$$

so that the final state x_T is not modified. The increase of the cost is: $\varepsilon^\alpha - \bar{u}_{t+1}^{\alpha-1}\alpha\gamma\varepsilon + o(\varepsilon)$, and is thus strictly positive for ε small enough (since $\alpha < 1$), meaning that the original point was not optimal. The case $\bar{u}_t > 0$ and $\bar{u}_{t+1} = 0$ can be treated similarly. If $x_T > \bar{x}$, it means that u_{T-1} (and thus the cost), therefore the point was not optimal.

Let (u_1, x_1) and (u_2, x_2) be two different optimal solutions. Since x_1 (resp. x_2) directly depends on u_1 (resp. u_2), it follows that $u_1 \neq u_2$ (otherwise we would have $x_1 = x_2$). The point $((u_1+u_2)/2, (x_1+x_2)/2)$ is feasible, since the constraints are convex, and has a strictly greater cost that (u_1, x_1) and (u_2, x_2), by strict convexity of $u \mapsto \sum_{t=0}^{T-1} c(u_t)$, the parameter α being strictly smaller than 1. Contradiction.

4. We denote by $(p_0, ..., p_{T-1})$ the dual variable associated with the dynamics of the system, by $\lambda \geq 0$ the one associated with $x_T \geq \bar{x}$. The Lagrangian is:

$$L(x, u, p, \lambda) = \sum_{t=0}^{T-1} -u_t^\alpha + p_t(x_{t+1} - \gamma x_t + u_t) + \lambda(\bar{x} - x_T).$$

We directly consider the non-trivial case $x_0\gamma^T > \bar{x}$, and therefore forget the inactive constraints $u_t \geq 0$. The KKT conditions are: there exist $\lambda \geq 0$, $(p_t)_{t=0,\dots,T-1}$ such that

$$\begin{aligned}
&D_{u_t} L(x, u, p, \lambda) = -\alpha u_t^{\alpha-1} + p_t = 0, \ \forall t = 0, ..., T-1 \\
&D_{x_t} L(x, u, p, \lambda) = -\gamma p_t + p_{t+1} = 0, \ \forall t = 1, ..., T-1 \\
&D_{x_T} L(x, u, p, \lambda) = p_{T-1} - \lambda = 0.
\end{aligned}$$

The complementarity condition is satisfied, since we know that $x_T = \bar{x}$.

5. We deduce that:

$$p_t = \lambda\gamma^{T-t-1}, \quad u_t = \left(\frac{p_t}{\alpha}\right)^{-1/(1-\alpha)} = \underbrace{\left(\frac{\lambda\gamma^{T-1}}{\alpha}\right)^{-1/(1-\alpha)}}_{\beta} \Big(\underbrace{\gamma^{1/(1-\alpha)}}_{\xi}\Big)^t.$$

Using the hint, we deduce that: $x_t = x_0\gamma^t - \beta\frac{\gamma^t - \xi^t}{\gamma - \xi}$. We know that $x_T = \bar{x}$, and deduce that $\beta = (x_0\gamma^T - \bar{x})\frac{\gamma-\xi}{\gamma^T - \xi^T}$, and finally: $\lambda = \frac{\alpha}{\gamma^{T-1}}\beta^{-(1-\alpha)}$. We have found a KKT point, which is therefore the optimal solution, since the problem is convex ($\alpha < 1$).

1.7.8 Robust optimization

We first recall an important result of linear programming. Let two vectors $b \in \mathbb{R}^p$ and $c \in \mathbb{R}^m$ and let a matrix $A \in \mathbb{R}^{p\times m}$ be given. Consider the following two linear optimization problems:

$$\inf_{z\in\mathbb{R}^p} \langle b, z\rangle, \quad \text{subject to: } A^\top z = c,\ z \geq 0$$
$$\sup_{y\in\mathbb{R}^m} \langle c, y\rangle, \quad \text{subject to: } Ay \leq b.$$

If one of the two problems possesses a solution, then the other one also has a solution. Moreover, the two problems have the same value.

We consider now the following *robust* optimization problem:

$$\inf_{x\in\mathbb{R}^n} f(x), \quad \text{subject to: } x \in \mathcal{F},$$

where $f\colon \mathbb{R}^n \to \mathbb{R}$ is given and where $\mathcal{F}$ is described as follows:

$$\mathcal{F} = \big\{x \in \mathbb{R}^n \,|\, \langle Uy + u, x\rangle + \langle v, y\rangle \leq w,\ \forall y \in \mathbb{R}^m \text{ such that } Ay \leq b\big\}.$$

Here, $U \in \mathbb{R}^{n\times m}$, $u \in \mathbb{R}^n$, $v \in \mathbb{R}^m$, $w \in \mathbb{R}$, $A \in \mathbb{R}^{p\times m}$, and $b \in \mathbb{R}^p$ are given. This kind of situation typically arises when a constraint, depending on an uncertain parameter y, has to be fulfilled for all possible values of y. In the above problem, the a priori knowledge of y is modelized by the constraint $Ay \leq b$.

We assume that the set $\{y \in \mathbb{R}^m \,|\, Ay \leq b\}$ is non-empty and bounded (thus compact). Prove that the robust optimization is equivalent to the following one:

$$\inf_{((x,z)\in\mathbb{R}^{n+p}} f(x), \quad \text{subject to: } \begin{cases} \langle b, z\rangle + \langle u, x\rangle \leq w \\ A^\top z = U^\top x + v \\ z \geq 0. \end{cases}$$

Solution We first observe that

$$\langle Uy + u, x\rangle + \langle v, y\rangle = \langle U^\top x + v, y\rangle + \langle u, x\rangle.$$

Let $x \in \mathbb{R}^n$. The following equivalences hold true:

$$\begin{aligned} x \in \mathcal{F} &\Longleftrightarrow \left(\sup_{Ax\leq b} \langle U^\top x + v, y\rangle\right) \leq w - \langle u, x\rangle \\ &\Longleftrightarrow \left(\inf_{\substack{A^\top z = U^\top x + v \\ z\geq 0}}\right) \leq w - \langle u, x\rangle \\ &\Longleftrightarrow \exists z \text{ such that } \begin{cases} \langle b, z\rangle + \langle u, x\rangle \leq w \\ A^\top z = U^\top x + v \\ z \geq 0. \end{cases} \end{aligned}$$

Note that the maximization problem has a solution since the feasible set is compact and non-empty. This justifies the application of the duality theorem for linear problems and also ensures that the minimization problem has a solution. This justifies the last equivalence.

Exercise 1.1 (Snell-Descartes law)**.** We study in this exercise one of the fundamental laws of light propagation. We consider a two-dimensional space, composed of two media. The interface is described by the equation $y = f(x)$. The medium 1 ($y > f(x)$) has a refractive index $n_1 > 0$, the medium 2 ($y < f(x)$) has a refractive index $n_2 > 0$.

Let A_1 and A_2 be two points in the media 1 and 2 respectively. A ray of light passing through these two points has a rectiline trajectory in each of the two media, but the direction may change at the crossing point M at the interface. Denoting by $\vec{n}$ the normal vector of the interface at M (pointing upwards) and by θ_1 (resp. θ_2) the oriented angles between $\vec{n}$ and $\overrightarrow{MA_1}$ (resp. $-\vec{n}$ and $\overrightarrow{MA_2}$), the Snell-Descartes law states that:

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2).$$

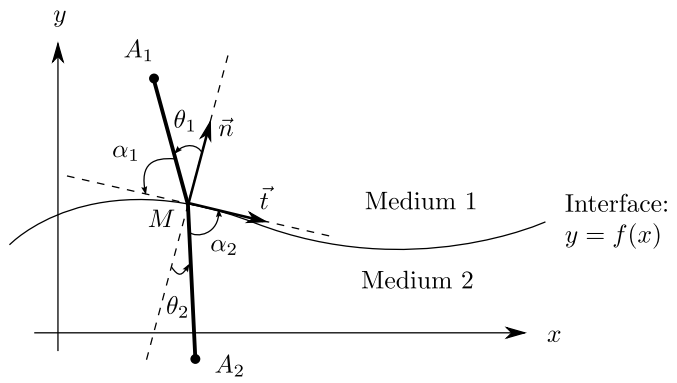


Figure 4: Refraction of light

The speed of light in the media 1 (resp. in the media 2) is c/n_1 (resp. c/n_2), where c is a constant. Fermat's principle states that the point M is such that the time needed by the light to go from A_1 to A_2 is minimized.

The tangential vector (pointing on the right) to the interface at point M is denoted by $\vec{t}$. We also set: $\alpha_1 = \pi/2 - \theta_1$ and $\alpha_2 = \pi/2 - \theta_2$.

1. Formulate the optimization problem corresponding to Fermat's principle, write the optimality conditions satisfied by M (we do not discuss in this exercise the existence of a minimizer).
2. Compute $\vec{t}$, $\cos(\alpha_1)$, and $\cos(\alpha_2)$.
3. Derive the Snell-Descartes law.

Correction of exercise 1.1. 1. Let (x_1, y_1), (x_2, y_2), $(x, f(x))$ be the coordinates of the points A_1, A_2, M. Let us denote by $f(x)$ the time needed by the ray of light to go from A_1 to A_2 through the point M:

$$f(x) = n_1\sqrt{(x-x_1)^2 + (f(x)-y_1)^2} + n_2\sqrt{(x-x_2)^2 + (f(x)-y_2)^2}.$$

The optimality condition reads:

$$f'(x) = \frac{n_1[(x-x_1) + f'(x)(f(x)-y_1)]}{\|A_1M\|} + \frac{n_2[(x-x_2) + f'(x)(f(x)-y_2)]}{\|A_2M\|} = 0.$$

2. We compute: $\vec{t} = \frac{(1, f'(x))}{\sqrt{1+f'(x)^2}}$ and:

$$\cos(\alpha_1) = \frac{\langle \overrightarrow{MA_1}, -\vec{t}\rangle}{\|MA_1\|} = \frac{(x-x_1) + (f(x)-y_1)f'(x)}{\|MA_1\| \cdot \sqrt{1+f'(x)^2}},$$
$$\cos(\alpha_2) = -\frac{(x-x_2) + (f(x)-y_2)f'(x)}{\|MA_2\| \cdot \sqrt{1+f'(x)^2}}.$$

3. Therefore,

$$f'(x) = \sqrt{1+f'(x)^2}[n_1\cos(\alpha_1) - n_2\cos(\alpha_2)] = \sqrt{1+f'(x)^2}[n_1\sin(\theta_1) - n_2\sin(\theta_2)].$$

The Snell-Descartes law follows from the optimality condition.

1.7.9 Inexact minimization

In this exercise, we assume that only an approximation $G(x)$ of the gradient can be computed. We prove that if the approximation if "good enough", then there exists a point $\tilde{x}$ in the neighborhood of a regular stationary point such that $G(\tilde{x}) = 0$.

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a twice continuously differentiable function. Let $\bar{x}$ be a regular stationary point, let $\alpha > 0$ be such that $\nabla^2 f(\bar{x}) \succeq \alpha I$. Let $G : \mathbb{R}^n \to \mathbb{R}^n$ be a continuously differentiable mapping. We assume that there exists $\delta \in (0, \alpha/3]$ such that for all $x \in \mathbb{R}^n$,

$$\|G(x) - \nabla f(x)\| \leq \delta \quad \text{and} \quad \|DG(x) - \nabla^2 f(x)\| \leq \delta.$$

1. By continuity of $x \mapsto \nabla^2 f(x)$, there exists $\varepsilon > 0$ such that for all $x \in \bar{B}(\bar{x}, \varepsilon)$, $\|\nabla^2 f(x) - \nabla f^2(\bar{x})\| \leq \alpha/3$. We set: $\lambda = \alpha/3$ and $\Lambda = \|\nabla f(\bar{x})\| + 2\alpha/3$. Prove that for all x_1 and $x_2 \in \bar{B}(\bar{x}, \varepsilon)$,

$$\langle x_2 - x_1, G(x_2) - G(x_1)\rangle \geq \lambda \|x_2 - x_1\|^2 \quad \text{and} \quad \|G(x_2) - G(x_1)\| \leq \Lambda \|x_2 - x_1\|.$$

2. Prove the existence of a value of $\gamma > 0$ such that the following mapping:

$$\phi : x \in \bar{B}(\bar{x}, \varepsilon) \mapsto x - \gamma G(x)$$

is Lipschitz-continuous with a modulus $\kappa \in [0, 1)$.

3. Prove that if $\delta > 0$ is small enough, then there exists $\tilde{x} \in \mathbb{R}^n$ such that $G(\tilde{x}) = 0$. Give an upper estimate of $\|\tilde{x} - \bar{x}\|$ and and upper estimate of $f(\tilde{x}) - f(\bar{x})$.

Solution

1. Let x_1 and $x_2 \in \bar{B}(\bar{x}, \varepsilon)$. It holds:

$$\begin{aligned}
\langle x_2 - x_1, G(x_2) - G(x_1)\rangle &= \left\langle x_2 - x_1, \int_0^1 DG(x_1 + \theta(x_2 - x_1))(x_2 - x_1)\,\mathrm{d}\theta \right\rangle \\
&\geq \langle x_2 - x_1, \nabla^2 f(\bar{x})(x_2 - x_1)\rangle - \underbrace{\sup_{x \in \bar{B}(\bar{x}, \varepsilon)} \|\nabla^2 f(x) - DG(x)\|}_{\leq \delta + \alpha/3 \leq 2\alpha/3} \cdot \|x_2 - x_1\|^2 \\
&\geq \frac{\alpha}{3} \|x_2 - x_1\|^2.
\end{aligned}$$

We also have:

$$\forall x \in \bar{B}(\bar{x}, \varepsilon),\ \|G(x)\| \leq \|\nabla f(x)\| + \|G(x) - \nabla f(x)\| \leq \|\nabla f(\bar{x})\| + \alpha/3 + \delta \leq \Lambda.$$

Therefore,

$$\|G(x_2) - G(x_1)\| \leq \int_0^1 \|DG(x_1 + \theta(x_2 - x_1))\| \cdot \|(x_2 - x_1)\|\,\mathrm{d}\theta \leq \Lambda \|x_2 - x_1\|.$$

2. Let $\gamma > 0$. Let x_1 and $x_2 \in \bar{B}(\bar{x}, \varepsilon)$, we have:

$$\begin{aligned}
\|(x_2 - \gamma G(x_2) - (x_1 - \gamma G(x_1))\|^2 &= \|x_2 - x_1\|^2 - 2\gamma \langle x_2 - x_1, G(x_2) - G(x_1)\rangle + \gamma^2 \|G(x_2) - G(x_1)\|^2 \\
&\leq (1 - 2\gamma\lambda + \gamma^2 \Lambda^2) \|x_2 - x_1\|^2.
\end{aligned}$$

Taking $\gamma = \frac{\lambda}{\Lambda^2}$, we obtain

$$\|\phi(x_2) - \phi(x_1)\| \leq \underbrace{\left(1 - \frac{\lambda^2}{\Lambda^2}\right)^{1/2}}_{=:\kappa \in [0,1)} \|x_2 - x_1\|.$$

3. Assume that $\delta' := \frac{\delta}{1-\kappa} \leq \varepsilon$. Then, for all $x \in \bar{B}(\bar{x}, \varepsilon)$,

$$\|\phi(x)\| \leq \|\phi(0)\| + \kappa \|x\| \leq \delta + \kappa \frac{\delta}{1 - \kappa} = \delta'.$$

Therefore, ϕ is a contraction on $\bar{B}(\bar{x}, \varepsilon)$ and $\bar{B}(\bar{x}, \varepsilon)$ is stable by ϕ. By the fix-point theorem, there exists $\tilde{x} \in \bar{B}(\bar{x}, \varepsilon)$ such that $\phi(\tilde{x}) = \tilde{x}$. Therefore, $G(\tilde{x}) = \tilde{x}$. Moreover,

$$f(\tilde{x}) - f(\bar{x}) \leq \int_0^1 (1 - \theta)(\tilde{x} - \bar{x}) \nabla^2 f(\bar{x} + \theta(\tilde{x} - \bar{x}))(\tilde{x} - \bar{x})\,\mathrm{d}\theta \leq \frac{1}{2} \Lambda \delta^2.$$

1.7.10 Proving local optimality

We consider a twice continuously differentiable function $f : \mathbb{R} \to \mathbb{R}$. We assume that the second-order derivative is Lipschitz continuous with modulus $M > 0$.

We assume that a point $x \in \mathbb{R}$ has been found (with a numerical procedure, for example) and that the following estimates are available:

(a) $|f'(x)| \leq \varepsilon$, for some $\varepsilon \geq 0$,

(b) $f''(x) \geq \alpha$, for some $\alpha > 0$.

Of course, if $\varepsilon = 0$, we know that x is locally optimal. We suppose that $\varepsilon > 0$. The goal of the exercise is to prove the existence of a local minimizer of f in the neighborhood of x if ε is small enough.

1. Using Taylor's theorem with remainder in integral form, prove that for all $y \in \mathbb{R}$,
$$f(y) \geq f(x) - \varepsilon|y - x| + \frac{\alpha}{2}|y - x|^2 - \frac{M}{6}|y - x|^3.$$

2. Consider the following condition (C): there exists $r > 0$ such that
$$-\varepsilon + \frac{\alpha}{2}r - \frac{M}{6}r^2 \geq 0.$$
Prove that there exists $\bar{\varepsilon}(\alpha, M) > 0$ such that condition (C) is satisfied if and only if $\varepsilon \leq \bar{\varepsilon}(\alpha, M)$. When this condition is satisfied, we set:
$$r(\varepsilon, \alpha, M) = \min \left\{ r \,|\, r \geq 0,\ -\varepsilon + \frac{\alpha}{2}r - \frac{M}{6}r^2 \geq 0 \right\}$$

3. Prove that if $\varepsilon \leq \bar{\varepsilon}(\alpha, M)$, then f has a local minimizer $\bar{x}$ such that $|\bar{x} - x| \leq r(\varepsilon, \alpha, M)$.

4. For fixed values of α and M, compute a first-order Taylor expansion of $r(\varepsilon, \alpha, M)$ with respect to ε.

Solution

1. Using Taylor's theorem with remainder in integral form, we obtain:
$$\begin{aligned} f(y) &= f(x) + f'(x)(y - x) + \int_0^1 f''(x + u(y - x))(y - x)^2(1 - u)\,\mathrm{d}u \\ &\geq f(x) - \varepsilon|y - x| + \int_0^1 (f''(x) - Mu|y - x|)|y - x|^2(1 - u)\,\mathrm{d}u. \end{aligned}$$
We compute:
$$\int_0^1 u(1 - u)\,\mathrm{d}u = \frac{1}{6}.$$
Combined together, we obtain the required estimate.

2. Let us set $h(r) = -\varepsilon + \frac{1}{2}\alpha r - \frac{1}{6}Mr^2$. One can easily check that the maximum of h is reached for $r = \frac{3\alpha}{2M}$ and take the value: $\frac{3\alpha^2}{8M} - \varepsilon$. This value is strictly positive if and only if:
$$\varepsilon < \bar{\varepsilon}(\alpha, M) := \frac{3\alpha^2}{8M}.$$
Moreover, $r(\varepsilon, \alpha, M)$ is in this case the smallest solution to the equation $h(r) = 0$ (which is always positive), given by:
$$r(\varepsilon, \alpha, M) = \frac{3}{M}\Big(\frac{1}{2}\alpha - \sqrt{\frac{1}{4}\alpha^2 - \frac{2M\varepsilon}{3}}\Big).$$

3. It holds:
$$r(\varepsilon, \alpha, M) = \frac{3\alpha}{2M}\Big(1 - \sqrt{1 - \frac{8M\varepsilon}{3\alpha^2}}\Big) = \frac{3\alpha}{2M}\frac{8M\varepsilon}{2 \cdot 3\alpha^2} + o(\varepsilon) = \frac{2\varepsilon}{\alpha} + o(\varepsilon).$$

4. Let us assume that $\varepsilon < \bar{\varepsilon}(\alpha, M)$, let us set $\bar{r} = r(\varepsilon, \alpha, M)$. Let $r > \bar{r}$ be such that $h(r) > 0$. Then, for all y such that $\bar{r} \geq |y - x| < r$,

$$f(y) > f(x) + |y - x|h(|y - x|) > f(x).$$

Since f is continuous, it reaches a minimum in the interval $[x - r, x + r]$, but belongs necessarily to $[x - \bar{r}, x + \bar{r}]$. This is therefore a local minimum (for the whole function).

1.7.11 Ekeland's variational principle

Let D be a closed and non-empty subset of $\mathbb{R}^n$, let $f : D \to \mathbb{R}$ be a continuous function on D. We assume that f is lower-bounded, that is to say: $\inf_{x \in D} f(x) > -\infty$. Note that f does not necessarily have a global minimizer. Let $\varepsilon > 0$, let $u \in D$ be an ε-optimal point, that is to say,

$$f(u) \leq \Big(\inf_{x \in D} f(x) \Big) + \varepsilon.$$

Note that the existence of u is always ensured. Let $\lambda > 0$, denoting by $\| \cdot \|$ the Euclidean norm, we define:

$$g : x \in \mathbb{R}^n \mapsto g(x) = f(x) + \frac{\varepsilon}{\lambda} \|x - u\|.$$

1. Ekeland's variational principle states that there exists $v \in D$ such that:

   (a) $f(v) \leq f(u)$

   (b) $\|v - u\| \leq \lambda$

   (c) $\forall x \in D,\ f(v) \leq f(x) + \frac{\varepsilon}{\lambda} \|x - v\|$.

   Prove that g has a global minimizer. Let v be a global minimizer of g, prove that v satisfies Ekeland's principle.

2. We assume now that v lies in the inner set of D and that f is differentiable at v. Deduce from inequality (c) that:

$$\|\nabla f(v)\| \leq \frac{\varepsilon}{\lambda}.$$

Solution

1. The function g has a minimizer since it is continuous, defined on a non-empty and closed domain, and since it is coercive. We have:

$$f(v) \leq g(v) \leq g(u) = f(u).$$

   Moreover,

$$f(v) + \frac{\varepsilon}{\lambda} \|v - u\| \leq f(u) \leq \inf_D f + \varepsilon \leq f(v) + \varepsilon.$$

   Thus, $\|v - u\| \leq \lambda$. Finally, for all $x \in D$,

$$f(v) \leq f(x) + \frac{\varepsilon}{\lambda} \big(\|x - u\| - \|v - u\| \big) \leq f(x) + \frac{\varepsilon}{\lambda} \|x - v\|.$$

2. For all $h \in \mathbb{R}^n$, for all $\alpha > 0$, we have:

$$\frac{f(v + \alpha h) - f(v)}{\alpha} \geq -\frac{\varepsilon}{\lambda} \|h\|.$$

   Passing to the limit when $\alpha \downarrow 0$, we obtain: $\nabla f(v)^\top h \geq -\frac{\varepsilon}{\lambda} \|h\|$. The result follows, taking $h = -\nabla f(v)$.

1.7.12 Applications of Ekeland's principle (1)

We study in this exercise two applications of Ekeland's principle.

1. Let D be a closed and non-empty subset of $\mathbb{R}^n$, let $T : D \to D$ be a μ-Lipschitz function with $\mu \in [0, 1)$. We define $f : x \in D \mapsto \|T(x) - x\|$. Using Ekeland's principle with $\lambda = 1$ and $\varepsilon < 1 - \mu$, prove that T has a fixed point.

2. Let $f : \mathbb{R}^n \to \mathbb{R}$ be a continuously differentiable function. We say that the Palais-Smale condition is satisfied if any sequence $(x_k)_{k\in\mathbb{N}}$ satisfying the following assumptions:

   (a) $(f(x_k))_{k\in\mathbb{N}}$ is bounded

   (b) $Df(x_k) \underset{k\to\infty}{\longrightarrow} 0$,

   has an accumulation point. Prove that if f is lower-bounded and satisfies the Palais-Smale condition, then f has a global minimizer. Use Ekeland's principle.

Solution

1. Let $v \in D$ satisfying the third condition of Ekeland's principle, for the function f (which is continuous and lower-bounded) and with $\lambda = 1$ and $\varepsilon < 1 - \mu$. Then, taking $x = T(v)$, we obtain:

$$\|T(v) - v\| \leq \|T(T(v)) - T(v)\| + \varepsilon\|T(v) - v\| \leq (\mu + \varepsilon)\|T(v) - v\|.$$

   Since $\mu + \varepsilon < 1$, we deduce that $\|T(v) - v\| = 0$, that is to say, $T(v) = v$.

2. For all n, Ekeland's principle (with $\varepsilon = \frac{1}{n}$ and $\lambda = 1$) ensures the existence of v_n such that:

$$f(v_n) \leq \inf f + \frac{1/n}{}\|Df(v_n)\| \leq \frac{1}{n}.$$

   By the Palais-Smale condition, the sequence $(v_n)_{n\in\mathbb{N}}$ has a accumulation point. By continuity of f, this point is a minimizer.

1.7.13 Applications of Ekeland's principle (2)

In this exercise, we prove the KKT conditions with Ekeland's principle. Let $f : \mathbb{R}^n \to \mathbb{R}$, $g : \mathbb{R}^n \to \mathbb{R}^m$, and $e : \mathbb{R}^n \to \mathbb{R}^p$ be three continuously differentiable functions, let $\bar{x}$ be a local solution to the following problem:

$$\inf_{x\in\mathbb{R}^n} f(x), \quad \text{subject to: } g(x) \leq 0,\ e(x) = 0.$$

Let $\bar{\varepsilon} > 0$ be such that $\bar{x}$ minimizes f on $\mathcal{F} \cap \bar{B}(\bar{x}, \bar{\varepsilon})$. We assume that $g(\bar{x}) = 0$. For a vector $z \in \mathbb{R}^q$, we define $z^+ \in \mathbb{R}^q$ as: $z_i^+ = \max(z_i, 0)$, for all $i = 1, ..., q$.

1. Let $h : \mathbb{R}^n \to \mathbb{R}^q$ be a differentiable function. Prove that the function $x \in \mathbb{R}^n \mapsto H(x) = \|h^+(x)\|^2$ is differentiable and that its derivative is given by $DH(x) = 2h^+(x)^\top Dh(x)$.

2. For all $k \in \mathbb{N}$, denoting by $\|\cdot\|$ the Euclidean norm, we define:

$$h_k : x \in \mathbb{R}^n \mapsto \Big(\big(f(x) - f(\bar{x}) + \bar{\varepsilon}2^{-k}\big)_+^2 + \|e(x)\|^2 + \|g_+(x)\|^2\Big)^{1/2}.$$

   Construct 4 sequences $(x_k)_{k\in\mathbb{N}} \subset \mathbb{R}^n$, $(\alpha_k)_{k\in\mathbb{N}} \subset \mathbb{R}$, $(\mu_k)_{k\in\mathbb{N}} \subset \mathbb{R}^p$, and $(\lambda_k)_{k\in\mathbb{N}} \subset \mathbb{R}^m$ such that

$$\forall k \in \mathbb{N}, \quad \alpha_k \geq 0, \quad \lambda_k \geq 0, \quad \text{and} \quad \alpha_k^2 + \|\mu_k\|^2 + \|\lambda_k\|^2 = 1$$

   and such that

$$\alpha_k Df(x_k) + \mu_k^\top De(x_k) + \lambda_k^\top Dg(x_k) \underset{k\to\infty}{\longrightarrow} 0 \quad \text{and} \quad x_k \underset{k\to\infty}{\longrightarrow} \bar{x}.$$

   To this purpose, observe that $\bar{x}$ is $h_k(\bar{x})$-optimal and apply Ekeland's principle. Pay attention to the fact that the square-root function is not differentiable at 0.

3. Prove the existence of $\alpha \in \mathbb{R}$, $\mu \in \mathbb{R}^p$ and $\lambda \in \mathbb{R}^m$ such that:

$$\alpha Df(\bar{x}) + \mu^\top De(\bar{x}) + \lambda^\top Dg(\bar{x}) = 0, \quad \alpha \geq 0, \quad \lambda \geq 0, \quad \alpha^2 + \|\mu\|^2 + \|\lambda\|^2 = 1.$$

   Assume that the LICQ is satisfied. Prove the KKT conditions (without using the result of the lecture).

Solution

1. One can easily check that $x \in \mathbb{R} \mapsto (x^+)^2$ is continuously differentiable with derivative $2x^+$. Therefore, applying the chain rule,

$$D_{x_i} H(x) = D_{x_i} \sum_{j=1}^{n} (h_j^+(x))^2 = 2 \sum_{j=1}^{n} h_j^+(x) D_{x_i} h_j(x) = \left(h^+(x)^\top D_x h(x)\right)_i.$$

It follows that $D_x H(x) = h^+(x)^\top D_x h(x)$.

2. We first notice that h_k is non-negative. Therefore, $\bar{x}$ is $h(\bar{x})$-optimal. We have: $h_k(\bar{x}) = \bar{\varepsilon} 2^{-k} := \varepsilon_k$. Let $\lambda_k = \bar{\varepsilon} 2^{-k/2}$. By Ekeland's principle, there exists x_k such that:

$$\|x_k - \bar{x}\| \leq \lambda_k \leq \bar{\varepsilon}, \quad h_k(x_k) \leq h_k(\bar{x}),$$

and such that the 3rd relation (...) holds. If x_k is not feasible, than $\|e(x)\|^2 > 0$ or $\|g^+(x)\|^2 > 0$. If x_k is feasible, then, since $\|x_k - \bar{x}\| \leq \bar{\varepsilon}$, $f(x_k) \geq f(\bar{x})$. Therefore, $(f(x_k) - f(\bar{x}) + \bar{\varepsilon} 2^{-k})_+ > 0$ and consequently, $h_k(x_k) > 0$. Therefore, h_k is differentiable at x_k, with $\|Dh_k(x_k)\| \leq \varepsilon_k / \lambda_k = 2^{-k/2}$. Moreover,

$$Dh_k(x_k) = \alpha_k Df(x_k) + \mu_k^\top De(x_k) + \lambda_k^\top Dg(x_k),$$

where:

$$\alpha_k = \frac{\left(f(x_k) - f(\bar{x}) + \varepsilon_k\right)_+}{h(x_k)}, \quad \mu_k = \frac{e(x_k)}{h(x_k)}, \quad \lambda_k = \frac{g^+(x_k)}{h(x_k)}.$$

One can easily check that $\alpha_k^2 + \|\mu_k\|^2 + \|\lambda_k\|^2 = 1$. Therefore, the sequence $(\alpha_k, \mu_k, \lambda_k)$ has an accumulation point (α, μ, λ), which is such that $\alpha \geq 0$, $\lambda \geq 0$ and such that:

$$\alpha^2 + \|\mu\|^2 + \|\lambda\|^2 = 1 \quad \alpha Df(\bar{x}) + \mu^\top De(\bar{x}) + \lambda^\top Dg(\bar{x}).$$

If $\alpha = 0$, then:

$$\left(\nabla e(\bar{x}) \ \nabla g(\bar{x})\right) \begin{pmatrix} \mu \\ \lambda \end{pmatrix} = 0.$$

The LICQ ensures that the matrix $\left(\nabla e(\bar{x}) \ \nabla g(\bar{x})\right)$ has full column-rank, therefore, $\mu = 0$ and $\lambda = 0$. Contradiction. Therefore, $\alpha > 0$. The KKT conditions are satisfied that the multipliers μ/α and λ/α.

2 Analysis of numerical methods

2.1 Gradient methods

2.1.1 Armijo and Wolfe's rules

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a differentiable function. We assume that its gradient is Lipschitz-continuous with modulus $L > 0$, that is to say:

$$\forall x_1 \in \mathbb{R}^n, \ \forall x_2 \in \mathbb{R}^n, \ \|\nabla f(x_2) - \nabla f(x_1)\| \leq L \|x_2 - x_1\|.$$

Let $x \in \mathbb{R}^n$ and let p be a descent direction. We set $\phi : \alpha \geq 0 \mapsto f(x + \alpha p)$ and consider the algorithm seen in the lecture for finding a step length satisfying Armijo and Wolfe's rules (see Algorithm 1 below). We assume that there exists $M \in \mathbb{R}$ such that for all $\alpha \geq 0$, $\phi(\alpha) \geq M$ and that $\phi'(0) < 0$.

In this exercise, we look for estimates on the number of iterations k_0 and k_1 needed for the two while-loops.

1. Prove that $\alpha \mapsto \phi'(\alpha)$ is Lipschitz-continuous. Let $L' > 0$ be such that for all α_1 and $\alpha_2 \geq 0$,

$$|\phi'(\alpha_2) - \phi'(\alpha_1)| \leq L' |\alpha_2 - \alpha_1|.$$

2. Let $\bar{\alpha} = \frac{\phi(0) - M}{-c_1 \phi'(0)}$. Prove that if $\alpha > \bar{\alpha}$, then Armijo's rule is not fulfilled.

3. Let k_0 be such $\alpha_0^{\max} = \hat{\alpha}_{k_0} = \beta^{k_0}$. Prove that

$$k_0 \leq \max\Big(0, \frac{\ln(\bar{\alpha})}{\ln(\beta)} + 1\Big).$$

4. Let k_1 be such that $\alpha = \alpha_{k_1}$. Prove that

$$k_1 \leq \ln\Big(\frac{L'\alpha_0^{\max}}{(c_2 - c_1)(-\phi'(0))}\Big)/\ln(2).$$

Hint: use the sequence $(\tilde{\alpha}_k)_{k=0,1,\ldots}$ introduced in the proof of termination of the algorithm.

Algorithm 2: Finding a step length satisfying Wolfe's rules

```
Input: β > 1, A : α ≥ 0 ↦ {true; false}, W : α ≥ 0 ↦ {true; false}.
Set: α̂_0 = 1 and k = 0;
while A(α̂_k) do
    Set: α̂_{k+1} = β α̂_k;
    Set: k = k + 1;
end
Set: α_0^min = 0, α_0^max = α̂_k, α_0 = (α_0^min + α_0^max)/2;
Set: k = 0;
while not [A(α_k) and W(α_k)] do
    if not (A(α_k)) then
        Set: α_{k+1}^min = α_k^min;
        Set: α_{k+1}^max = α_k;
    else
        Set: α_{k+1}^min = α_k;
        Set: α_{k+1}^max = α_k^max;
    end
    Set: α_{k+1} = (α_{k+1}^min + α_{k+1}^max)/2;
    Set: k = k + 1;
end
Output: α_k.
```

Solution

1. For all α_1 and $\alpha_2 \geq 0$,

$$|\phi'(\alpha_2) - \phi'(\alpha_1)| \leq \|p\| \cdot \|\nabla f(x + \alpha_2 p) - \nabla f(x + \alpha_1 p)\| \leq L\|p\|^2 \cdot |\alpha_2 - \alpha_1|.$$

2. For all $\alpha > \bar{\alpha}$, we have:

$$\phi(\alpha) - \phi(0) - c\phi'(0)\alpha > M - \phi(0) - c\phi'(0)\bar{\alpha} = 0.$$

Therefore, for $\alpha > \bar{\alpha}$, Armijo's rule is not satisfied.

3. If $k_0 = 0$, then the result inequality is trivially satisfied. Otherwise, before the last iteration, Armijo's rule is still satisfied, which means that $\beta^{k_0 - 1} \leq \bar{\alpha}$. It follows that:

$$k_0 \leq \ln(\bar{\alpha})/\ln(\beta) + 1.$$

4. At the last iteration k_1, $\alpha_{k_1}^{\min}$ does not satisfy Wolfe's rule, thus $\phi'(\alpha_{k_1}^{\min}) \leq c_2\phi'(0)$, whereas $\phi'(\tilde{\alpha}_{k_1}) = c_1\phi'(0)$. Therefore,

$$-(c_2 - c_1)\phi'(0) \leq \phi'(\tilde{\alpha}_{k_1}) - \phi'(\alpha_{k_1}^{\min}) \leq L'(\tilde{\alpha}_{k_1} - \alpha_{k_1}^{\min}) = 2^{-k_1}\alpha_0^{\max}.$$

Therefore,

$$k_1 \leq \ln\Big(\frac{L'\alpha_0^{\max}}{(c_2 - c_1)(-\phi'(0))}\Big)/\ln(2).$$

2.1.2 Strong Wolfe's rules

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a continuously differentiable function to be minimized. Let $x \in \mathbb{R}^n$, let $p \in \mathbb{R}^n$ be a descent direction. We define $\phi : \alpha \geq 0 \mapsto f(x + \alpha p)$. The function ϕ is continuously differentiable. Note that $\phi'(0) < 0$. Let $0 < c_1 < c_2 < 1$. In this exercise, consider the three following rules:

$$\begin{aligned} A(\alpha) \text{ is satisfied } &\iff \phi(\alpha) \leq \phi(0) + c_1 \alpha \phi'(0) \\ W^-(\alpha) \text{ is satisfied } &\iff \phi'(\alpha) \geq c_2 \phi'(0) \\ W^+(\alpha) \text{ is satisfied } &\iff \phi'(\alpha) \leq -c_2 \phi'(0). \end{aligned}$$

Consider the Algorithm 1 below for finding a steplength satisfying the three rules.

Algorithm 3: Finding a step length satisfying strong Wolfe's rules

```
Input: β > 1, A : α ≥ 0 ↦ {true; false}, α ≥ 0 ↦ W⁻ ↦ {true; false}, α ≥ 0 ↦ W⁺ ↦ {true; false}
Set: α̂_0 = 1 and k = 0;
while A(α̂_k) do
    Set: α̂_{k+1} = β α̂_k;
    Set: k = k + 1;
end
Set: α_0^min = 0, α_0^max = α̂_k, α_0 = (α_0^min + α_0^max)/2;
Set: k = 0;
while not [A(α_k) and W⁻(α_k) and W⁺(α_k)] do
    if not (A(α_k)) or not (W⁺(α_k)) then
        Set: α_{k+1}^min = α_k^min;
        Set: α_{k+1}^max = α_k;
    else
        Set: α_{k+1}^min = α_k;
        Set: α_{k+1}^max = α_k^max;
    end
    Set: α_{k+1} = (α_{k+1}^min + α_{k+1}^max)/2;
    Set: k = k + 1;
end
Output: α_k.
```

1. On a graph, show the set of points satisfying each of the three rules. Explain the behaviour of the second while loop of the algorithm.
2. Proof the termination of the second while loop. Look first for properties satisfied by $\alpha_k^{\min}$ and $\alpha_k^{\max}$ at each iteration of the loop and adapt the proof given in the lecture.

Solution By induction, one can prove that for all k, [not($A(\alpha_k^{\max})$) or not($W^+(\alpha_k^{\max})$)] and $A(\alpha_k^{\min})$ and not($W^-(\alpha_k^{\min})$).

Proof by contradiction: assume that the second loop does not terminate. There exists a subsequence $(\alpha_{k_\ell}^{\max})_{\ell \in \mathbb{N}}$ such that for all ℓ, for all ℓ, not($A(\alpha_{k_\ell}^{\max})$) or for all ℓ, not($W^+(\alpha_{k_\ell}^{\max})$).

In the first situation, one can argue as in the lecture and obtain a contradition. In the second situation, we obtain to the limit:

$$\phi'(\bar{\alpha}) \leq c_2 \phi'(0) < -c_2 \phi'(0) = \phi'(\bar{\alpha}).$$

Contradiction also.

2.1.3 Goldstein's rule

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a differentiable function. We assume that its gradient is Lipschitz-continuous with modulus $L > 0$. Let $x \in \mathbb{R}^n$, let $p \in \mathbb{R}^n$ be a descent direction, let $c \in (0, 1/2)$. We set $\phi(\alpha) = f(x + \alpha p)$ for $\alpha \geq 0$. We consider the following rules:

$$\begin{aligned} A(\alpha) &\iff \phi(\alpha) \leq \phi(0) + c\phi'(0)\alpha \\ B(\alpha) &\iff \phi(\alpha) > \phi(0) + (1 - c)\phi'(0)\alpha. \end{aligned}$$

1. Show the existence of $\bar{\alpha} > 0$ such that for all $\alpha \in [0, \bar{\alpha}]$, $B(\alpha)$ does not hold.
2. Prove that the decay condition is satisfied for all α satisfying both rules.
3. Propose an algorithm for finding a steplength satisfying both rules and justify that it terminates.

Solution

1. We know that ϕ' is Lipschitz continuous with modulus $\tilde{L} = L\|p\|^2$. We set
$$\tilde{\alpha} = \frac{-2c\phi'(0)}{\tilde{L}} = \frac{2c\langle \nabla f(x), p\rangle}{L\|p\|^2}.$$
We have seen in the lecture that for all $\alpha \in [0, \bar{\alpha}]$,
$$\phi(\alpha) \leq \phi(0) + (1-c)\phi'(0)\alpha,$$
thus $B(\alpha)$ does not hold.
2. Let α satisfy both rules. Then, necessarily, $\alpha \geq \bar{\alpha}$. By Armijo's rule, we have
$$\phi(\alpha) \leq \phi(0) + c\phi'(0)\alpha \leq \phi(0) + c\phi'(0)\bar{\alpha} = \phi(0) - \frac{2c^2\langle \nabla f(x), p\rangle^2}{\|p\|^2}.$$
3. Consider the algorithm below. By induction, $A(\alpha_k^{\min})$, $not(B(\alpha_k^{\min}))$, and $not(A(\alpha_k^{\max}))$. Necessarily, $\alpha_k^{\max} \geq \bar{\alpha}$ (since $not(B(\alpha)) \Rightarrow A(\alpha)$), thus the limit point $\bar{\alpha}$ is such that $\bar{\alpha} > 0$ and such that
$$\phi(\bar{\alpha}) \geq \phi(0) + c\phi'(0)\bar{\alpha}, \quad \text{and} \quad \phi(\bar{\alpha}) \leq \phi(0) + (1-c)\phi'(0)\bar{\alpha}.$$
Thus, $c\phi'(0)\bar{\alpha} \leq (1-c)\phi'(0)\bar{\alpha}$, thus $c \geq (1-c)$ and finally $c \geq 1/2$. Contradiction.

Algorithm 4: Finding a step length satisfying Goldstein's rules

```
Input: β > 1, A : α ≥ 0 ↦ {true; false}, B : α ≥ 0 ↦ {true; false} and α_0^max such that A(α_0^max) is
  not fulfilled.
Set: α_0^min = 0, α_0 = (α_0^min + α_0^max)/2;
Set: k = 0;
while not [A(α_k) and B(α_k)] do
    if not (A(α_k)) then
        Set: α_{k+1}^min = α_k^min;
        Set: α_{k+1}^max = α_k;
    else
        Set: α_{k+1}^min = α_k;
        Set: α_{k+1}^max = α_k^max;
    end
    Set: α_{k+1} = (α_{k+1}^min + α_{k+1}^max)/2;
    Set: k = k + 1;
end
Set: α = α_k;
Output: α.
```

2.1.4 A simple gradient-descent method

In this exercise, $\|\cdot\|$ denotes the Euclidean norm. We recall first two results, which will be used for this exercise.

- Let D be a closed subset of $\mathbb{R}^n$. Let $F : D \to D$ be a contraction, that is to say, a mapping such that there exists $\lambda \in [0, 1)$ such that for all x and $y \in D$,
$$\|F(y) - F(x)\| \leq \lambda\|y - x\|.$$
Then, there exists a unique $\bar{x} \in D$ such that $F(\bar{x}) = \bar{x}$. Moreover, all sequences $(x_k)_{k\in\mathbb{N}}$ satisfying $x_{k+1} = F(x_k)$ (for all $k \in \mathbb{N}$) converge to $\bar{x}$.

- Let S_n denote the set of symmetric matrices of size n. For all $M \in S_n$, denote by $\sigma^-(M)$ its smallest eigenvalue and by $\sigma^+(M)$ its largest eigenvalue. The operator norm of a symmetric matrix M (associated with the Euclidean norm) is given by:

$$\|M\| = \max\big(|\sigma^-(M)|, |\sigma^+(M)|\big).$$

We consider now a twice continuously differentiable function f. We assume that there exists $\bar{x}$ such that $Df(\bar{x}) = 0$ and such that $D^2 f(\bar{x})$ is positive definite. We aim at proving the following result: there exist $\delta > 0$ and $\rho > 0$ such that for all $x \in \mathbb{R}^n$ with $\|x - \bar{x}\| \leq \delta$, the sequence $(x_k)_{k\in\mathbb{N}}$ defined below converges to $\bar{x}$:

$$x_0 = x, \quad x_{k+1} = x_k - \rho\nabla f(x_k), \ \forall x \in \mathbb{N}.$$

1. It is possible to show that there exist $\delta > 0$, $a > 0$ and $A \geq a$ such that for all $x \in \mathbb{R}^n$ with $\|x - \bar{x}\| \leq \delta$,

$$a \leq \sigma^-(M) \leq \sigma^+(M) \leq A.$$

   Prove the existence of $\rho > 0$ and $\lambda \in [0, 1)$ such that for all $x \in \mathbb{R}^n$ with $\|x - \bar{x}\| \leq \delta$, we have $\|\mathrm{Id} - \rho D^2 f(x)\| \leq \lambda$.

2. Prove the main result of the exercise, by applying the fixed-point theorem.

Solution

1. For all $x \in \mathbb{R}^n$ with $\|x - \bar{x}\| \leq \delta$, we have

$$\begin{aligned}\sigma^+(I - \rho\nabla^2 f(x)) &= 1 - \rho\sigma^-(\nabla^2 f(x)) \leq 1 - \rho a\\ \sigma^-(I - \rho\nabla^2 f(x)) &= 1 - \rho\sigma^+(\nabla^2 f(x)) \geq 1 - \rho A.\end{aligned}$$

   By choosing $\rho = 1/A$, we obtain:

$$\begin{aligned}\sigma^+(I - \rho\nabla^2 f(x)) &\leq 1 - a/A\\ \sigma^-(I - \rho\nabla^2 f(x)) &\geq 0.\end{aligned}$$

   Therefore, $\|I - \rho\nabla^2 f(x)\| \leq 1 - a/A =: \lambda$.

2. We define now D as the ball of center $\bar{x}$ and radius δ and

$$F\colon x \in D \mapsto x - \rho\nabla f(x).$$

   We have: $F(\bar{x}) = \bar{x}$. For all x_1 and $x_2 \in D$, we have:

$$\nabla f(x_2) - \nabla f(x_1) = \int_0^1 \nabla^2 f(x_1 + \theta(x_2 - x_1))(x_2 - x_1)\,\mathrm{d}\theta.$$

   It follows that:

$$\begin{aligned}\|F(x_2) - F(x_1)\| &= \|\int_0^1 (\mathrm{Id} - \rho\nabla^2 f(x_1 + \theta(x_2 - x_1))(x_2 - x_1)\,\mathrm{d}\theta\|\\ &= \int_0^1 \|\mathrm{Id} - \rho\nabla^2 f(x_1 + \theta(x_2 - x_1))(x_2 - x_1)\| \cdot \|x_2 - x_1\|\,\mathrm{d}\theta\\ &\leq \lambda\|x_2 - x_1\|.\end{aligned}$$

   We deduce that for all $x \in D$,

$$\|F(x) - \bar{x}\| = \|F(x) - F(\bar{x})\| \leq \lambda\|x - \bar{x}\| \leq \lambda\delta \leq \delta.$$

   Therefore, $F(x) \in D$. The result follows with the fixed point theorem.

2.1.5 Decay condition

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a continuously differentiable function. We assume that its gradient is Lipschitz-continuous with modulus $L > 0$. Let $x \in \mathbb{R}^n$, let $p \in \mathbb{R}^n$ be a descent direction, and let $\delta > 0$. We set

$$\alpha = -\delta \frac{\langle \nabla f(x), p \rangle}{\|p\|^2} > 0.$$

1. Prove the following relation: for all x and $y \in \mathbb{R}^n$,

$$f(y) \leq f(x) + \langle \nabla f(x), y - x) \rangle + \frac{1}{2} L \|y - x\|^2.$$

2. Prove that for an appropriate choice of δ, there exists $\theta > 0$ such that the decay condition holds:

$$f(x) - f(x + \alpha p) \geq \theta \Big(\frac{\langle \nabla f(x), p \rangle}{\|p\|} \Big)^2.$$

Solution

1. For all x and $y \in \mathbb{R}^n$, it holds:

$$\begin{aligned}
f(y) &= f(x) + \int_0^1 \langle \nabla f(x + \theta(y - x)), y - x \rangle \, \mathrm{d}\theta \\
&= f(x) + \langle \nabla f(x), y - x \rangle + \int_0^1 \langle \nabla f(x + \theta(y - x)) - \nabla f(x), y - x \rangle \, \mathrm{d}\theta \\
&\leq f(x) + \langle \nabla f(x), y - x \rangle + \int_0^1 \|\nabla f(x + \theta(y - x)) - \nabla f(x)\| \cdot \|y - x\| \, \mathrm{d}\theta \\
&\leq f(x) + \langle \nabla f(x), y - x \rangle + L \|y - x\|^2 \Big(\int_0^1 \theta \, \mathrm{d}\theta \Big) \\
&= f(x) + \langle \nabla f(x), y - x \rangle + \frac{1}{2} L \|y - x\|^2.
\end{aligned}$$

2. We have:

$$\begin{aligned}
f(x + \alpha p) - f(x) &\leq \alpha \langle \nabla f(x), p \rangle + \frac{1}{2} L \alpha^2 \|p\|^2 \\
&= -\delta \Big(1 - \frac{1}{2} L \delta \Big) \Big(\frac{\langle \nabla f(x), p \rangle}{\|p\|} \Big)^2.
\end{aligned}$$

   For $\delta = \frac{1}{L}$, the result holds with $\theta = \frac{1}{2L}$.

2.1.6 Decay condition (2)

We consider a uniformly convex function $f : \mathbb{R}^n \to \mathbb{R}$, with associated parameter $m > 0$ (see Exercise 2.3). We also assume that ∇f is Lipschitz with modulus $L > 0$. We consider a steepest gradient descent method of the form:

$$\forall k \in \mathbb{N}, \quad x_{k+1} = x_k - \alpha_k \nabla f(x_k),$$

where the step length α_k is computed in such a way that for all k,

$$-c_1 \|\nabla f(x_k)\|^2 \leq \langle \nabla f(x_{k+1}), \nabla f(x_k) \rangle \leq c_2 \|\nabla f(x_k)\|^2,$$

for some constants c_1 and $c_2 > 0$.

1. Prove that $\alpha_k \leq (1 + c_1)/m$.
2. We set $\phi_k(\alpha) = f(x_k - \alpha \nabla f(x_k))$. Prove that ϕ_k' is Lipschitz-continuous, compute $\phi_k'(0)$ and $\phi_k'(\alpha_k)$, and deduce that $\alpha_k \geq (1 - c_2)/L$.
3. Prove that there exist values of c_1 and $c_2 > 0$ such that there exists $\theta > 0$ such that for all k, the decay condition is satisfied:

$$f(x_k) - f(x_{k+1}) \geq \theta \|\nabla f(x_k)\|^2.$$

4. Prove that if the decay condition is satisfied, then the sequence $(x_k)_{k \in \mathbb{N}}$ converges to the unique minimizer of f.

Solution

1. By uniform convexity, we have:

$$\langle \nabla f(x_{k+1}) - \nabla f(x_k), x_{k+1} - x_k \rangle \geq m\|x_{k+1} - x_k\|^2. \tag{22}$$

We have:

$$m\|x_{k+1} - x_k\|^2 = m\alpha_k^2 \|\nabla f(x_k)\|^2. \tag{23}$$

We also have

$$\begin{aligned}\langle \nabla f(x_{k+1}) - \nabla f(x_k), x_{k+1} - x_k \rangle = & -\alpha_k \langle \nabla f(x_{k+1}) - \nabla f(x_k), \nabla f(x_k) \rangle \\ &= \alpha_k \|\nabla f(x_k)\|^2 - \alpha_k \langle \nabla f(x_{k+1}), \nabla f(x_k) \rangle \\ &\leq \alpha_k (1 + c_1) \|\nabla f(x_k)\|^2. \end{aligned} \tag{24}$$

Combining (22), (23), and (24) together, we obtain that $\alpha_k \leq (1 + c_1)/m$.

2. We have already proved that ϕ_k' is Lipschitz continuous with modulus $L\|\nabla f(x_k)\|^2$. Moreover, we have:

$$\phi_k'(0) = -\|\nabla f(x_k)\|^2 \quad \text{and} \quad \phi_k'(\alpha_k) = -\langle \nabla f(x_{k+1}), \nabla f(x_k) \rangle \geq -c_2 \|\nabla f(x_k)\|^2.$$

Therefore,

$$\phi_k'(\alpha_k) - \phi_k'(0) \geq (1 - c_2)\|\nabla f(x_k)\|^2.$$

Moreover,

$$\phi_k'(\alpha_k) - \phi_k'(0) \leq \alpha_k L \|\nabla f(x_k)\|^2.$$

Therefore, $\alpha_k \geq (1 - c_2)/L$.

3. We have:

$$\begin{aligned} f(x_k) - f(x_{k+1}) &\geq \langle \nabla f(x_{k+1}), x_k - x_{k+1} \rangle + \frac{m}{2}\|x_k - x_{k+1}\|^2 \\ &= \alpha_k \langle \nabla f(x_{k+1}), \nabla f(x_k) \rangle + \frac{m}{2}\alpha_k^2 \|\nabla f(x_k)\|^2 \\ &\geq \Big(-\alpha_k c_2 + \frac{m}{2}\alpha_k^2 \Big) \|\nabla f(x_k)\|^2 \\ &\geq \underbrace{\Big(\frac{m}{2}\Big(\frac{1 - c_2}{L}\Big)^2 - \Big(\frac{1 + c_1}{m}\Big) c_1 \Big)}_{=:\theta} \|\nabla f(x_k)\|^2. \end{aligned}$$

The constant θ defined above is positive if c_1 and c_2 are sufficiently small.

4. We know that any accumulation point of $(x_k)_{k\in\mathbb{N}}$ is a stationary point. Since f is uniformly convex, it has a unique stationary point $\bar{x}$, which is a global minimizer. Therefore, there is only one possible accumulation point for $(x_k)_{k\in\mathbb{N}}$, the point $\bar{x}$.

By the decay condition, the sequence $f(x_k)$ is decreasing. We know from (...) that f is coercive and that $f^{-1}((\infty, f(x_0)])$ is bounded. Therefore, the sequence $(x_k)_{k\in\mathbb{N}}$ is bounded. Assume that it does not converge to $\bar{x}$. Then, there exists $\varepsilon > 0$ and a subsequence $(x_{k_\ell})_{\ell\in\mathbb{N}}$ which is such that $\|x_{k_\ell} - \bar{x}\| \geq \varepsilon$. This subsequence is bounded, thus possesses an accumulation point $\tilde{x} \neq \bar{x}$, which is also an accumulation point of $(x_k)_{k\in\mathbb{N}}$, contradiction.

2.1.7 A descent method for constrained problems

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a continously differentiable function. We assume that there exists $L > 0$ such that for all x_1, $x_2 \in \mathbb{R}^n$, $\|\nabla f(x_2) - \nabla f(x_1)\| \leq L\|x_2 - x_1\|$. Let $\mathcal{F}$ be a compact, convex, and non-empty subset of $\mathbb{R}^n$. We consider the problem:

$$\inf_{x\in\mathbb{R}^n} f(x), \quad \text{subject to: } x \in \mathcal{F}. \tag{P}$$

We consider Algorithm 5 below for solving problem (P). In the whole exercise, we assume that the algorithm never stops, that is to say, we assume that for all $k \in \mathbb{N}$, $z_k \neq 0$.

Part 1. We first justify that the algorithm is well-defined.

1. Let $k \in \mathbb{N}$. Assume that $x_k \in \mathcal{F}$.
   (a) Prove that problem (SP_k) has a solution and that $z_k < 0$.
   (b) Let us define: $\phi_k : \alpha \geq 0 \mapsto f(x_k + \alpha(y_k - x_k))$. Prove that ϕ_k is continuously differentiable and that ϕ_k' is Lipschitz-continuous with modulus $\tilde{L}$.
   (c) Let us define $\alpha_k^{\min} = 2(1-\sigma)|z_k|/(L\|y_k - x_k\|^2)$. Using the following inequality: $\forall \alpha \geq 0$, $\phi'(\alpha) \leq \phi'(0) + \tilde{L}\alpha$, prove that for all $\alpha \in [0, \alpha_k^{\min}]$,
   $$\phi_k(\alpha) \leq \phi_k(0) + \sigma\alpha\phi_k'(0).$$
   (d) Deduce that the inner while-loop (lines 9-11) stops in finitely many iterations.
2. Prove by induction that for all $k \in \mathbb{N}$, $x_k \in \mathcal{F}$.

Part 2. We now analyse the method from a theoretical point of view.

1. Let $\bar{x} \in \mathcal{F}$ be a solution to problem (P). Prove that the value $\bar{V}$ of the following problem is equal to 0:
   $$\inf_{y \in \mathbb{R}^n} \langle \nabla f(\bar{x}), y - \bar{x} \rangle, \quad \text{subject to: } y \in \mathcal{F}.$$
2. Let $\hat{x}$ be such that the value $\hat{V}$ of the following problem is equal to 0:
   $$\inf_{y \in \mathbb{R}^n} \langle \nabla f(\hat{x}), y - \hat{x} \rangle, \quad \text{subject to: } y \in \mathcal{F}.$$
   Assume moreover (in this question only), that $\mathcal{F} = \{x \in \mathbb{R}^n \,|\, e(x) = 0 \text{ and } g(x) \leq 0\}$, where $e : \mathbb{R}^p \to \mathbb{R}^p$ and $g : \mathbb{R}^m \to \mathbb{R}^m$ are continuously differentiable. Assume finally that the LICQ hols at $\hat{x}$. Prove that $\hat{x}$ satisfies the KKT conditions (associated with problem (SP_k)).
3. Let $\tilde{x}$ be such that the value $\tilde{V}$ of the following problem is equal to 0:
   $$\inf_{y \in \mathbb{R}^n} \langle \nabla f(\tilde{x}), y - \tilde{x} \rangle, \quad \text{subject to: } y \in \mathcal{F}.$$
   Assume (in this question only) that f is convex. Prove that $\tilde{x}$ is a solution to (P).

Part 3. We now prove the convergence of the algorithm.

1. Let $k \in \mathbb{N}$. Assume that $\alpha_k < 1$. Prove the existence of a constant $A > 0$, independent of k, such that: $f(x_{k+1}) \leq f(x_k) - Az_k^2$.
2. Prove that $z_k \underset{k \to \infty}{\longrightarrow} 0$.
3. Let x_∞ be a limit point of $(x_k)_{k \in \mathbb{N}}$ (there exists at least one, since $\mathcal{F}$ is bounded). Prove that the value V_∞ of the following problem is equal to 0:
   $$\inf_{y \in \mathbb{R}^n} \langle \nabla f(x_\infty), y - x_\infty \rangle, \quad \text{subject to: } y \in \mathcal{F}.$$

Algorithm 5: Descent method with Armijo's rule for a constrained problem

Imput: $x_0 \in \mathcal{F}$, $\beta \in (0,1)$, $\sigma \in (0, 1/2)$;
Set $k = 0$; set stopping_test= wrong;
while *stopping_test= wrong* **do**
  Find a solution y_k to the following problem:

$$\inf_{y \in \mathbb{R}^n} \langle \nabla f(x_k), y - x_k \rangle, \quad \text{subject to: } y \in \mathcal{F}; \tag{SP_k}$$

  Set $z_k = \langle \nabla f(x_k), y_k - x_k \rangle$;
  if $z_k = 0$ **then**
    Set stopping_test= true;
  else
    Set $\alpha = 1$;
    while $f(x_k + \alpha(y_k - x_k) > f(x_k) + \sigma \alpha z_k$ **do**
      Set $\alpha = \beta\alpha$;
    end
    Set $\alpha_k = \alpha$; set $x_{k+1} = x_k + \alpha_k(y_k - x_k)$, set $k = k + 1$;
  end
end

Solution Part 1.

1. (a) The feasible set is compact and non-empty, the cost function is continuous (because affine). Therefore, there exists a solution. For all $y \in \mathcal{F}$, $z_k \leq \langle \nabla f(x_k), y - x_k \rangle$. Taking $y = x_k$, we obtain that $z_k \leq 0$. By assumption $z_k \neq 0$, thus $z_k < 0$.

   (b) The function ϕ_k is continuously differentiable, since f is itself continuously differentiable. For all $\alpha \geq 0$, we have:
$$\phi_k'(\alpha) = \langle \nabla f(x_k + \alpha(y_k - x_k)), y_k - x_k \rangle.$$
   Note that $\phi_k'(0) = z_k$. For all α_1 and $\alpha_2 \geq 0$, we have:
$$|\phi_k'(\alpha_2) - \phi_k'(\alpha_1)| = |\langle \nabla f(x_k + \alpha_2(y_k - x_k)) - \nabla f(x_k + \alpha_1(y_k - x_k)), y_k - x_k)| \leq L\|y_k - x_k\|^2 \cdot |\alpha_2 - \alpha_1|.$$
   Therefore, ϕ_k' is Lipschitz continuous, with modulus $\tilde{L} = L\|y_k - x_k\|^2$.

   (c) For $\alpha \in [0, \alpha_k^{\min}]$, we have:
$$\phi_k(\alpha) \leq \phi_k(0) + \int_0^\alpha \phi_k'(\theta)\,\mathrm{d}\theta \leq \phi_k(0) + \int_0^\alpha \phi'(0) + \tilde{L}\theta\,\mathrm{d}\theta = \phi_k(0) + \alpha\Big(\phi'(0) + \tilde{L}\alpha/2\Big).$$
   We conclude by observing that $\phi'(0) + \tilde{L}\alpha/2 \leq \sigma z_k$.

   (d) The inner-while loop stop when $\phi_k(\alpha) \leq \phi_k(0) + \alpha\phi_k'(0)$, which occurs at the latest when $\alpha \in [0, \alpha_k^{\min}]$, and therefore in finitely many iterations, since α "converges to 0".

2. We have assumed that $x_0 \in \mathcal{F}$. Let $k \in \mathbb{N}$, assume that $x_k \in \mathcal{F}$. We have: $x_{k+1} = x_k + \alpha_k(y_k - x_k) = (1 - \alpha_k)x_k + \alpha_k y_k \in \mathcal{F}$. Since $\alpha_k \in [0, 1]$, $x_k \in \mathcal{F}$ and $y_k \in \mathcal{F}$ and since $\mathcal{F}$ is convex, $x_{k+1} \in \mathcal{F}$.

Part 2.

1. Let $\varepsilon > 0$ be such that for all $x \in \bar{B}(\bar{x}, \varepsilon) \cap \mathcal{F}$, $f(x) \geq f(\bar{x})$. Let $y \in \mathcal{F}$. For $\alpha > 0$ small enough, $\bar{x} + \alpha(y - \bar{x}) \in \bar{B}(\bar{x}, \varepsilon) \cap \mathcal{F}$ (by convexity of $\mathcal{F}$) and therefore,
$$0 \leq \frac{f(\bar{x} + \alpha(y - \bar{x})}{\alpha} \to \langle \nabla f(\bar{x}), y - \bar{x} \rangle.$$
   Therefore, $\bar{V} \geq 0$. Taking $y = \bar{x}$, we obtain that $\bar{V} \leq 0$. Thus, $\bar{V} = 0$.

2. Since $\hat{V} = 0$, $\tilde{x}$ is a solution to the problem. Since the LICQ holds, the KKT conditions states that:
$$\nabla f(\hat{x}) + \nabla e(\hat{x})\mu + \nabla g(\hat{x})\lambda = 0$$
   and that $\lambda \geq 0$ and that for all $i = 1, ..., m$, $g_i(\bar{x}) = 0 \Rightarrow \lambda_i = 0$. These conditions are the KKT conditions as the ones for the original problem.

3. Let $y \in \mathcal{F}$. Since f is convex, we have:

$$f(y) \geq f(\tilde{x}) + \nabla f(\tilde{x}), y - \tilde{x} \geq f(\tilde{x}) + \tilde{V} = f(\tilde{x}).$$

Therefore, $\tilde{x}$ is optimal.

Part 3.

1. If $\alpha_k < 1$, then the previous iterate does not satisfy Armijo's rule and therefore, $\alpha/\beta \geq \alpha_k^{\min}$. We deduce that:

$$\alpha_k \geq \frac{2(1-\sigma)|z_k|}{L\|y_k - x_k\|^2} \geq \frac{2(1-\sigma)|z_k|}{LD^2},$$

where D is the diameter of $\mathcal{F}$, which is finite, since $\mathcal{F}$ is bounded. We deduce from Armijo's rule that:

$$f(x_{k+1}) \leq f(x_k) + \sigma\alpha_k z_k \leq f(x_k) - \frac{2(1-\sigma)|z_k|^2}{LD^2}.$$

This proves the result, with $A = \frac{2(1-\sigma)}{LD^2}$.

2. If $\alpha_k = 1$, we deduce from Armijo's rule that $f(x_{k+1}) \leq f(x_k) - \sigma|z_k|$. It follows that for all $k \in \mathbb{N}$,

$$f(x_{k+1}) - f(x_k) \leq -\min\big(A|z_k|^2, \sigma|z_k|\big),$$

therefore, since the sequence $(f(x_k)_{k\in\mathbb{N}})$ is bounded from below, $\sum_{k=0}^{\infty} \min\big(A|z_k|^2, \sigma|z_k|\big) < +\infty$, thus $\min\big(A|z_k|^2, \sigma|z_k|\big) \to 0$ and finally, $z_k \to 0$.

3. Let $(x_{k_\ell})_{l\in\mathbb{N}}$ be a convergent subsequence with limit x_*. For all $y \in \mathcal{F}$, for all $\ell \in \mathbb{N}$, we have:

$$\langle \nabla f(x_{k_\ell}), y - x_{k_\ell}\rangle \geq z_{k_\ell},$$

thus, passing to the limit:

$$\langle \nabla f(x_*), y - x_*\rangle \geq \lim_{\ell\to\infty} z_{k_\ell} = 0.$$

We deduce that $V_\infty \geq 0$. Since $y = x_\infty$ is feasible, $V_\infty \leq 0$, thus $V_\infty = 0$.

2.2 Newton's method

2.2.1 Armijo's rule and Newton's method

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a twice differentiable function with a Lipschitz-continuous second-order derivative. Let $\bar{x}$ be a stationary point. We assume that there exists $\alpha > 0$ such that $h^\top \nabla^2 f(\bar{x}) h \geq \alpha\|h\|^2$, for all $h \in \mathbb{R}^n$.

We already know from the lecture that there exists $\varepsilon_0 > 0$ such that for all $x \in \bar{B}(\bar{x}, \varepsilon_0)$, $\nabla^2 f(x)$ is positive definite, thus invertible, and such that

$$\|x + p - \bar{x}\| \leq \|x - \bar{x}\|,$$

where $p = -\nabla^2 f(x)^{-1}\nabla f(x)$.

Let $c_1 \in (0, 1/2)$. Armijo's rule is satisfied (with steplength 1) if

$$f(x+p) \leq f(x) + c_1\langle \nabla f(x), p\rangle.$$

Let us set

$$\delta = \frac{\left(\frac{1}{2} - c_1\right)\alpha}{\frac{3}{2} - c_1},$$

let $\varepsilon \in (0, \varepsilon_0]$ be such that for all $x \in \bar{B}(\bar{x}, \varepsilon)$, $\|\nabla^2 f(x) - \nabla^2 f(\bar{x})\| \leq \delta$.

Prove that for all $x \in \bar{B}(\bar{x}, \varepsilon)$, Armijo's rule holds. To this purpose, prove first that for all $x \in \bar{B}(\bar{x}, \varepsilon)$, for all $\theta \in [0, 1]$, the three following inequalities hold:

$$\frac{1}{2}p^\top \nabla^2 f(x + \theta p)p \leq \frac{1}{2}\left[p^\top \nabla^2 f(\bar{x})p + \delta\|p\|^2\right] \leq (1 - c_1)\left[p^\top \nabla^2 f(\bar{x})p - \delta\|p\|^2\right] \leq (1 - c_1)p^\top \nabla^2 f(x)p.$$

Then, conclude with a second-order Taylor expansion of $f(x + p)$.

Solution Since $\|x-\bar{x}\| \leq \varepsilon$ and $\|x+p-\bar{x}\| \leq \varepsilon$, we have: $\|x+\theta p-\bar{x}\| \leq \varepsilon$. Therefore,

$$\begin{aligned} p^\top \nabla^2 f(x+\theta p)p &\leq p^\top \nabla^2 f(\bar{x})p + |p^\top(\nabla f(x+\theta p) - \nabla f(\bar{x}))p| \\ &\leq p^\top \nabla^2 f(\bar{x})p + \|p\| \cdot \|\nabla f(x+\theta p) - f(\bar{x})\| \cdot \|p\| \\ &\leq p^\top \nabla^2 f(\bar{x})p + \delta\|p\|^2. \end{aligned}$$

The first inequality follows. The third inequality can be proved similarly. It remains to check that:

$$\frac{1}{2}\big[p^\top \nabla^2 f(\bar{x})p + \delta\|p\|^2\big] \leq (1-c_1)\big[p^\top \nabla^2 f(\bar{x})p - \delta\|p\|\big],$$

which is equivalent to:

$$\Big(\frac{3}{2} - c_1\Big)\delta\|p\| \leq \Big(\frac{1}{2} - c_1\Big)p^\top \nabla^2 f(\bar{x})p.$$

and to:

$$\Big(\frac{1}{2} - c_1\Big)\alpha\|p\|^2 \leq \Big(\frac{1}{2} - c_1\Big)p^\top \nabla^2 f(\bar{x})p,$$

by definition of δ. The last inequality is true, by definition of α.

Let us prove Armijo's rule. Let $\theta \in [0,1]$ be such that:

$$f(x+p) = f(x) + \nabla f(x)^\top p + \frac{1}{2}p^\top \nabla^2 f(x+\theta p)p.$$

Armijo's rule is then equivalent to:

$$\frac{1}{2}p^\top \nabla^2 f(x+\theta p)p \leq -(1-c_1)\nabla f(x)^\top p = (1-c_1)p^\top \nabla^2 f(x)p.$$

Armijo's rule follows.

2.2.2 Convergence of an approximate Newton's method

We investigate in this exercise the convergence rate of an approximate local Newton's method described as follows: $x_{k+1} = x_k + p_k$, where p_k is an approximate solution to the linear system $\nabla^2 f(x_k)p + \nabla f(x_k) = 0$. We denote by r_k the residual (i.e. $r_k = \nabla^2 f(x_k)p_k + \nabla f(x_k)$). We assume that the linear system is solved at a certain precision, so that

$$\|r_k\| \leq \beta\|\nabla f(x_k)\|,$$

for some constant $\beta > 0$. The used norm is the Euclidean norm.

We consider a twice differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ which is such that the mapping $x \in \mathbb{R}^n \mapsto \nabla^2 f(x)$ is Lipschitz continuous with modulus $L > 0$. We also consider a local minimizer $\bar{x}$ of f. We assume that there exists $\varepsilon_0 > 0$, $\alpha > 0$, and $A > 0$ such that for all $x \in \mathbb{R}^n$, if $\|x-\bar{x}\| \leq \varepsilon_0$, then

$$\alpha\|h\|^2 \leq h^\top \nabla^2 f(x)h \leq A\|h\|^2.$$

1. Assume that at the iteration k, $\|x_k - \bar{x}\| \leq \varepsilon$, for some $\varepsilon \in (0, \varepsilon_0]$. Prove that x_{k+1} is well-defined and that
$$\|x_{k+1} - \bar{x}\| \leq \frac{1}{\alpha}\Big(\frac{L}{2}\varepsilon + A\beta\Big)\|x_k - \bar{x}\|.$$
2. Deduce that if $\varepsilon > 0$ and $\beta > 0$ are chosen sufficiently small and that $\|x_0 - \bar{x}\| \leq \varepsilon$, then the whole sequence $(x_k)_{k\in\mathbb{N}}$ is well-defined and converges linearly to $\bar{x}$.

Solution

1. We have
$$\begin{aligned} \|\nabla f(x_k)\| &= \Big\|\nabla f(\bar{x}) + \int_0^1 \nabla^2 f(\bar{x} + \theta(x_k - \bar{x}))(x_k - \bar{x})\,\mathrm{d}\theta\Big\| \\ &\leq \int_0^1 \|\nabla^2 f(\bar{x} + \theta(x_k - \bar{x}))\| \cdot \|(x_k - \bar{x})\|\,\mathrm{d}\theta \\ &\leq A\|x_k - \bar{x}\|. \end{aligned}$$

2. We have

$$\begin{aligned} x_{k+1} - \bar{x} &= x_k - \bar{x} - \nabla^2 f(x_k)^{-1}(\nabla f(x_k) - r_k) \\ &= \nabla^2 f(x_k)^{-1}\big(\nabla^2 f(x_k)(x_k - \bar{x}) - \nabla f(x_k) + r_k\big). \end{aligned}$$

Since $\|\nabla^2 f(x_k)^{-1}\| \leq 1/\alpha$, we deduce that

$$\|x_{k+1} - \bar{x}\| \leq \frac{1}{\alpha}\big(\|\nabla^2 f(x_k)(x_k - \bar{x}) - \nabla f(x_k)\| + \|r_k\|\big).$$

Moreover, we have:

$$\begin{aligned} \|\nabla^2 f(x_k)(x_k - \bar{x}) - \nabla f(x_k)\| &= \Big\| \int_0^1 (\nabla^2 f(x_k) - \nabla^2 f(\bar{x} + \theta(x_k - \bar{x}))(x_k - \bar{x})\, \mathrm{d}\theta \Big\| \\ &\leq L\|x_k - \bar{x}\|^2 \int_0^1 \theta\, \mathrm{d}\theta \\ &\leq \frac{L\varepsilon}{2}\|x_k - \bar{x}\|. \end{aligned}$$

The inequality follows then easily.

3. If β and ε are chosen sufficiently small, then

$$\frac{1}{\alpha}\Big(\frac{L}{2}\varepsilon + A\beta\Big) \leq \frac{1}{2}$$

and thus the whole sequence converges.

Comments for the students.

- One can think to variants to this exercise, where the coefficent β depends on k and converges to 0, for example, so that a superlinear rate of convergence can be obtained.
- Practical interest: if an iterative method is used for solving the linear system $\nabla^2 f(x_k)p_k + \nabla f(x_k) = 0$, such as a conjugate gradient, then one can use the above analysis for designing an appropriate stopping criterion.

2.3 Quasi-Newton methods

2.3.1 On the invariance of the Quasi-Newton method by an affine transformation

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a twice continuously differentiable function. Consider the quasi-Newton method with the DFP formula (see Algorithm 6 below) for minimizing f. In this algorithm, the step length α_k satisfies Armijo and Wolfe's rule; is computed with the algorithm seen in the lecture.

Algorithm 6: Quasi-Newton method with DFP formula

```
Input: x_0 ∈ ℝ^n, M_0 ∈ S_n (M_0 is positive definite), 0 < c_1 < c_2 < 1.
Set: k = 0;
while ∇f(x_k) ≠ 0 do
    Set: p_k = −M_k^{−1} ∇f(x_k);
    Compute α_k satisfying Armijo and Wolfe's rules;
    Set: x_{k+1} = x_k + α_k p_k;
    Set: s_k = x_{k+1} − x_k and y_k = ∇f(x_{k+1}) − ∇f(x_k);
    Set:
        M_{k+1} = M_k + ((y_k − M_k s_k) y_k^⊤ + y_k (y_k − M_k s_k)^⊤) / ⟨s_k, y_k⟩ − ⟨y_k − M_k s_k, s_k⟩ / ⟨s_k, y_k⟩^2 · y_k y_k^⊤;
    Set: k = k + 1;
end
```

Let $g : \mathbb{R}^n \to \mathbb{R}$ be a twice continuously differentiable function. Let $A \in \mathbb{R}^{n\times n}$ be an invertible matrix and let $b \in \mathbb{R}^n$. We define: $\tilde{g}(x) = g(Ax + b)$.

1. Compute the gradient and Hessian matrix of $\tilde{g}$.

2. Let $x_0 \in \mathbb{R}^n$, let $\tilde{x}_0 = A^{-1}(x_0 - b)$. Let $M_0 \in \mathbb{R}^{n\times n}$ be a symmetric positive definite matrix and let $\widetilde{M}_0 = A^\top M_0 A$. Let $(x_k)_{k=0,1,...}$ and $(M_k)_{k=0,1,...}$ (resp. $(\tilde{x}_k)_{k=0,1,...}$ and $(\widetilde{M}_k)_{k=0,1,...}$) be the sequences generated by Algorithm 6 for the minimization of g with starting point x_0 and initial matrix M_0 (resp. for the minimization of $\tilde{g}$ with starting point $\tilde{x}_0$ and initial matrix $\widetilde{M}_0$). Prove that for all $k = 0, 1, ...$
$$\tilde{x}_k = A^{-1}(x_k - b) \quad \text{and} \quad \widetilde{M}_k = A^\top M_k A.$$

Solution

1. By the chain rule, we have $D\tilde{g}(x) = Dg(Ax + b)A$ and therefore, $\nabla\tilde{g}(x) = A^\top \nabla g(Ax + b)$. The Hessian is obtained as follows:
$$\nabla^2\tilde{g}(Ax + b) = D\big[\nabla\tilde{g}(Ax + b)\big] = A^\top \nabla g(Ax + b)A,$$
applying once again the chain rule.

2. We prove the result by induction (it is clearly satisfied at $k = 0$). Assume that it holds for some $k \geq 0$. In what follows, we denote with "tilde" all vectors/matrices/functions related to the minimization of $\tilde{g}$. We obtain first:
$$\tilde{p}_k = -\widetilde{M}_k^{-1}\nabla\tilde{g}(\tilde{x}_k) = -(A^\top M_k A)^{-1}A^\top \nabla g(A\tilde{x}_k + b) = -A^{-1}M_k\nabla g(x_k) = A^{-1}p_k.$$
Then, we obtain, for all $\alpha \geq 0$,
$$\tilde{\phi}_k(\alpha) := \tilde{g}(\tilde{x}_k + \alpha\tilde{p}_k) = g(A(\tilde{x}_k + \alpha\tilde{p}_k) + b) = g(x_k + \alpha p_k) =: \phi_k(\alpha).$$
The line-search methods are entirely based on ϕ_k (resp. $\tilde{\phi}_k$). Therefore, they generate the same step length in the two situations: $\tilde{\alpha}_k = \alpha_k$. We deduce easily that $\tilde{x}_{k+1} = A^{-1}(x_{k+1} - b)$. Finally we have: $\tilde{s}_k = A^{-1}s_k$, $\tilde{y}_k = A^\top y_k$, and therefore
$$\tilde{y}_k - \widetilde{M}_k\tilde{s}_k = A^\top y_k - A^\top M_k A A^{-1}s_k = A^\top(y_k - M_k s_k)$$
and
$$\langle\tilde{s}_k, \tilde{y}_k\rangle = \langle s_k, y_k\rangle \quad \text{and} \quad \tilde{y}_k\tilde{y}_k^\top = A^\top y_k y_k^\top A.$$
It follows then that $\widetilde{M}_{k+1} = A^\top M_{k+1}A$.

Comments for the students:

- This "invariance property" is not satisfied for "first-order" methods (e.g. steepest gradient method). As an example: take $g(x) = \|x\|^2$ and $\tilde{g}(x) = g(Ax)$ where A is diagonal with positive coefficients. The steepest gradient method with exact search converges in one step, when minimizing g. If A is ill-conditioned, then the same method will be slow (for the minimization of $\tilde{g}$, zig-zag effect seen in the lecture).

- This "invariance property" is not satisfied by all the Quasi-Newton methods in general. It has been seen in the lecture that M_{k+1} is the solution to:
$$\inf_{M\in\mathcal{S}_n} \|C(M - M_k)C^\top\|_F^2, \quad Ms_k = y_k,$$
where $\|\cdot\|_F$ is the Frobenius norm and for a regular matrix C which is such that $y = C^{-\top}C^{-1}s$. The matrix C enables to "scale" in an appropriate way the Frobenius norm so that the invariance property holds.

2.3.2 SR1-Update Formula (1)

We consider an inverse quasi-Newton method: at the iteration k, an approximation $H \in S_n$ of $\nabla^2 f(x_k)^{-1}$ has been computed. After computation of the next iterate, x_{k+1}, a new approximation H_+ has to be computed.

The SR1 (Symmetric rank 1) Update Formula consists in finding $\gamma \in \mathbb{R}$ and $u \in \mathbb{R}^n$ such that

$$H_+ = H + \gamma u u^\top, \quad H_+ y = s, \tag{25}$$

where $s = x_{k+1} - x_k$ and $y = \nabla f(x_{k+1}) - \nabla f(x_k)$. We assume that $\langle s - Hy, y\rangle \neq 0$.

1. Motivate the constraint $H_+ y = s$.
2. Prove that for $u = s - Hy$ and $\gamma = \frac{1}{\langle u,y\rangle}$, the conditions (25) are fulfilled.
3. Let $\tilde{u} \in \mathbb{R}^n$ and $\tilde{\gamma} \in \mathbb{R}$ be such that the conditions (25) are fulfilled. Prove that
$$H + \gamma u u^\top = H + \tilde{\gamma}\tilde{u}\tilde{u}^\top.$$
4. We assume now that H is invertible. Give a necessary and sufficient condition for the invertibility of H_+. Using the lemma of Shermann-Woodbury-Morrison, prove that if this condition is satisfied, then
$$H_+^{-1} = M + \frac{(y - Ms)(y - Ms)^\top}{\langle y - Ms, s\rangle},$$
where $M = H^{-1}$.

Solution

1. We have: $\nabla^2 f(x_k)s \approx y$ (first-order Taylor expansion of the gradient). Thus, the matrix H_+ should be such that $H_+ y = s$, since it approximates $\nabla^2 f(x_k)$.
2. We have
$$H_+ y = Hy + \gamma u u^\top y = Hy + \frac{\langle u, y\rangle}{\langle u, y\rangle}(s - Hy) = s.$$
3. Let $\tilde{\gamma} \in \mathbb{R}$ and $\tilde{u} \in \mathbb{R}^n$ be such that (25) holds. We have
$$\tilde{\gamma}\tilde{u}\tilde{u}^\top y = s - Hy = u \neq 0.$$
We also have:
$$\tilde{\gamma}\tilde{u}\tilde{u}^\top y = \underbrace{(\tilde{\gamma}\langle \tilde{u}, y\rangle)}_{=:\alpha}\tilde{u}.$$
Therefore, $u = \alpha\tilde{u}$ and
$$\gamma = \frac{1}{\langle u, y\rangle} = \frac{1}{\alpha\langle \tilde{u}, y\rangle} = \frac{\tilde{\gamma}}{\alpha\langle \tilde{u}, y\rangle\tilde{\gamma}} = \frac{\tilde{\gamma}}{\alpha^2}.$$
It follows that
$$\gamma u u^\top = \tilde{\gamma}\tilde{u}\tilde{u}^\top.$$
4. By the SWM lemma, H_+ is invertible if and only if
$$1 + \langle \gamma u, Mu\rangle \neq 0.$$
In this case,
$$H_+^{-1} = M - \frac{\gamma M u u^\top M}{1 + \langle \gamma u, Mu\rangle}.$$
We have: $Mu = Ms - MHy = Ms - y$. Moreover,
$$\frac{1 + \langle \gamma u, Mu\rangle}{\gamma} = \langle u, y\rangle + \langle u, Mu\rangle = \langle u, y + Mu\rangle = \langle u, Ms\rangle = \langle Mu, s\rangle = \langle Ms - y, s\rangle.$$
Therefore,
$$H_+^{-1} = M - \frac{(Ms - y)(Ms - y)^\top}{\langle Ms - y, s\rangle} = M + \frac{(y - Ms)(y - Ms)^\top}{\langle y - Ms, s\rangle}.$$

Remarks for the students. There is no guaranty that H_+ is positive definite, even if H is positive definite and x_{k+1} is computed with a "good" step-length rule (as it is the case with the DFP formula with Wolfe's rules, seen in the lecture), which can be problematic for line-search methods. However, the SR1 formula can be useful for trust-region methods.

2.3.3 SR1-Update Formula (2)

We consider the inverse quasi-Newton method described in the previous exercise. The method generates sequences $(x_k)_{k\in\mathbb{N}}$ and $(H_k)_{k\in\mathbb{N}}$ given by

$$x_{k+1} = x_k - H_k \nabla f(x_k), \quad H_{k+1} = H_k + \frac{(s_k - H_k y_k)(s_k - H_k y_k)^\top}{\langle s_k - H_k y_k, y_k\rangle},$$

where $s_k = x_{k+1} - x_k$, $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$. We consider the case of a linear quadratic function:

$$f(x) = \langle b, x\rangle + \frac{1}{2} x^\top A x,$$

where A is symmetric positive definite. We assume that the condition $\langle s_k - H_k y_k, y_k\rangle \neq 0$ is satisfied for all k.

1. Find a relation between s_k and y_k.
2. Prove by induction that for all $k \geq 1$, for all $i = 0, ..., k-1$,
$$H_k y_i = s_i.$$
3. We assume that the vectors y_0,...,y_{n-1} are linearly independent. Prove that $H_n = A^{-1}$. What can we say about x_{n+1}?

Solution

1. For all k, $y_k = \nabla f(x_{k+1}) - \nabla f(x_k) = A(x_{k+1} - x_k) = As_k$.
2. We already know from the previous exercise that for all k, $H_k y_{k-1} = s_{k-1}$. Therefore, the property to be proved holds for $k = 0$. Assume that it is satisfied for a given $k \geq 0$. We already know that $H_k y_{k-1} = s_{k-1}$. Let $i \in \{0, ..., k-2\}$. By the induction assumption, $H_{k-1} y_i = s_i$. Therefore, we just need to prove that
$$(s_k - H_k y_k)^\top y_i = 0.$$
We have:
$$(s_k - H_k y_k)^\top y_i = \langle s_k, y_i\rangle - \langle H_k y_k, y_i\rangle = \langle s_k, y_i\rangle - \langle y_k, H_k y_i\rangle = \langle s_k, y_i\rangle - \langle y_k, s_i\rangle,$$
by the induction assumption. Then, we have:
$$\langle s_k, y_i\rangle - \langle y_k, s_i\rangle = \langle s_k, As_i\rangle - \langle As_k, s_i\rangle = \langle As_k, s_i\rangle - \langle As_k, s_i\rangle = 0.$$
The property holds at $k+1$.
3. If the vectors y_0,...,y_{n-1} are linearly independent, they form a basis of $\mathbb{R}^n$. We have, for all $i =$ 0,...,$n-1$,
$$H_n y_i = s_i = A^{-1} y_i.$$
The linear mappings associated with the matrices H_n and A^{-1} are equal on a basis of $\mathbb{R}^n$, therefore, $H_n = A^{-1}$. Denoting by $\bar{x}$ the unique minimizer of f, we have for all $x \in \mathbb{R}^n$ that $\nabla f(x) = A(x - \bar{x})$ and we deduce that
$$x_{n+1} = x_n - H_n \nabla f(x_n) = x_n - A^{-1} A(x_n - \bar{x}) = \bar{x}.$$

2.3.4 Broyden's formula

In this exercise and the next one, we give formulas used to approximate the Hessian of a function f to minimize without computing it. At the iteration k, the gradients $\nabla f(x_k)$ and $\nabla f(x_{k-1})$ have been computed, as well as an approximation B_{k-1}. We set $s = x_k - x_{k-1}$ and $y = \nabla f(x_k) - \nabla f(x_{k-1})$ and assume that $s \neq 0$. Before computing x_{k+1} (by solving (**??**)), we need to compute an approximation of $Df(x_k)$.

A first idea is to use the solution to the following problem:

$$\min_{B \in \mathbb{R}^{n \times n}} \frac{1}{2}\|B - B_{k-1}\|_F^2, \quad \text{subject to: } Bs = y. \tag{26}$$

The norm $\|\cdot\|_F$ is the norm associated with the Frobenius scalar product $\langle \cdot, \cdot \rangle_F$, defined by:

$$\langle B, B' \rangle_F = \sum_{i=1}^{n} \sum_{j=1}^{n} B_{ij} B'_{ij}.$$

Motivate the constraint $Bs = y$ and prove that the solution to (26) is:

$$B^* = B_{k-1} + \frac{zs^\top}{\|s\|^2}.$$

where $z = y - B_{k-1}s$. To this purpose, prove the following steps:

1. The rank of the mapping: $B \in \mathbb{R}^{n \times n} \mapsto Bs \in \mathbb{R}^n$ is equal to n.
2. For all $\lambda \in \mathbb{R}^n$, for all $B \in \mathbb{R}^{n \times n}$, $\lambda^\top Bs = \langle \lambda s^\top, B \rangle_F$.
3. State the KKT conditions (denote by $\lambda \in \mathbb{R}^n$ the dual variable).
4. Prove that there exists a unique pair $(B^*, \lambda^*) \in \mathbb{R}^{n \times n} \times \mathbb{R}^n$ satisfying the KKT conditions, by computing first λ^*, then B^* (similarly to exercise 2, sheet 3). Conclude.

Solution

1. It suffices to check that the mapping $B \in \mathbb{R}^{n \times n} \mapsto Bs \in \mathbb{R}^n$ is surjective. Let $v \in \mathbb{R}^n$, let us define: $B = \frac{vs^\top}{\|s\|^2}$. We easily check that $Bs = y$.
2. For all $\lambda \in \mathbb{R}^n$, for all $B \in \mathbb{R}^{n \times n}$,
$$\lambda^\top Bs = \sum_{i=1}^{n} \lambda_i (Bs)_i = \sum_{i=1}^{n} \sum_{j=1}^{n} \lambda_i B_{ij} s_j = \sum_{i=1}^{n} \sum_{j=1}^{n} (\lambda s^\top)_{ij} B_{ij} = \langle \lambda s^\top, B \rangle_F.$$
3. The Lagrangian is as follows:
$$\frac{1}{2}\|B - B_{k-1}\|^2 + \langle \lambda s^\top, B \rangle_F - \lambda^\top y,$$
thus, the KKT conditions read: $\exists \lambda \in \mathbb{R}^n$,
$$B - B_{k-1} + \lambda s^\top = 0 \quad \text{and} \quad Bs = y.$$
4. Multiplying by s on the right, we find that $z + \lambda \|s\|^2 = 0$, thus $\lambda = -\frac{z}{\|s\|^2}$. Finally,
$$B = B_{k-1} + \lambda s^\top = B_{k-1} + \frac{zs^\top}{\|s\|^2}.$$
We know from exercise 2, sheet 3, that the problem has a unique optimal solution. By the full-rank condition, this solution satisfies the KKT conditions, and we have just proved that B^* is the only matrix satisfying these conditions: therefore, it is the optimal solution. (We can also argue with the convexity of the problem.)

2.3.5 PSB formula

A second approach consists in finding the solution to the following problem:

$$\min_{B \in \mathcal{S}_n} \frac{1}{2}\|B - B_{k-1}\|_F^2, \quad \text{subject to: } Bs = y, \tag{27}$$

where $\mathcal{S}_n$ is the set of symmetric matrices of size n. We assume here that B_{k-1} is symmetric.

Prove that the solution to (27) is:

$$B^{**} = B_{k-1} + \frac{sz^\top + zs^\top}{\|s\|^2} - \frac{(s^\top z)ss^\top}{\|s\|^4},$$

where z is still equal to $y - B_{k-1}s$. To this purpose, prove the following steps:

1. Find a linear mapping $g : \mathbb{R}^{n\times n} \to \mathbb{R}^{n\times n}$ such that for all $B \in \mathbb{R}^{n\times n}$, $g(B) = 0$ if and only if B is symmetric.
2. Using the mapping g to reformulate the problem and denoting by $\Omega \in \mathbb{R}^{n\times n}$ the associated dual variable, state the KKT conditions.
3. Prove that any matrix $B \in \mathbb{R}^{n\times n}$ satisfies the KKT conditions if and only if there exist $\lambda \in \mathbb{R}^n$ such that:
$$B - B_{k-1} = -\frac{1}{2}(\lambda s^\top + s\lambda^\top). \tag{28}$$
4. Multiplying (28) by s on the right, then by $s^\top$ on the left, deduce the expression of $\lambda^\top s$, of λ, and finally, prove that B^{**} is the optimal solution.

Solution

1. We simply define $g(B) = B - B^\top$.
2. The Lagrangian is now:
$$L(B, \lambda, \Omega) = \frac{1}{2}\|B - B_{k-1}\|_F^2 + \langle \lambda s^\top, B\rangle_F - \lambda^\top y + \langle \Omega, B - B^\top\rangle_F.$$
We observe that $\langle \Omega, B - B^\top\rangle_F = \langle \Omega - \Omega^\top, B\rangle_F$. Therefore, the optimality conditions read:
$$B - B_{k-1} + \lambda s^\top + \Omega - \Omega^\top = 0. \tag{29}$$
3. Taking the transpose of equation (29), we obtain that:
$$B - B_{k-1} + s^\top\lambda + \Omega^\top - \Omega = 0.$$
The result follows by additioning the last two equalities.
4. Multiplying by s on the right, we find that:
$$z = -\frac{1}{2}(\lambda\|s\|^2 + s\lambda^\top s). \tag{30}$$
Multiplying by $s^\top$ on the left, we find that:
$$s^\top z = -\frac{1}{2}(s^\top\lambda\|s\|^2 + \|s\|^2\lambda^\top s) = -\|s\|^2\lambda^\top s.$$
We deduce that: $\lambda^\top s = -\frac{s^\top z}{\|s\|^2}$ and using (30),
$$\lambda = \frac{-2z - s\lambda^\top s}{\|s\|^2} = \frac{-2z}{\|s\|^2} + \frac{(s^\top z)s}{\|s\|^4}.$$
Finally, we obtain that:
$$B = B_{k-1} - \frac{1}{2}(\lambda s^\top + s\lambda^\top) = B^{**}.$$
The optimality of B^{**} follows, since the problem is convex.

2.3.6 Limited BFGS method

Consider the BFGS method, a quasi-Newton methods which provides at each iteration k an approximation H_k of $\nabla^2 f(x_k)^{-1}$. The next iterate x_{k+1} is computed with the following formula:

$$x_{k+1} = x_k - \alpha_k H_k \nabla f(x_k),$$

where α_k satisfies appropriate step length rules. The approximation of the inverse of the Hessian matrix is updated with the following formula:

$$H_{k+1} = V_k^\top H_k V_k + \rho_k s_k s_k^\top,$$

where

$$s_k = x_{k+1} - x_k, \quad y_k = \nabla f(x_{k+1}) - \nabla f(x_k), \quad \rho_k = \frac{1}{y_k^\top s_k}, \quad V_k = I - \rho_k y_k s_k^\top.$$

We do not discuss the well-posedness of the sequences $(x_k)_{k\in\mathbb{N}}$, $(H_k)_{k\in\mathbb{N}}$, and $(\alpha_k)_{k\in\mathbb{N}}$ in this exercise.

0. Let $m \in \mathbb{N}\backslash\{0\}$ be given. Prove that for $k \geq m$,

$$\begin{aligned} H_k = {} & (V_{k-1}^\top ... V_{k-m}^\top) H_{k-m} (V_{k-m} ... V_{k-1}) \\ & + \rho_{k-m} (V_{k-1}^\top ... V_{k-m+1}^\top) s_{k-m} s_{k-m}^\top (V_{k-m+1} ... V_{k-1}) \\ & + \rho_{k-m+1} (V_{k-1}^\top ... V_{k-m+2}^\top) s_{k-m+1} s_{k-m+1} (V_{k-m+2} ... V_{k-1}) \\ & + ... \\ & + \rho_{k-2} V_{k-1}^\top s_{k-2} s_{k-2}^\top V_{k-1} \\ & + \rho_{k-1} s_{k-1} s_{k-1}^\top. \end{aligned}$$

The limited BFGS-method consists in computing directly the descent direction p_k with the following formula:

$$\begin{aligned} -p_k = {} & (V_{k-1}^\top ... V_{k-m}^\top) H_k^0 (V_{k-m} ... V_{k-1}) \nabla f(x_k) \\ & + \rho_{k-m} (V_{k-1}^\top ... V_{k-m+1}^\top) s_{k-m} s_{k-m}^\top (V_{k-m+1} ... V_{k-1}) \nabla f(x_k) \\ & + \rho_{k-m+1} (V_{k-1}^\top ... V_{k-m+2}^\top) s_{k-m+1} s_{k-m+1}^\top (V_{k-m+2} ... V_{k-1}) \nabla f(x_k) \\ & + ... \\ & + \rho_{k-2} V_{k-1}^\top s_{k-2} s_{k-2}^\top V_{k-1} \nabla f(x_k) \\ & + \rho_{k-1} s_{k-1} s_{k-1}^\top \nabla f(x_k), \end{aligned}$$

where H_k^0 is given (typically, a multiple of the identity matrix). In this way, the descent direction can be obtained without computing (and storing) explicitely the matrix H_k. Consider the Algorithm 7 below for computing p_k.

The goal of this exercise is to prove that the output of Algorithm (7) indeed satisfies the above formula, and that the number of multiplications required for computing the descent direction is small.

1. Prove that for all $i = k-m, ..., k-1$,

$$q_i = V_i V_{i+1} ... V_{k-1} \nabla f(x_k).$$

2. Deduce that

$$-p_k = s_{k-1}\alpha_{k-1} + V_{k-1}^\top \Big(s_{k-2}\alpha_{k-2} + V_{k-2}^\top \big(... + V_{k-m+1}^\top (s_{k-m}\alpha_{k-m} + V_{k-m}^\top H_k^0 q_{k-m}) ... \big) \Big).$$

3. Prove that for all $i = k-m, ...k-1$,

$$-p_k = s_{k-1}\alpha_{k-1} + V_{k-1}^\top \Big(s_{k-2}\alpha_{k-2} + V_{k-2}^\top \big(... + V_{i+1}^\top (s_i \alpha_i + V_i^\top r_i) ... \big) \Big).$$

   Conclude.

4. How many multiplications have to be performed in the two loops (lines 3-6 and 8-11)?

Algorithm 7: L-BFGS method for computing a descent direction

1 Imput: $\nabla f(x_k)$, $s_{k-m},...,s_{k-1}$, $y_{k-m},...,y_{k-1}$, $\rho_{k-m},...,\rho_{k-1}$;
2 Set: $q_k = \nabla f(x_k)$;
3 **for** $i = k-1, ..., k-m$ **do**
4 | Set: $\alpha_i = \rho_i s_i^\top q_{i+1}$;
5 | Set: $q_i = q_{i+1} - \alpha_i y_i$;
6 **end**
7 Set: $r_{k-m} = H_k^0 q$;
8 **for** $i = k-m, ..., k-1$ **do**
9 | Set: $\beta_i = \rho_i y_i^\top r_i$;
10 | Set: $r_{i+1} = r_i + s_i(\alpha_i - \beta_i)$;
11 **end**
12 Output: r_k.

Solution

0. The formula is clearly true for $m = 1$. Let assume that it is satisfied for a given $m \geq 1$. We have:

$$H_{k-m} = V_{k-m-1}^\top H_{k-m-1} V_{k-m-1} + \rho_{k-m-1} s_{k-m-1} s_{k-m-1}^\top.$$

Therefore,

$$\begin{aligned}(V_{k-1}^\top ... V_{k-m}^\top) H_{k-m} (V_{k-m} ... V_{k-1}) = {} & (V_{k-1}^\top ... V_{k-m-1}^\top) H_{k-m-1} (V_{k-m-1} ... V_{k-1}) \\ & + \rho_{k-m-1} (V_{k-1}^\top ... V_{k-m}^\top) s_{k-m-1} s_{k-m-1}^\top (V_{k-m} ... V_{k-1}).\end{aligned}$$

Combining the formula (true for the index m) with the above relation, we obtain the formula for the index $m+1$.

1. Combining lines 4 and 5, we obtain that:

$$q_i = q_{i+1} - \rho_i y_i s_i^\top q_{i+1} = (\mathrm{Id} - \rho_i y_i s_i^\top) q_{i+1} = V_i q_{i+1}.$$

The result follows immediately by induction.

2. We deduce that:

$$\alpha_i = \rho_i s_i^\top q_{i+1} = \rho_i s_i^\top V_{i+1} ... V_{k-1} \nabla f(x_k)$$

and therefore that:

$$\begin{aligned}-p_k = {} & (V_{k-1}^\top ... V_{k-m}^\top) H_k^0 q_{k-m} \\ & + (V_{k-1}^\top ... V_{k-m+1}^\top) s_{k-m} \alpha_{k-m} \\ & + (V_{k-1}^\top ... V_{k-m+2}^\top) s_{k-m+1} \alpha_{k-m+1} \\ & + ... \\ & + V_{k-1}^\top s_{k-2} \alpha_{k-2} \\ & + s_{k-1} \alpha_{k-1}.\end{aligned}$$

Factorizing successively the different terms, we obtain the result.

1. The formula is already proved for $i = k-m$, since $r_{k-m} = q_{k-m}$. It can be proved by induction, observing that

$$r_{i+1} = r_i + s_i(\alpha_i - \rho_i y_i r_i^\top) = V_i^\top r_i + s_i \alpha_i.$$

We obtain therefore (for $i = k-1$) that

$$-p_k = s_{k-1} \alpha_{k_1} + V_{k-1}^\top r_{k-1} = r_k,$$

which proves that the output of the algorithm is indeed $-p_k$.

2. There are $n-1$ multiplications to perform at lines (...).

2.4 Penalty methods

2.4.1 Lagrange relaxation

Let f and $g : \mathbb{R}^n \to \mathbb{R}$ be two continuous functions, let X be a compact set of $\mathbb{R}^n$. We consider the following family of optimization problems:

$$V(u) = \inf_{x \in X} f(x), \quad \text{subject to: } g(x) \leq u. \tag{$P(u)$}$$

Here, $u \in \mathbb{R}$ is not an optimization variable, but only a parameter. The optimal value is therefore a function of u, denoted by $V(u)$. We also introduce another family of problems:

$$W(\lambda) = \inf_{x \in X} f(x) + \lambda g(x), \tag{$D(\lambda)$}$$

where λ is a real parameter. These problems are called *penalized problems.*

1. Prove that V is a non-increasing function.
2. Prove the existence of u^- and u^+ in $\mathbb{R}$ such that for all $u \in \mathbb{R}$,
$$u < u^- \Longrightarrow V(u) = +\infty, \quad u \geq u^+ \Longrightarrow V(u) = V(u^+).$$
3. Prove that W is concave.
4. Prove that for all u and λ in $\mathbb{R}_+$,
$$V(u) \geq W(\lambda) - \lambda u. \tag{31}$$
5. Let $\lambda^* \in \mathbb{R}_+$, let x^* be an optimal solution to $D(\lambda)$ for the value $\lambda = \lambda^*$.
   (a) Prove that inequality 31 is an equality for the values $u^* := g(x^*)$ and λ^* and prove that x^* is an optimal solution to $P(u)$ for $u = u^*$.
   (b) Prove that if V is finite in the neighborhood of u^* and differentiable at u^*, then its derivative at u^* is $-\lambda^*$.
6. Let λ_1 and λ_2 in $\mathbb{R}_+$ be such that $\lambda_1 < \lambda_2$. Let x_1 and x_2 be respectively two optimal solutions to problem $D(\lambda)$ with the values $\lambda = \lambda_1$ and $\lambda = \lambda_2$, respectively. Let $u_1 = g(x_1)$ and $u_2 = g(x_2)$. Prove that $u_1 \geq u_2$.

Solution Note first that the compactness of X and the continuity of f and g imply the existence of an optimal to solution to problem $D(\lambda)$ for all $\lambda \in \mathbb{R}$, and to problem $P(u)$, for all $u \in \mathbb{R}$ such that $P(u)$ if feasible.

1. Let $u_1 \geq u_2$. The feasible set of $P(u)$ with $u = u_1$ is included into the feasible set of $P(u)$ with $u = u_2$. Therefore, $V(u_1) \geq V(u_2)$.
2. The values u^- and u^+ can be defined as follows:
$$u^- = \inf_{x \in X} g(x), \quad u^+ = g(x^+), \text{ where: } f(x^*) = \inf_{x \in X} f(x).$$
For all $u < u^-$, the problem is infeasible, therefore $V(u) = +\infty$, for all $u \geq u^+$, x^* is feasible and therefore optimal.
3. Let λ_1 and $\lambda_2 \in \mathbb{R}$, let $\lambda = (\lambda_1 + \lambda_2)/2$. Then, for all $x \in X$,
$$f(x) + \lambda g(x) = \frac{1}{2}(f(x) + \lambda_1 g(x)) + \frac{1}{2}(f(x) + \lambda_2 g(x)) \geq \frac{1}{2}(W(\lambda_1) + W(\lambda_2)).$$
Minimizing the left-hand-side, we obtain that $W(\lambda) \geq (W(\lambda_1) + W(\lambda_2))/2$. The concavity follows.
4.

5. Let u and λ in $\mathbb{R}$. If $P(u)$ is infeasible, the inequality is satisfied. Otherwise, for all feasible point x,
$$f(x) \geq f(x) + \lambda(g(x) - u) \geq W(\lambda) - \lambda u,$$
since $\lambda \geq 0$ and $g(x) - u \leq 0$. We obtain the inequality by minimizing the left-hand side.

6. (a) Let x be feasible for problem $P(u)$, with $u = g(x^*)$, i.e. $g(x) \leq g(x^*)$. We obtain:
$$f(x^*) = f(x^*) + \lambda g(x^*) - \lambda u = W(\lambda) - \lambda u \leq V(u),$$
by question 4. Therefore, x^* is an optimal solution to $P(u)$ with $u = g(x^*)$, and the above inequality is an equality.

   (b) Consider the function $h(u) = V(u) - (W(\lambda^*) - \lambda^* u)$. For all u, $h(u) \geq 0$ (by question 4) and $h(u^*) = 0$ by question 5a. If V is differentiable at u^*, then h is differentiable with a derivative which is equal to 0 (at u^*). Thus, $V'(u^*) = -\lambda^*$.

7. We have:
$$f(x_1) + \lambda_2 g(x_1) \geq f(x_2) + \lambda_2 g(x_2) \quad \text{and} \quad f(x_2) + \lambda_1 g(x_2) \geq f(x_1) + \lambda_1 g(x_1).$$
Additioning these inequalities, we obtain that
$$(\lambda_2 - \lambda_1) g(x_1) \geq (\lambda_2 - \lambda_1) g(x_2)$$
and thus $g(x_1) \geq g(x_2)$.

2.4.2 Quadratic penalty

Consider the following optimization problem:
$$\inf_{x \in \mathbb{R}^n} f(x), \quad \text{subject to: } e(x) = 0,$$
where $f\colon \mathbb{R}^n \to \mathbb{R}$ and $e\colon \mathbb{R}^n \to \mathbb{R}^p$ are twice continuously differentiable. We consider a feasible point $\bar{x} \in \mathbb{R}^n$ and a Lagrange multiplier $\bar{\mu} \in \mathbb{R}^p$ such that the KKT conditions hold true. We assume that the LICQ and the sufficient second-order optimality conditions are satisfied.

For a given $\delta > 0$, we define the quadratic penalty function $Q(x, \delta)$ as follows:
$$Q(x, \delta) = f(x) + \frac{1}{2\delta} \|e(x)\|^2.$$

1. Prove that there exist $\bar{\delta} > 0$ and two continuously differentiable mappings $\delta \in [0, \bar{\delta}] \mapsto X(\delta)$ and $\delta \in [0, \bar{\delta}] \mapsto M(\delta)$ such that for all $\delta \in (0, \bar{\delta}]$,
$$\nabla_x Q(X(\delta), \delta) = \nabla_x L(X(\delta), M(\delta)) = 0$$
and such that $X(0) = \bar{x}$ and $M(0) = \bar{\mu}$. To this purpose, you can use the mapping
$$\phi\colon (x, \mu, \delta) \in \mathbb{R}^{n+p+1} \mapsto \begin{pmatrix} \nabla_x L(x, \mu) \\ e(x) - \delta\mu \end{pmatrix}.$$

2. Prove that for all $\delta \in [0, \bar{\delta}]$,
$$\nabla^2_{xx} Q(X(\delta), \delta) = \nabla_{xx} L(X(\delta), M(\delta)) + \frac{1}{\delta} \nabla e(X(\delta)) \nabla e(X(\delta))^\top.$$

3. We now prove that there exists $\tilde{\delta} \in (0, \bar{\delta}]$ such that for all $\delta \in [0, \tilde{\delta}]$, $X(\delta)$ is a local minimizer of the mapping $x \mapsto Q(x, \delta)$. We do a proof by contradiction and therefore assume that there exists a sequence $(\delta_k)_{k \in \mathbb{N}}$ in $[0, \bar{\delta}]$, converging to 0, and such that for all $k \in \mathbb{N}$, $X(\delta_k)$ is not a local minimizer of $Q(\cdot, \delta_k)$. The sufficient second-order optimality conditions are therefore not satisfied, which implies that for all k, there exists a vector $h_k \in \mathbb{R}^n$ such that
$$\|h_k\| = 1 \quad \text{and} \quad h_k^\top \nabla^2_{xx} Q(X(\delta_k), \delta_k) h_k \leq 0.$$
Extracting if necessary a subsequence, we assume that $(h_k)_{k \in \mathbb{N}}$ converges to some vector $h \in \mathbb{R}^n$.

   (a) Prove that $h^\top \nabla^2_{xx} L(\bar{x}, \bar{\mu}) h \leq 0$ and that $\|h\| = 1$. What can be deduced?

   (b) Prove that
$$h_k^\top \nabla^2_{xx} Q(X(\delta_k), \delta_k) h_k \underset{k \to \infty}{\longrightarrow} +\infty$$
and conclude.

Solution

1. We apply the implicit function theorem to the mapping ϕ, which is continuously differentiable. We have $\phi(\bar{x}, \bar{\mu}, 0) = 0$ and
$$D_{(x,\mu)}\phi(\bar{x}, \bar{\mu}, 0) = \begin{pmatrix} \nabla^2_{xx} L(\bar{x}, \bar{\mu}) & \nabla e(\bar{x}) \\ \nabla e(\bar{x})^\top & 0 \end{pmatrix}.$$
Thus, $D_{(x,\mu)}\phi(\bar{x}, \bar{\mu}, 0)$ is the KKT matrix, known to be invertible under the sufficient second-order optimality conditions. There exist therefore $\bar{\delta} > 0$ and two continuously differentiable mapping $X\colon [0, \bar{\delta}] \to \mathbb{R}^n$ and $M\colon [0, \bar{\delta}] \to \mathbb{R}^p$ such that for all $\delta \in [0, \bar{\delta}]$,
$$\phi(X(\delta), M(\delta), \delta) = 0.$$
For $\delta > 0$, we have $M(\delta) = e(X(\delta))/\delta$ and therefore,
$$\nabla_x Q(X(\delta), \delta) = \nabla f(X(\delta)) + \frac{1}{\delta}\nabla e(X(\delta))e(X(\delta)) = \nabla_x L(X(\delta), M(\delta)).$$

2. We have for all x and $\delta > 0$ that
$$\nabla^2_{xx} Q(x, \delta) = \Big(\nabla^2 f(x) + \sum_{i=1}^p \frac{e_i(x)}{\delta}\nabla^2 e(x)\Big) + \frac{1}{\delta}\sum_{i=1}^p \nabla e_i(x)\nabla e_i(x)^\top.$$
The result follows.

3. For all $k \in \mathbb{N}$,
$$0 \geq h_k^\top \nabla^2_{xx} Q(X(\delta_k), \delta_k)h_k \geq h_k^\top \nabla^2_{xx} L(X(\delta_k), M(\delta_k))h_k.$$
Passing to the limit, we obtain that $\|h\| = 1$ and that
$$0 \geq h^\top \nabla^2_{xx} L(\bar{x}, \bar{\lambda})h.$$
Therefore, h is not in the critical cone, meaning that $\nabla e(\bar{x})^\top h \neq 0$. It follows that
$$h_k^\top \nabla_{xx} Q(X(\delta_k), \delta_k)h_k = \underbrace{h_k^\top \nabla_{xx} L(X(\delta_k), M(\delta_k))h_k}_{\longrightarrow h^\top \nabla^2_{xx} L(\bar{x},\bar{\mu})h} + \underbrace{\frac{1}{\delta_k}}_{\longrightarrow\infty} \underbrace{\|\nabla e(X(\delta_k))^\top h_k\|^2}_{\longrightarrow \|\nabla e(\bar{x})^\top h\|^2 > 0} \longrightarrow \infty.$$
Contradiction with $h_k^\top \nabla_{xx} Q(X(\delta_k), \delta_k)h_k \leq 0$.

2.4.3 Non-smooth penalty functions

Consider the following optimization problem:
$$\inf_{x \in \mathbb{R}^n} f(x) \quad \text{subject to: } e(x) = 0,$$
where $f\colon \mathbb{R}^n \to \mathbb{R}$ and $e\colon \mathbb{R}^p \to \mathbb{R}$ are twice continuously differentiable. Let $\bar{x} \in \mathbb{R}^n$ be a feasible point, satisfying the LICQ, the KKT conditions (with associated multiplier $\mu \in \mathbb{R}^p$), and the second-order sufficient optimality conditions.

For a given penalty parameter $c > 0$, we consider the penalty function Θ_2 defined as follows:
$$\Theta_2(x, c) = f(x) + c\|e(x)\|,$$
where $\|\cdot\|$ is the Euclidean norm.

1. Prove that the mapping $y \in \mathbb{R}^p \mapsto \|y\|$ is directionally differentiable and compute its directional derivative (two cases have to be considered).

2. Prove that Θ_2 is directionally differentiable and compute its directional derivative for a feasible point x.

3. Prove that if $c > \|\mu\|$, then $\bar{x}$ is a local minimizer of $\Theta_2(\cdot, c)$. To this purpose, you can adapt the proof given in the lecture.

Solution

1. Let $y \neq 0$. Then, the mapping $\|\cdot\|$ is continuously differentiable in the neighborhood of y, with the chain rule. We have
$$D\|y\| = \frac{D\Big(\sum_{i=1}^p y_i^2\Big)}{2\|y\|} = \frac{y^\top}{\|y\|}.$$
For $y = 0$, we have for all $h \in \mathbb{R}^p$ that
$$\frac{\|y + \varepsilon h\| - \|y\|}{\varepsilon} = \frac{\|\varepsilon h\|}{\varepsilon} = \|h\|.$$
Therefore, the mapping $\|\cdot\|$ is directionally differentiable, with
$$D(\|\cdot\|)(y;h) = \begin{cases} \frac{\langle y,h\rangle}{\|y\|} & \text{if } y \neq 0 \\ \|h\| & \text{if } y = 0. \end{cases}$$

2. The directional differentiability of Θ_2 follows from the Lemmas given above. For a feasible x, we obtain:
$$D\Theta_2(x,c;h) = Df(x)h + c\|De(x)h\|.$$

3. For all $x \in \mathbb{R}^n$, we have
$$L(x,\mu) = f(x) + \langle \mu, e(x)\rangle \leq f(x) + \|\mu\| \cdot \|e(x)\| = f(x) + c\|e(x)\| - (c - \|\mu\|)\|e(x)\|.$$
It follows that
$$\Theta_2(x,c) \geq L(x,\mu) + (c - \|\mu\|)\|e(x)\|.$$
The rest of the proof is the same as the one of the lecture.

2.4.4 Proximal method

We analyse in this exercise the proximal method for minimizing a real function f defined on a domain $X \subseteq \mathbb{R}^n$. Given a starting point $x_0 \in X$ and a positive number $c > 0$, the methods consists in computing a sequence $(x_k)_{k\in\mathbb{N}}$ as follows:
$$x_{k+1} \in \arg\min_{x\in X} \Big\{ f(x) + \frac{1}{2}c\|x - x_k\|^2 \Big\}. \tag{32}$$

In this exercise, we prove the convergence of the sequence $(x_k)_{k\in\mathbb{N}}$ to a minimizer of f in the following framework: $X = \mathbb{R}^n$, f is convex and continuously differentiable and posseses exactly one stationary point (which is then the unique global minimizer), denoted by $\bar{x}$.

We set $q_k = \nabla f(x_k)$, for all $k \in \mathbb{N}$.

1. Prove that the sequence $(x_k)_{k\in\mathbb{N}}$ is well-defined (i.e. there exists a sequence $(x_k)_{k\in\mathbb{N}}$ satisfying (32)).

2. Prove that for all $k \in \mathbb{N}$,
$$\|x_k - \bar{x}\|^2 = \|x_{k+1} - \bar{x}\|^2 + \frac{2}{c}\langle q_{k+1}, x_{k+1} - \bar{x}\rangle + \frac{1}{c^2}\|q_{k+1}\|^2.$$

3. Prove that for all $k \in \mathbb{N}$, $\langle q_{k+1}, x_{k+1} - \bar{x}\rangle \geq 0$.

4. Prove that $q_k \to 0$ and then that $x_k \to \bar{x}$.

Solution

1. The function f is lower bounded and $x \mapsto \|x - x_k\|^2$ is coercive, thus $f(\cdot) + \frac{1}{2}c\|x - x_k\|^2$ is coercive. Since it is also continuous, it posseses a minimizer (which is actually unique, since the function to minimize is strictly convex).

2. The optimality conditions read: $q_{k+1} + c(x_{k+1} - x_k) = 0$. Thus,
$$x_k - \bar{x} = (x_{k+1} - \bar{x}) + \frac{1}{c}q_{k+1}.$$
The inequality follows by computing the square of the norm of both sides.

3. It holds:
$$0 \geq f(\bar{x}) - f(x_{k+1}) \geq \langle q_{k+1}, \bar{x} - x_{k+1}\rangle.$$

4. We deduce that:
$$\|q_{k+1}\|^2 \leq c^2\big(\|x_k - \bar{x}\|^2 - \|x_{k+1} - \bar{x}\|^2\big)$$
and then: for all $K \in \mathbb{N}$,
$$\sum_{k=1}^{K} \|q_{k+1}\|^2 \leq c^2\big(\|x_0 - \bar{x}\|^2 - \|x_{K+1} - \bar{x}\|^2\big) \leq c^2\|x_0 - \bar{x}\|^2.$$
The sequence is therefore summable and converges to 0.

5. We deduce from question 2:
$$\|x_{k+1} - \bar{x}\|^2 \leq \|x_k - \bar{x}\|^2 - \frac{1}{c^2}\|q_{k+1}\|^2 \leq \|x_k - \bar{x}\|^2.$$
Therefore, the sequence $(x_k)_{k\in\mathbb{N}}$ is bounded. To prove its convergence, it suffices to prove that $\bar{x}$ is the unique accumulation point. Let $(x_{k_q})_{q\in\mathbb{N}}$ be a subsequence converging to $y \in \mathbb{R}^n$. Then, $Df(x_{k_q}) \to Df(y)$, which proves that $Df(y) = 0$ and finally, that $y = \bar{x}$.

2.4.5 Non-smooth optimization

Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be a continuously differentiable function. We analyze in this exercise the following optimization problem:
$$\inf_{x\in\mathbb{R}^n} F(x) := f(x) + c\|x\|_1,$$
where $c > 0$ is given and $\|x\|_1 = \sum_{i=1}^n |x_i|$. This kind of problem has been intensively studied for the following reason: if c is chosen in an appropriate way, a minimizer is likely to be *sparse*, that is, to have many coordinates equal to zero, which is of interest in some practical situations.

1. Prove that if c is sufficiently large, then 0 is a local minimizer of F.

2. Consider now a local minimizer $\bar{x}$ of F. Prove the existence of $\lambda \in \mathbb{R}^n$ such that
$$\nabla f(\bar{x}) + c\lambda = 0$$
and such that the following complementarity condition holds true: for all $i = 1, ..., n$,
$$\begin{cases} x_i > 0 & \Rightarrow \lambda_i = 1 \\ x_i = 0 & \Rightarrow \lambda_i \in [-1, 1] \\ x_i < 0 & \Rightarrow \lambda_i = -1. \end{cases}$$

3. For given real numbers $a > 0$ and b, prove the existence and uniqueness of a minimizer to the function $y \in \mathbb{R} \mapsto \frac{1}{2}ay^2 + by + |y|$. Compute it explicitely with the help of the optimality conditions formulated above.

4. Proximal methods are appropriate methods for minimizing F. At the iteration k, given the iterate x_k, the next iterate x_{k+1} is obtained as the solution to the following problem:
$$\inf_{x\in\mathbb{R}^n} f(x_k) + \langle \nabla f(x_k), x - x_k\rangle + a_k\|x\|_1 + \frac{\alpha_k}{2}\|x - x_k\|_2^2,$$
where $a_k > 0$ is given and $\|\cdot\|_2$ denotes the Euclidean norm. Justify the existence and uniqueness of x_{k+1} and compute it explicitely.

Solution

1. The claim holds true for $c > \|\nabla f(0)\|_\infty$. Consider the problem: $\inf_{x\in\mathbb{R}^n} f(x)$, subject to: $x = 0$. It has a unique feasible point, thus a unique solution, 0. The LICQ is satisfied, the KKT conditions $\nabla f(0)+\lambda = 0$ are also satisfied with $\lambda = -\nabla f(0)$. The sufficient second-order optimality conditions are trivially satisfied, since $C(0) = \{0\}$. Now, we observe that F is the non-smooth penalty function corresponding to the constrained problem given above. Then, by theorem, 0 is a local minimizer of F if $c > \|\lambda\|_\infty = \|\nabla f(0)\|_\infty$.

2. We first note that λ must be equal to $-\nabla f(x)/c$. We prove the result coordinate-wise. Let $i \in \{1, ..., n\}$. If $\bar{x}_i > 0$, then F is differentiable with respect to x_i in the neighborhood of $\bar{x}$, and thus
$$\frac{\partial F}{\partial x_i}(\bar{x}) = \frac{\partial f}{\partial x_i}(\bar{x}) + c = 0$$
and finally $\lambda_i = 1$. We proceed similarly when $\bar{x}_i < 0$. In the case $\bar{x}_i = 0$, we use the fact that F is only directionally differentiable with respect to x_i. We have
$$0 \le D_{x_i}F(\bar{x}; 1) = \frac{\partial f}{\partial x_i}(\bar{x}) + c \quad \text{and} \quad 0 \le D_{x_i}F(\bar{x}; -1) = \frac{\partial f}{\partial x_i}(\bar{x}) - c.$$
It follows that $-c \le \frac{\partial f}{\partial x_i}(\bar{x}) \le c$ and finally that $-1 \le \lambda_i \le 1$.

3. The function to be minimized is obviously continuous, coercive, and strictly convex. Thus, it has a unique minimizer. Let $\bar{y}$ be this unique minimizer and let λ be the associated multiplier.
   - If $\bar{y} > 0$, then $\lambda = 1$, $y = -(b + 1)/a$, which is possible only if $b < -1$.
   - If $\bar{y} < 0$, then $\lambda = -1$, $y = -(b - 1)/a$, which is possible only if $b > 1$.
   - If $\bar{y} = 0$, then $b + \lambda = 0$ and $\lambda \in [-1, 1]$, which is possible only if $b \in [-1, 1]$.

   We obtain now the following:
$$\bar{y} = \begin{cases} -(b+1)/a & \text{if } b < -1 \\ -(b-1)/a & \text{if } b > 1 \\ 0 & \text{otherwise.} \end{cases}$$

4. We denote now by $y(a, b)$ the minimizer obtained in the last point. The problem to be minimized now is equivalent to minimizing
$$\sum_{i=1}^{n} \frac{\alpha_k}{2} x_i^2 - \alpha_k x_{k,i} x_i + \frac{\partial f}{\partial x_i}(x_k) x_i + a_k |x_i|.$$
For minimizing the above cost function, it suffices to minimize each term with respect to x_i. Dividing the whole cost function by a_k, we are back to point 3 and finally obtain
$$x_{k+1,i} = y\Big(\frac{\alpha_k}{a_k}, \frac{1}{a_k}\Big(\frac{\partial f}{\partial x_i}(x_k) - \alpha_k x_{k,i}\Big)\Big).$$

2.4.6 Motivation for the augmented Lagrangian method

1. Let $A \in \mathbb{R}^{n\times n}$ be a symmetric matrix and let $B \in \mathbb{R}^{m\times n}$. We assume that the following property holds:
$$\forall h \in \mathbb{R}^n \backslash \{0\},\ Bh = 0 \Longrightarrow h^\top A h > 0.$$
Prove the existence of $c > 0$ such that $A + cB^\top B$ is positive definite.

2. Let $f : \mathbb{R}^n \to \mathbb{R}$ and $g : \mathbb{R}^n \to \mathbb{R}^m$ be continuously twice differentiable functions, let $\bar{x}$ be a feasible KKT point of the following optimization problem:
$$\inf_{x\in\mathbb{R}^n} f(x), \quad \text{subject to: } g(x) = 0, \tag{33}$$
with associated Lagrange multiplier $\bar{\lambda} \in \mathbb{R}^n$. We assume that the second-order sufficient optimality conditions are satisfied at $\bar{x}$. Prove the existence of $c > 0$ such that $\bar{x}$ is a local minimizer to the following function:
$$x \in \mathbb{R}^n \mapsto L(x, \bar{\lambda}) + \frac{c}{2}\|g(x)\|^2 := f(x) + \langle \lambda, g(x)\rangle + \frac{c}{2}\|g(x)\|^2.$$

3. Derive the local optimality of $\bar{x}$ (for problem (33)) from the above question.

Solution

1. We prove the result by contradiction. We assume that for all $n \in \mathbb{N}$, there exists $h_n \neq 0$ such that:
$$h_n^\top (A + nB^\top B) h_n \leq 0. \tag{34}$$
Without loss of generality, we assume that $\|h_n\| = 1$. We extract a converging subsequence of $(h_n)_{n\in\mathbb{N}}$ that we still denote $(h_n)_{n\in\mathbb{N}}$ for simplicity. Dividing (34) by n and passing to the limit, we obtain:
$$\bar{h}B^\top B\bar{h} = \|B\bar{h}\|^2 \leq 0.$$
Thus, $B\bar{h} = 0$ and then $\bar{h}^\top A\bar{h} > 0$. For n large enough, $h_n A h_n > 0$ and therefore
$$h_n^\top (A + nB^\top B) h_n \geq h_n^\top A h_n > 0,$$
which is a contradiction to (34).

2. We set $F(x) = L(x, \bar{\lambda}) + \frac{c}{2}\|g(x)\|^2$. We observe that:
$$D\|g(x)\|^2 = 2\sum_{i=1}^m g_i(x) Dg_i(x), \quad \nabla^2\|g(x)\|^2 = 2\sum_{i=1}^m g_i(x) D^2 g_i(x) + 2\sum_{i=1} Dg_i(x) Dg_i(x)^\top.$$
Therefore,
$$D_x F(\bar{x}) = 0 \quad \text{and} \quad D_{xx}F(\bar{x}) = D_{xx}L(x, \bar{\lambda}) + cD_x g(x)^\top D_x g(x).$$
For $c > 0$ large enough, $\nabla_{xx}F(\bar{x})$ is positive definite, as a consequence of question 1 and of the sufficient optimality conditions.

3. Let $\varepsilon > 0$ be small enough so that $\bar{x}$ minimizes F on $B(\bar{x}, \varepsilon)$. Then, for all feasible x with $\|x - \bar{x}\| \leq \varepsilon$, $f(x) = F(x) \geq F(\bar{x}) = f(\bar{x})$, which proves the local optimality of $\bar{x}$.

2.4.7 Augmented Lagrangian method

In this exercise, we analyse the augmented Lagrangian method for solving problems with equality constraints (only) of the form:
$$\inf_{x\in\mathbb{R}^n} f(x), \quad \text{subject to: } e(x) = 0,$$
where $f : \mathbb{R}^n \to \mathbb{R}$ and $e : \mathbb{R}^n \to \mathbb{R}^m$ are given continuously differentiable functions. The Lagrangian is denoted by $L(x, \mu) = f(x) + \langle \mu, e(x)\rangle$.

Given $x_0 \in \mathbb{R}^n$, $\mu_0 \in \mathbb{R}^m$ and $c > 0$, the method consists in computing iteratively two sequences $(x_k)_{k\in\mathbb{N}}$ and $(\mu_k)_{k\in\mathbb{N}}$ as follows: for all $k \in \mathbb{N}$:
$$x_{k+1} \in \arg\min_{x\in\mathbb{R}^n} \Big[L(x, \mu_k) + \frac{c}{2}\|e(x)\|^2\Big], \quad \mu_{k+1} = \mu_k + ce(x_{k+1}).$$
Note that no update mechanism for the penalty parameter c is employed here. We prove the convergence of the method in the following case: f is strongly convex, $e(x) = Ax - b$ where $A \in \mathbb{R}^{m\times n}$ has rank m and $b \in \mathbb{R}^m$. In this framework, there exists a unique optimal solution $\bar{x}$, which is then the unique KKT point. Let $\mu \in \mathbb{R}^m$ be the unique associated Lagrange multiplier.

To solve this exercise, the following result can be used: since f is uniformly convex and continuously differentiable, the mapping $x \in \mathbb{R}^n \mapsto \nabla f(x) \in \mathbb{R}^n$ is a bijection, with a continuous inverse function.

1. Justify that the sequence $(x_k)_{k\in\mathbb{N}}$ and $(\mu_k)_{k\in\mathbb{N}}$ are well-defined.

2. Write the KKT conditions for $\bar{x}$ and x_{k+1}. Find a relation between x_{k+1} and μ_{k+1}.

3. Prove that for all $k \in \mathbb{N}$,
$$\|\mu_k - \bar{\mu}\|^2 = \|\mu_{k+1} - \bar{\mu}\|^2 + c^2\|Ax^{k+1} - b\|^2 - 2c\langle Ax^{k+1} - b, \mu^{k+1} - \bar{\mu}\rangle.$$

4. Prove that $\langle Ax^{k+1} - b, \mu^{k+1} - \bar{\mu}\rangle \leq 0$. Hint: $b = A\bar{x}$.

5. Prove that $(\mu_k)_{k\in\mathbb{N}}$ is bounded and that $(x_k)_{k\in\mathbb{N}}$ is also bounded.

6. Prove that $\mu_k \to \bar{\mu}$ and $x_k \to \bar{x}$.

Solution

1. Link with previous exercise. Comments on the update of the λ.
2. Observe that $x \mapsto L(x, \lambda^k)$ is convex. Thus, for all $x \in \mathbb{R}^n$, $L(x, \lambda_k) \geq L(0, \lambda_k) + D_x L(0, \lambda_k)x$ and thus, by Cauchy-Schwarz inequality:
$$L(x, \lambda_k) + \frac{c}{2}\|x - x_k\|^2 \geq L(0, \lambda_k) + \|x_k\|^2 - (\|D_x L(0, \lambda_k)\| + 2\|x_k\|) \cdot \|x\| + \frac{c}{2}\|x\|^2 \underset{\|x\|\to\infty}{\longrightarrow} +\infty.$$
The coercivity property is satisfied, thus, there exists a minimizer x_{k+1}.
3. The optimality conditions are:
$$\nabla f(\bar{x}) + A^\top \bar{\lambda} = 0, \quad \nabla f(x_{k+1}) + A^\top \lambda_k + cA^\top(Ax_{k+1} - b) = 0.$$
It follows that: $\nabla f(x_{k+1}) + A^\top \lambda_{k+1} = 0$.
4. Direct consequence of $\lambda_{k+1} = \lambda_k + c(Ax_{k+1} - b)$.
5. It holds:
$$\langle Ax^{k+1} - b, \lambda^{k+1} - \bar{\lambda}\rangle = \langle x^{k+1} - \bar{x}, A^\top \lambda^{k+1} - A^\top \bar{\lambda}\rangle = \langle x^{k+1} - \bar{x}, -\nabla f(x_{k+1}) + \nabla f(\bar{x})\rangle \leq 0$$
by convexity of f.
6. The boundedness of $(\lambda_k)_k$ follows, as well as the boundedness of $(x_k)_{k\in\mathbb{N}}$, since:
$$x_k = (\nabla f)^{-1}(-A^\top \lambda_k).$$
From the summability of $\|Ax^k - b\|$, we derive that $Ax_k - b \to 0$. Then, every accumulation point of $(x_k, \lambda_k)_{k\in\mathbb{N}}$ satisfies the KKT conditions, thus is equal to $(\bar{x}, \bar{\lambda})$. This proves the convergence.

2.5 Interior point method

2.5.1 Interior-point method

We analyse in this exercise the interior-point method to solve the following convex quadratic problem:
$$\min_{x\in\mathbb{R}^n} f(x) := \frac{1}{2}x^\top Hx + \langle c, x\rangle, \quad \text{subject to: } Ax = b,\ x \geq 0, \tag{P}$$
where $H \in \mathbb{R}^{n\times n}$ is symmetric positive semi-definite and $A \in \mathbb{R}^{m\times n}$ has rank m.

We assume that the feasible set is bounded and that the optimal solution of (P) is unique. We denote it by $\bar{x}$ and assume that the KKT conditions are satisfied, with Lagrange multipliers $\bar{\lambda}$ and $\bar{s}$.

Finally, we assume that there exists (at least) one feasible point $\tilde{x}$ with strictly positive coordinates and introduce the following mapping F:
$$F : (x, \lambda, s) \in \mathbb{R}^{n+m+n} \mapsto \begin{pmatrix} \nabla f(x) + A^\top \lambda - s \\ Ax - b \\ x \times s \end{pmatrix} \in \mathbb{R}^{n+m+n},$$
where the multiplication $\times$ must be considered coordinate by coordinate.

1. For all $\mu > 0$, prove the existence and uniqueness of $(x(\mu), \lambda(\mu), s(\mu)) \in \mathbb{R}^{n+m+n}$ such that:
$$F(x(\mu), \lambda(\mu), s(\mu)) = \begin{pmatrix} 0 \\ 0 \\ \mu\mathbf{1} \end{pmatrix}, \quad x > 0, \quad s > 0,$$
where $\mathbf{1} \in \mathbb{R}^n$ has all its coordinates equal to 1. The mapping: $\mu > 0 \mapsto (x(\mu), \lambda(\mu), s(\mu))$ is called *central path*.

   To this purpose, observe first that $s(\mu)$ can be replaced by $\left(\frac{\mu}{s_1(\mu)}, ..., \frac{\mu}{s_n(\mu)}\right)$ in the above system. The point $x(\mu)$ is then a KKT point of an optimization problem with associated Lagrange multiplier $\lambda(\mu)$.
2. Prove that the following set: $\{(x(\mu), \lambda(\mu), s(\mu)) \,|\, \mu \in (0, 1]\}$ is bounded.
3. Deduce that: $(x(\mu), \lambda(\mu), s(\mu)) \underset{\mu\to 0}{\longrightarrow} (\bar{x}, \bar{\lambda}, \bar{s})$.
4. Prove that for all $x > 0$, $s > 0$, and $\lambda \in \mathbb{R}^m$, $DF(x, \lambda, s)$ is invertible.

Solution

1. We define: $\tilde{f}(x) = f(x) - \mu\big(\sum_{i=1}^n \ln(x_i)\big)$ and consider the following optimization problem:

$$\inf_{x\in\mathbb{R}^n} \tilde{f}(x), \quad \text{subject to: } Ax = b,\ x > 0. \tag{35}$$

We first prove the existence of an optimal solution. Let us define: $y_i = \max_{x\in\mathbb{R}^n} x_i$, subject to: $Ax = b$, $x \geq 0$. The number y_i is finite, since the feasible set is bounded. Then, let x be a feasible point such that for some $j \in \{1, ..., n\}$,

$$x_j < \exp\Big(\frac{1}{\mu}(f(\bar{x}) - f(\tilde{x})) - \sum_{i\neq j} \ln(y_i)\Big) := \hat{x}_j.$$

Then, it holds:

$$\tilde{f}(x) \geq f(\bar{x}) - \mu\Big(\sum_{i\neq j} \ln(y_i)\Big) - \mu \ln(\hat{x}_j) > f(\tilde{x}).$$

Therefore, x is not optimal. Therefore, (35) is equivalent to:

$$\inf_{x\in\mathbb{R}^n} \tilde{f}(x), \quad \text{subject to: } Ax = b,\ x \geq \hat{x}.$$

This problem posseses an optimal solution, satisfying the KKT conditions (the constraint $x \geq 0$ is not active).

2. It suffices to prove that the central path is bounded (for $\mu \in (0, 1]$) and to prove that the only possible accumulation point is $(\bar{x}, \bar{\lambda}, \bar{s})$. The boundedness of $(x(\mu))_{\mu\in(0,1]}$ derives from the feasibility. The boundedness of $(s(\mu))_{\mu\in(0,1]}$ can be established by finding a uniform lower-bound for $x(\mu)$. Boundedness of $\lambda(\mu)$ derives from the surjectivity condition.

3. The derivative is given by:

$$DF(x, \lambda, s) = \begin{pmatrix} H & A^\top & -\mathrm{Id} \\ A & 0 & 0 \\ S & 0 & X \end{pmatrix}$$

Let $(\delta x, \delta\lambda, \delta s) \in \mathrm{Ker}(DF(x, \lambda, s))$. It holds: $\delta s = -X^{-1}S\delta x$, thus:

$$(H + X^{-1}S)\delta x + A^\top \lambda = 0.$$

Thus,

$$A\delta x + A(H + X^{-1}S)^{-1}A^\top \lambda = 0.$$

Since $A\delta x = 0$, and since $A(H + X^{-1}S)^{-1}A^\top$ is positive definite, $\lambda = 0$. Thus, $(H + X^{-1}S)\delta x = 0$ and finally $\delta x = 0$ and $\delta s = 0$.

2.5.2 Convergence of the interior-point method

This exercise is the continuation of the previous one. We introduce the following notations:

$$\delta(x, \lambda, s, \mu) = \Big\|\frac{x \times s}{\mu} - \mathbf{1}\Big\|,$$
$$\mathcal{V}_\alpha = \big\{(x, \lambda, s, \mu) \,|\, Hx + A^\top + c - s = 0,\ x \geq 0,\ s \geq 0,\ Ax = b,\ \delta(x, \lambda, s, \mu) \leq \alpha\big\},$$

where $\alpha \in \big(0, \frac{1}{2}\big]$ is fixed, and where the norm $\|\cdot\|$ is the Euclidean norm. The set $\mathcal{V}_\alpha$ can be seen as a neighborhood of the central path.

We analyse in this exercise the interior-point method. The method computes a sequence $(x_k, \lambda_k, s_k, \mu_k) \in \mathbb{R}^{n+m+n} \times \mathbb{R}_+$ by induction.

(a) Find $(x_0, \lambda_0, s_0, \mu_0) \in \mathcal{V}_\alpha$, with $\mu_0 > 0$. Set $k = 0$.

(b) *Centering step.* Solve the equation:

$$F(x_k, \lambda_k, s_k) + DF(x_k, \lambda_k, s_k) \begin{pmatrix} \delta x \\ \delta \lambda \\ \delta s \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \mu \mathbf{1} \end{pmatrix}$$

and set $(\tilde{x}, \tilde{\lambda}, \tilde{s}) = (x_k, \lambda_k, s_k) + (\delta x, \delta \lambda, \delta s)$.

(c) *Affine step.* Solve the equation:

$$F(\tilde{x}, \tilde{\lambda}, \tilde{s}) + DF(\tilde{x}, \tilde{\lambda}, \tilde{s}) \begin{pmatrix} \delta x \\ \delta \lambda \\ \delta s \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

and compute:

$$\theta_k = \max_{\theta \in [0,1]} \theta \quad \text{subject to: } \big(\tilde{x} + \theta \delta x, \tilde{\lambda} + \theta \delta \lambda, \tilde{s} + \theta \delta s, (1-\theta)\mu\big) \in \mathcal{V}_\alpha.$$

Set: $(x_{k+1}, \lambda_{k+1}, s_{k+1}) = (\tilde{x}, \tilde{\lambda}, \tilde{s}) + \theta(\delta x, \delta \lambda, \delta s)$.

(d) Set $k = k + 1$ and go back to step (b).

1. Let $(x, s, \lambda, \mu) \in \mathcal{V}_\alpha$, with $\mu > 0$. Prove the following estimates:

$$f(x) \leq f(\bar{x}) + \langle s, x \rangle \leq f(\bar{x}) + \mu(\sqrt{n}\alpha + n).$$

To this purpose, prove first the following relations:

$$f(\bar{x}) - f(x) = \langle s - A^\top \lambda, \bar{x} - x \rangle + \frac{1}{2}(x - \bar{x})^\top H (x - \bar{x}) \quad \text{and} \quad \langle s, x \rangle = \langle s \times x, \mathbf{1} \rangle.$$

2. We analyse now the step (b) of the numerical method. Our analysis is based on the following result: for all u and $v \in \mathbb{R}^n_+$, for all $\alpha \in \mathbb{R}^n_+$ with $\alpha > 0$,

$$\|u \times v\| \leq \frac{1}{\sqrt{8}} \Big\| (\alpha \times u) + \Big(\frac{1}{\alpha} \times v\Big) \Big\|^2,$$

where $\frac{1}{\alpha}$ is the vector of coordinates $\frac{1}{\alpha_i}$. We prove that:

$$\delta(\tilde{x}, \tilde{s}, \mu) \leq \frac{1}{\sqrt{8}} \cdot \frac{\delta(x_k, s_k, \mu)^2}{1 - \delta(x_k, s_k, \mu)}.$$

To this purpose, prove first the following relations:

(a) $\delta(\tilde{x}, \tilde{s}, \mu) = \frac{1}{\mu} \|\delta x \times \delta s\|$

(b) $\sqrt{\frac{s_k}{x_k}} \times \delta x + \sqrt{\frac{x_k}{s_k}} \times \delta s = \frac{\mu}{\sqrt{s_k \times x_k}} \mathbf{1} - \sqrt{s_k \times x_k}$.

(c) $\delta(\tilde{x}, \tilde{s}, \mu) \leq \frac{1}{\sqrt{8}} \|\frac{\mu}{x_k \times s_k}\|_\infty \delta(x_k, s_k, \mu)^2$.

The square root and the division must be here considered coordinate by coordinate.

Solution

1. Moreover, it holds:

$$\langle s, x \rangle = \langle s \times x, \mathbf{1} \rangle = \mu \langle \frac{s \times x}{\mu} - \mathbf{1}, \mathbf{1} \rangle + \mu \langle \mathbf{1}, \mathbf{1} \rangle \leq \mu \| \frac{x \times s}{\mu} - \mathbf{1} \| \cdot \|\mathbf{1}\| + \mu n \leq \mu \alpha \sqrt{n} + \mu n.$$

The derivative is given by:

$$DF(x, \lambda, s) = \begin{pmatrix} H & A^\top & -\mathrm{Id} \\ A & 0 & 0 \\ S & 0 & X \end{pmatrix}$$

Let $(\delta x, \delta\lambda, \delta s) \in \mathrm{Ker}(DF(x, \lambda, s))$. It holds: $\delta s = -X^{-1}S\delta x$, thus:

$$(H + X^{-1}S)\delta x + A^\top \lambda = 0.$$

Thus,

$$A\delta x + A(H + X^{-1}S)^{-1}A^\top \lambda = 0.$$

Since $A\delta x = 0$, and since $A(H + X^{-1}S)^{-1}A^\top$ is positive definite, $\lambda = 0$. Thus, $(H + X^{-1}S)\delta x = 0$ and finally $\delta x = 0$ and $\delta s = 0$.

2. Let us prove (a).

$$\delta(\tilde{x}, \tilde{s}, \mu) = \left\| \frac{x_k \times s_k - \delta x \times s_k - \delta s \times x_k - \mu \mathbf{1}}{\mu} + \frac{\delta x \times \delta s}{\mu} \right\|$$

We prove (b). It holds:

$$s_k \times \delta x + x_k \times \delta s = \mu \mathbf{1} - x_k \times s_k.$$

We divide coordinate by coordinate by $\sqrt{x_k \times s_k}$. We prove (c).

2.6 Projected gradient

2.6.1 Projected gradient

Let f be a continuously differentiable function, let K be a non-empty closed and convex set. We consider the projected gradient method for solving the following optimization problem:

$$\inf_{x \in \mathbb{R}^n} f(x) \quad \text{subject to: } x \in K.$$

At the iteration k, a point $x_k \in K$ has been obtained. For all $\alpha \geq 0$, we set

$$x_k(\alpha) = \mathrm{Proj}_K(x_k - \alpha \nabla f(x_k))$$

and for a fixed constant $\sigma \in (0, 1)$, we say that Armijo's rule holds if

$$f(x_k(\alpha)) \leq f(x_k) + \sigma \langle \nabla f(x_k), x_k(\alpha) - x_k \rangle.$$

The aim of the exercise is to prove that there exists $\bar{\alpha} > 0$ such that for all $\alpha \in [0, \bar{\alpha}]$, Armijo's rule holds.

We assume here that x_k does not satisfy the optimality conditions, that is to say,

$$\mathrm{Proj}_K(x_k - \nabla f(x_k)) - x_k = x_k(1) - x_k \neq 0.$$

For this exercise, you can use the following inequality without proof:

$$\|x_k(\alpha) - x_k\| \geq \alpha \|x_k(1) - x_k\|, \quad \forall \alpha \in [0, 1].$$

A direct consequence of this inequality is that for $\alpha \in (0, 1]$, $\|x_k(\alpha) - x_k\| > 0$.

1. Prove the following inequalities:

$$\alpha \|x_k(\alpha) - x_k\| \cdot \|x_k(1) - x_k\| \leq \|x_k(\alpha) - x_k\|^2 \leq \alpha \langle \nabla f(x_k), x_k - x_k(\alpha) \rangle.$$

2. For all $\alpha > 0$, we denote by $\tilde{x}_k(\alpha)$ a point on the segment $[x_k, x_k(\alpha)]$. Prove that if α is sufficiently small, then

$$\frac{(1 - \sigma)\langle \nabla f(x_k), x_k - x_k(\alpha) \rangle}{\|x_k(\alpha) - x_k\|} \geq \frac{\langle \nabla f(x_k) - \nabla f(\tilde{x}_k(\alpha)), x_k - x(\alpha) \rangle}{\|x_k(\alpha) - x_k\|}.$$

To this purpose, you can look for a lower bound of the left-hand side and prove that the right-hand side converges to 0, when $\alpha \to 0$. Conclude.

Solution

1. Since $x_k \in K$ and $x_k(\alpha) = \mathrm{Proj}_K(x_k - \alpha \nabla f(x_k))$, we have

$$\langle x_k - \alpha \nabla f(x_k) - x_k(\alpha), x_k - x_k(\alpha) \rangle \leq 0.$$

   It follows that

$$\|x_k - x_k(\alpha)\|^2 \leq \alpha \langle \nabla f(x_k), x_k - x_k(\alpha) \rangle.$$

2. Combining the obtained inequality with $\|x_k(\alpha) - x_k\| \geq \alpha \|x_k(1) - x_k\|$, we obtain that

$$\alpha^2 \|x_k(1) - x_k\|^2 \leq \alpha \langle \nabla f(x_k), x_k - x_k(\alpha) \rangle.$$

   It follows that

$$\langle \nabla f(x_k), x_k(\alpha) - x_k \rangle \leq -\|x_k(\alpha) - x_k\| \cdot \|x_k(1) - x_k\|$$

3. Clearly, the left-hand side is bounded by $(1 - \sigma)\|x_k(1) - x_k\| < 0$. Moreover,

$$\frac{|\langle \nabla f(\tilde{x}_k(\alpha)) - \nabla f(x_k), x(\alpha) - x_k \rangle|}{\|x_k(\alpha) - x_k\|} \leq \|\nabla f(\tilde{x}_k(\alpha)) - \nabla f(x_k)\| \underset{\alpha \to 0}{\longrightarrow} 0.$$

   Therefore, the right-hand side converges to 0. The inequality is satisfied, for α sufficiently small.

4. Let $\alpha \geq 0$ be sufficiently small, so that the inequality is satisfied. Let $\tilde{x}_k(\alpha)$ be such that

$$f(x_k(\alpha)) = f(x_k) + \langle \nabla f(\tilde{x}_k(\alpha)), x_k(\alpha) - x_k \rangle.$$

   It is easy to see that Armijo's rule holds if and only if the inequality is satisfied, which is the case.